MBG 1201 Introduction to Modern Biology

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Lipids are a diverse group of hydrophobic molecules

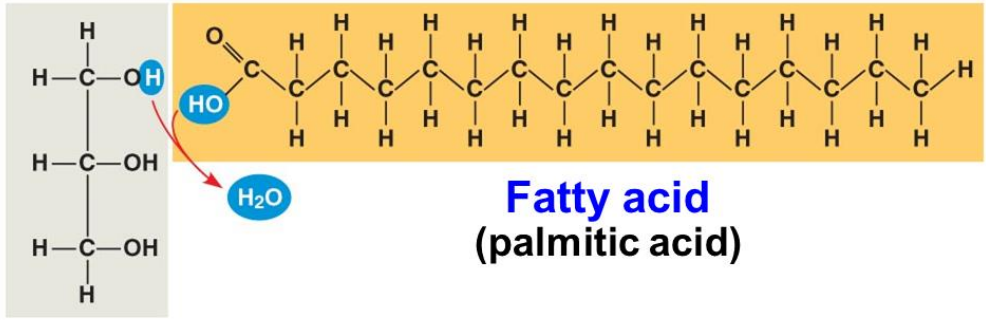
- Lipids are the one class of large biological molecules that do not form polymers.
- They are generally not big enough to be considered macromolecules.
- Their common property is that they are hydrophobic.
- Although they may have some polar bonds associated with oxygen, lipids consist mostly of hydrocarbon regions with relatively non-polar C—H bonds.
- The most biologically important lipids are fats, phospholipids, and steroids.

Fats (triglycerides)

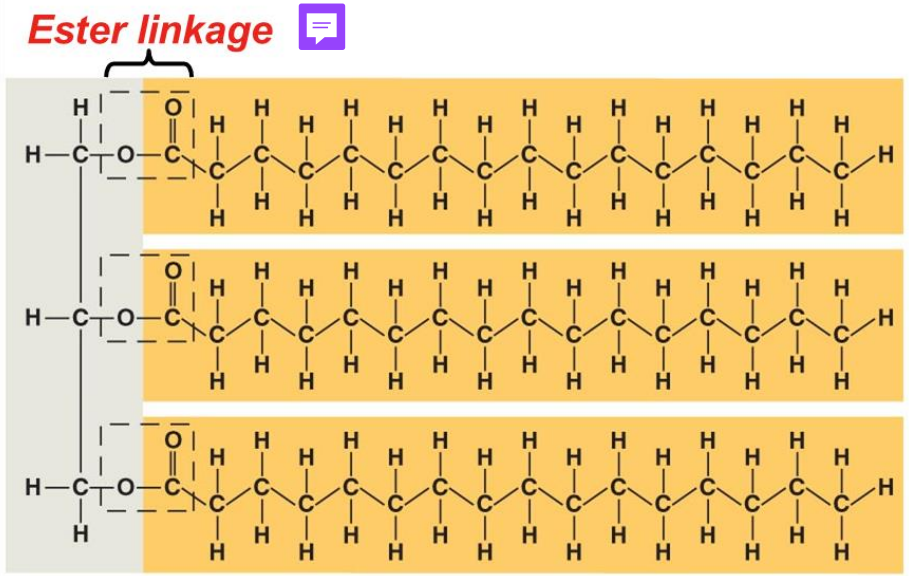
- Fats are composed of two types of smaller molecules: glycerol and fatty acids joined together by dehydration reactions.
- Glycerol is a three-carbon alcohol with a hydroxyl group attached to each carbon.
- A fatty acid consists of a carboxyl group attached to a long hydrocarbon chain (16-18 atoms in length).
- This fatty acid hydrocarbon can be either saturated or unsaturated.
- In a fat, three fatty acids are joined to glycerol by an ester linkage creating a triacylglycerol, or triglyceride. ester linkage is a covalent bond between a hydroxyl group and a carboxyl group.
- Fats separate from water because the water molecules hydrogen-bond to one another and exclude the fats.

Fats (triglycerides)

The Synthesis and Structure of a fat = triacylglycerol



(a) Dehydration reaction in the synthesis of a fat



(b) Fat molecule (*Triglyceride*)

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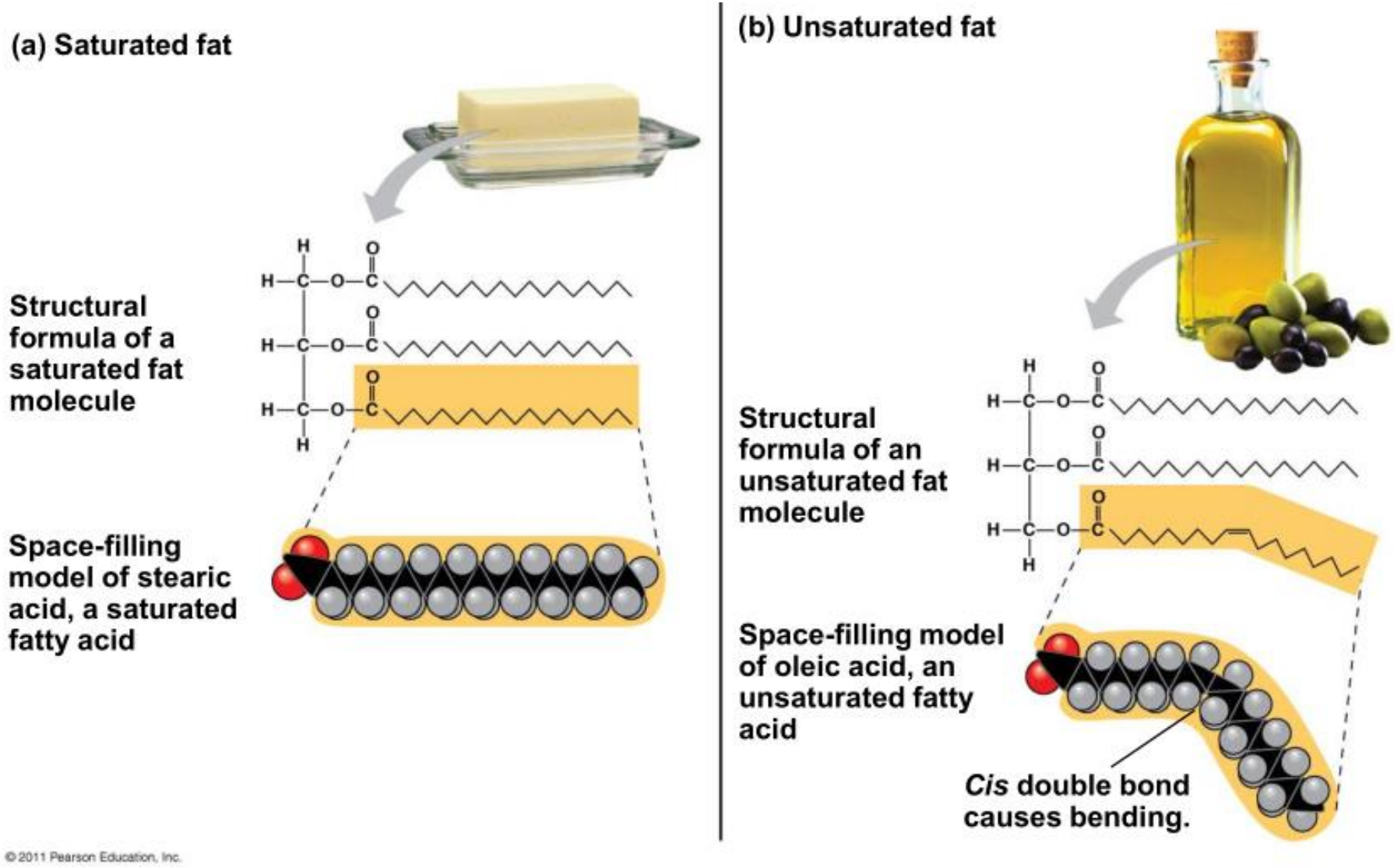
Fats (triglycerides)

Fatty acids vary in length (number of carbons) and in the number and locations of double bonds.

A **saturated fat** has three saturated fatty acid tails which can pack together tightly. This is why substances with a high proportion of saturated fats are solid at room temperature.
Saturated fatty acids have the maximum number of hydrogen atoms possible and no double bonds.
All C - C bonds are single. Most animal fats are saturated.

Unsaturated fatty acids have one or more **double bonds** C = C. Each of these bonds makes a bend, and **bent tails** cannot pack together tightly. This is why most substances with a high proportion of unsaturated fats are oils—liquid at room temperature. Plant fats and fish fats are usually unsaturated.

Fats (triglycerides)



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Fats (triglycerides)

A diet rich in saturated fats may contribute to cardiovascular disease through plaque deposits.
Hydrogenation is the process of converting unsaturated fats to saturated fats by adding hydrogen.
Hydrogenating vegetable oils also creates unsaturated fats with trans double bonds = trans fats. They are solid at room temperature because the special trans double bond keeps the tails straight, allowing them to pack just as tightly as saturated fats.
These trans fats may contribute more than saturated fats to cardiovascular disease.
Certain unsaturated fatty acids are in the human body. These must be supplied in the diet. These essential fatty acids include the omega-3 fatty acids, required for normal growth, and thought to provide protection against cardiovascular disease.

Fats (triglycerides)

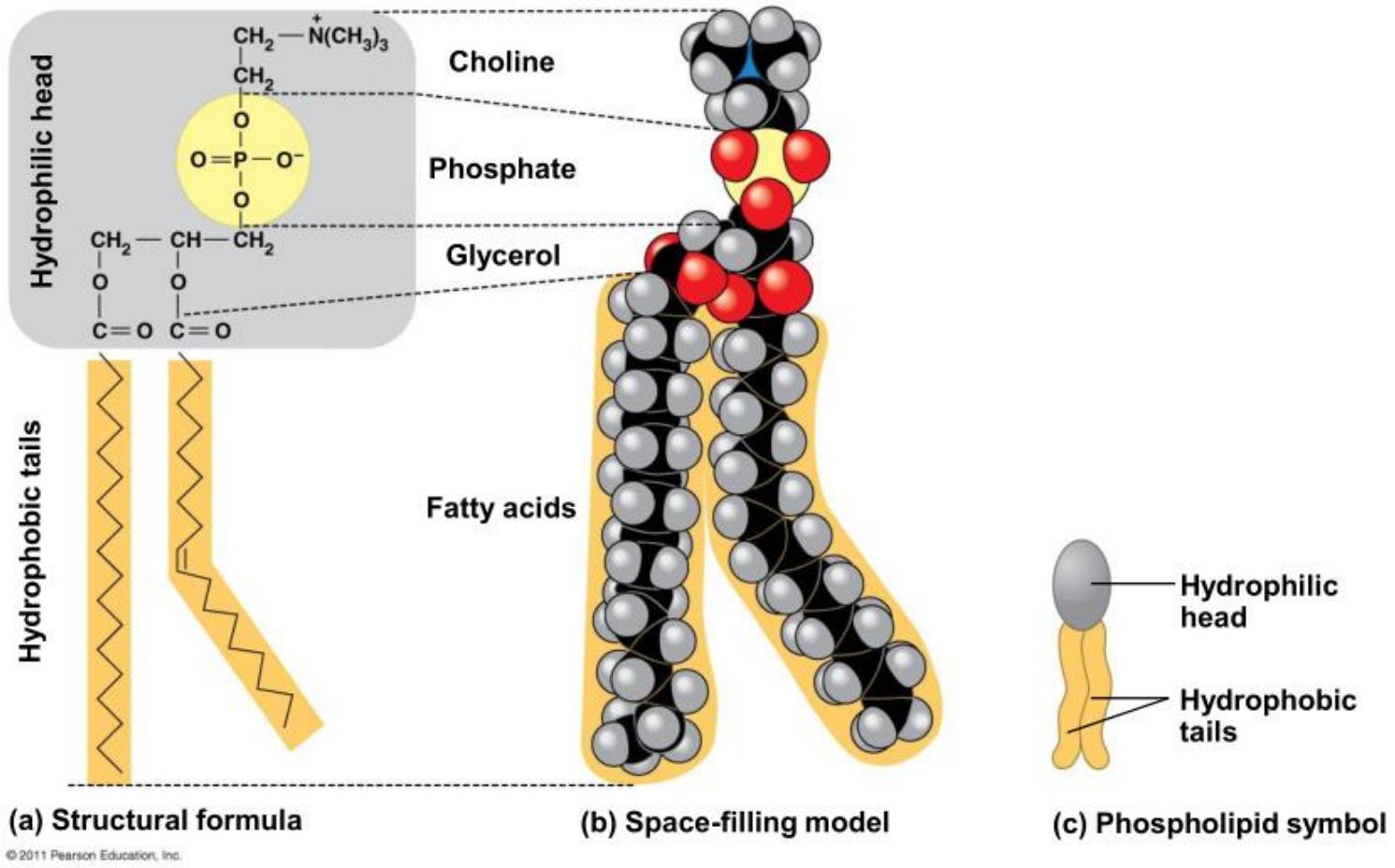
The major function of fats is energy storage.

Humans and other mammals stock their long-term food reserves in adipose cells which swell and shrink as fat is deposited and withdrawn from storage..

Adipose tissue also cushions vital organs and a layer of fat beneath the skin (subcutaneous layer) insulates the body.

Phospholipids

In a phospholipid, two fatty acids and a phosphate group are attached to glycerol.
The two fatty acid tails are hydrophobic, but the phosphate group and its attachments
form a hydrophilic head.
A phospholipid is an amphipathic molecule: hydrophillic head and hydrophobic tails.

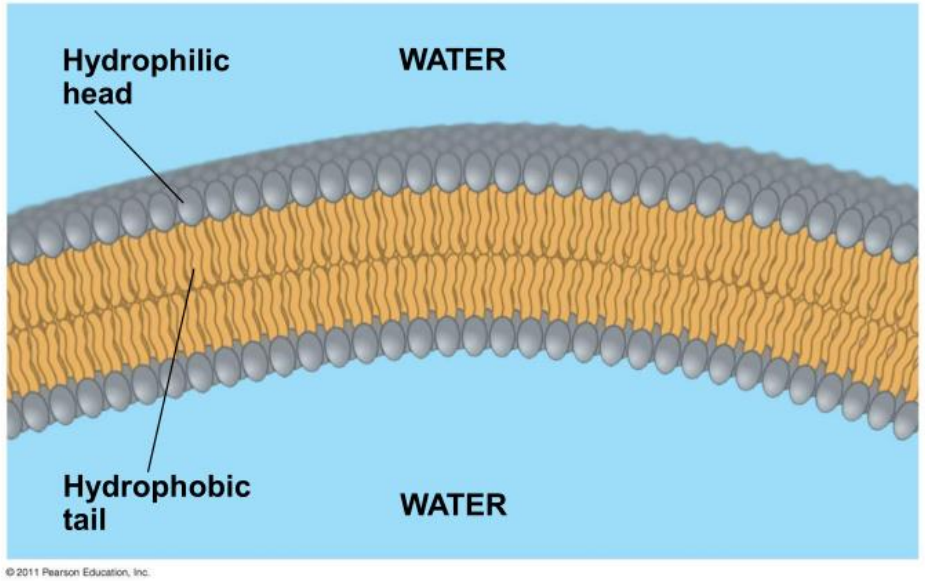


Phospholipids

When phospholipids are added to water, they self-assemble into a bilayer, with the hydrophobic tails pointing toward the interior.

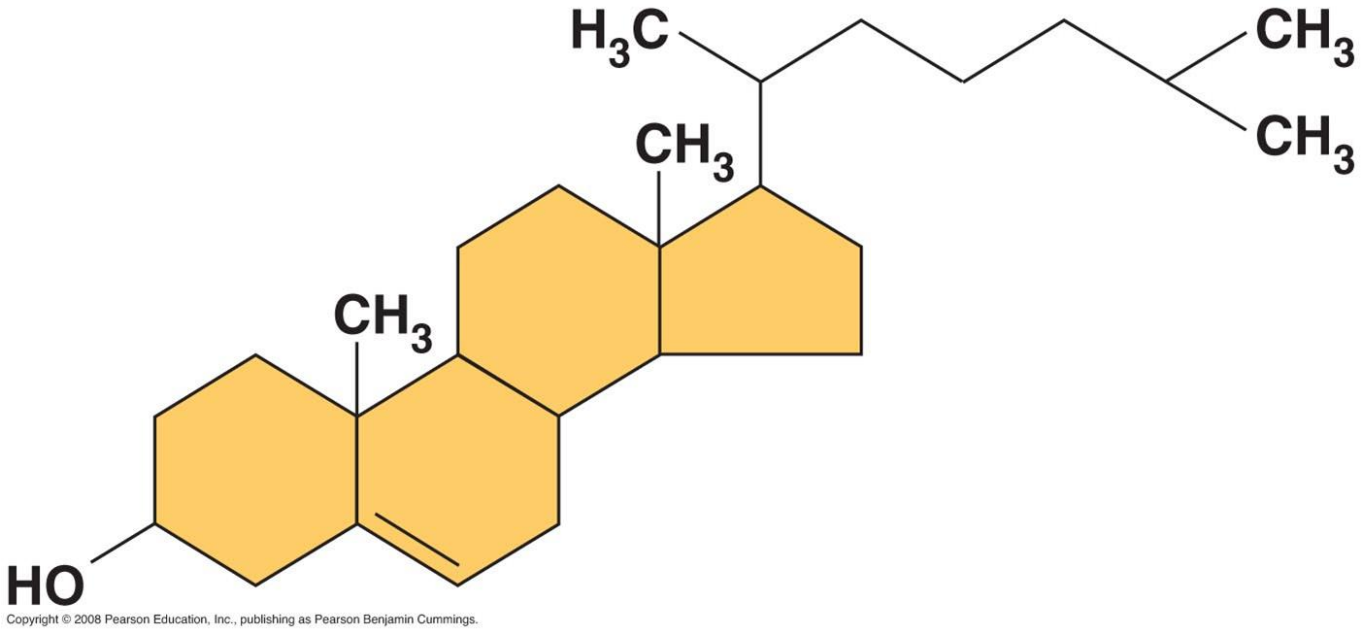
- The structure of phospholipids results in a bilayer arrangement found in cell membranes.
- Phospholipids are the major component of all cell membranes.

The existence of cells depends on the properties of phospholipids.



Steroids

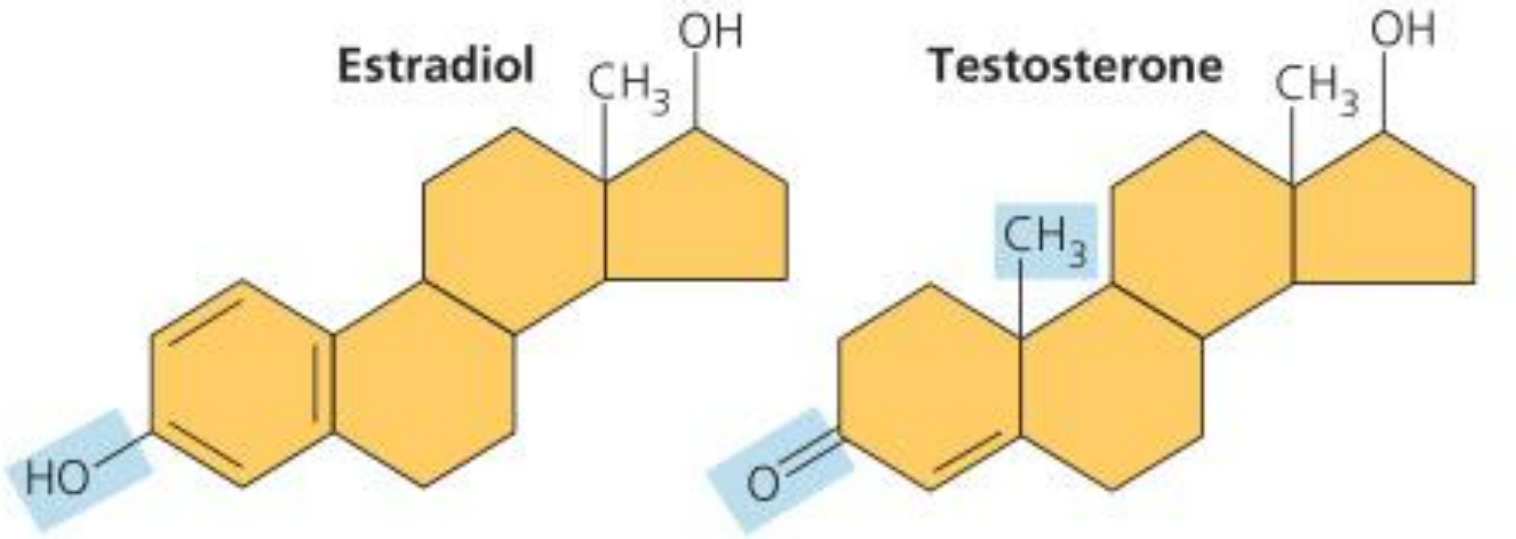
Steroids are lipids characterized by a carbon skeleton consisting of four fused rings.
Cholesterol, an important steroid, is a component in animal cell membranes.
Although cholesterol is essential in animals, high levels in the blood may contribute to cardiovascular disease.



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Steroids

Different chemical groups attached to the 4 rings of steroids give molecules different characteristics. The chemical groups affect molecular shape, contributing to function.



Proteins have many structures, resulting in a wide range of functions

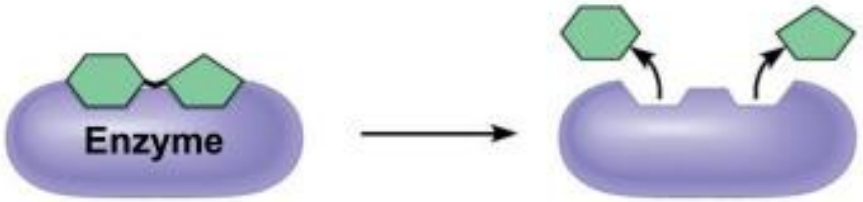
Proteins account for more than 50% of the dry mass of most cells.
Protein functions include structural support, storage, transport, cellular communications, movement, speeding up chemical reactions and defense against foreign substances.
Proteins are polymers called polypeptides.
Amino acids are the monomers used to build proteins.

Proteins have many structures, resulting in a wide range of functions

Enzymatic proteins

Function: Selective acceleration of chemical reactions

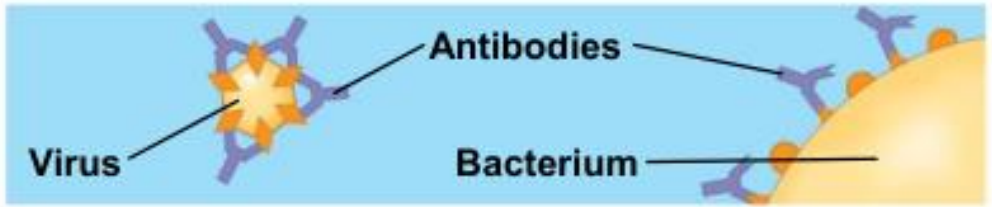
Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.



Defensive proteins

Function: Protection against disease

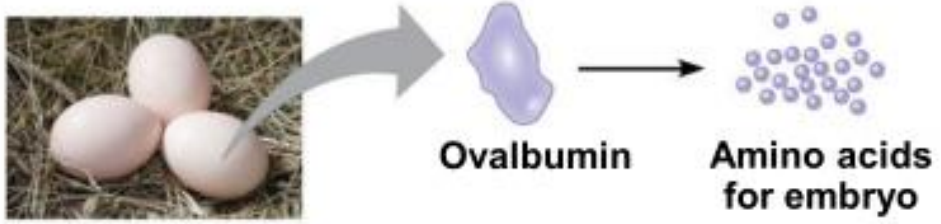
Example: Antibodies inactivate and help destroy viruses and bacteria.



Storage proteins

Function: Storage of amino acids

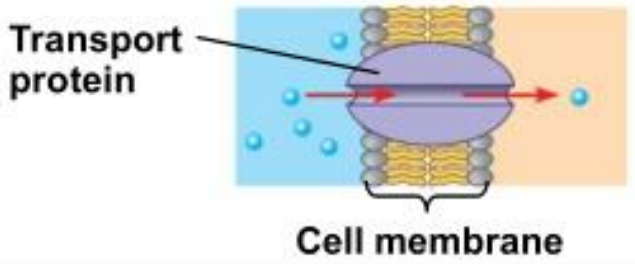
Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.



Transport proteins

Function: Transport of substances

Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across cell membranes.



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Enzymes

Enzymes are LARGE proteins that act as catalysts to speed up the rate of chemical reactions in cells.

Enzymes are specific. They must have a shape-match with molecules in the chemical reaction.

Enzymes can perform their functions repeatedly, working constantly to carry out the processes of life.

Proteins have many structures, resulting in a wide range of functions

Hormonal proteins

Function: Coordination of an organism's activities

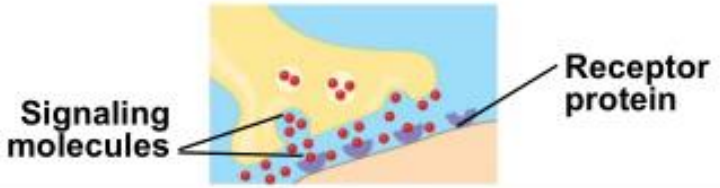
Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration



Receptor proteins

Function: Response of cell to chemical stimuli

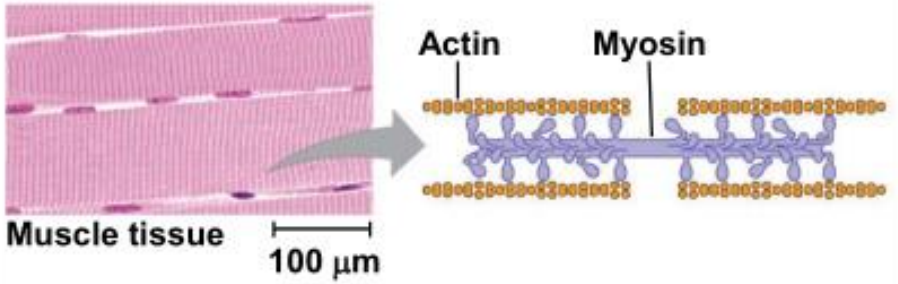
Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.



Contractile and motor proteins

Function: Movement

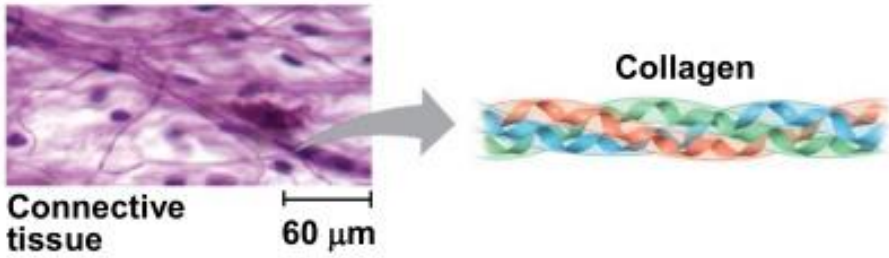
Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.



Structural proteins

Function: Support

Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.



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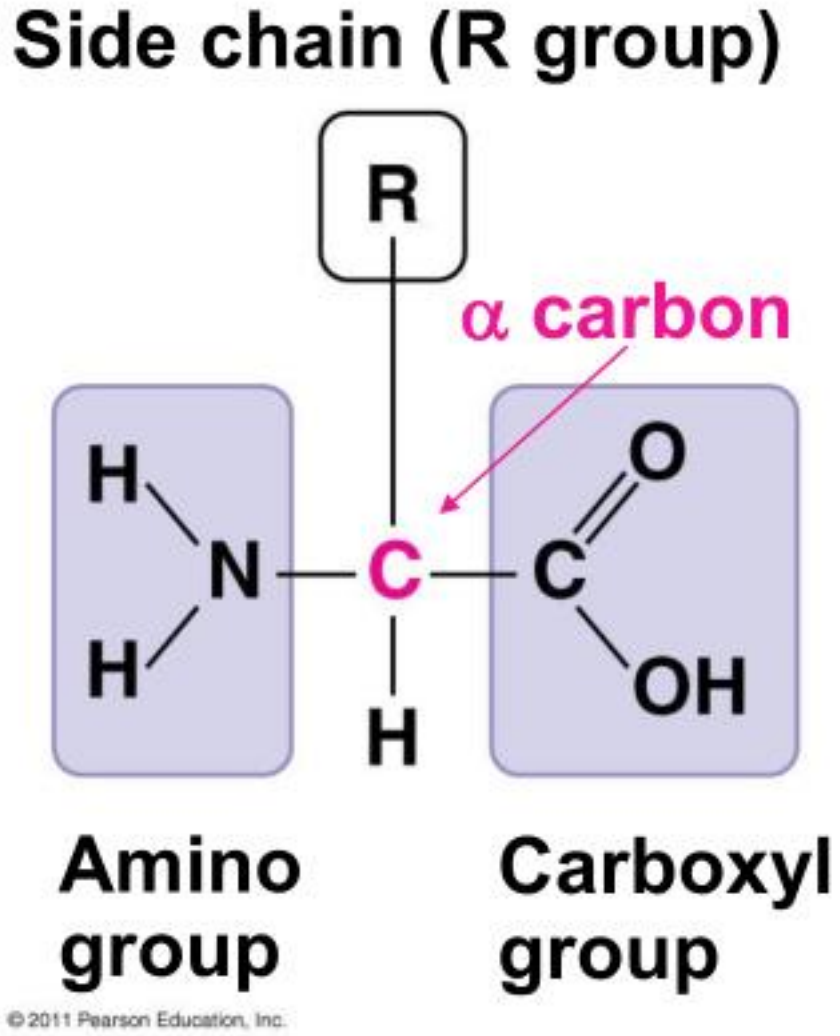
Proteins have many structures, resulting in a wide range of functions

Polypeptides are unbranched polymers built from a set of 20 amino acids (monomers).
A protein is a biologically functional molecule that consists of one or more polypeptides
The sequence of amino acids determines a protein's 3D structure.
A protein's structure determines its function.
A wide variety of proteins can be made from a few monomers by varying the amino acid sequence.

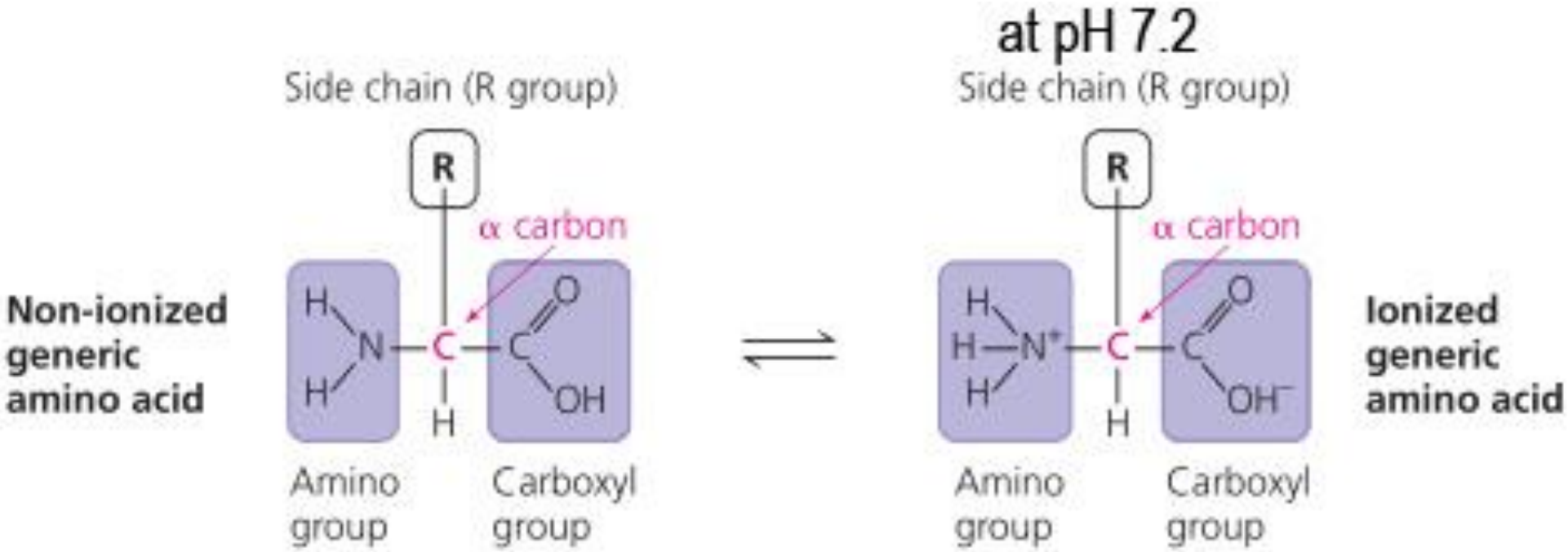
Amino Acid Monomers

- Amino acids are organic molecules with carboxyl and amino groups
- Amino acids differ in their properties due to differing side chains, called R groups

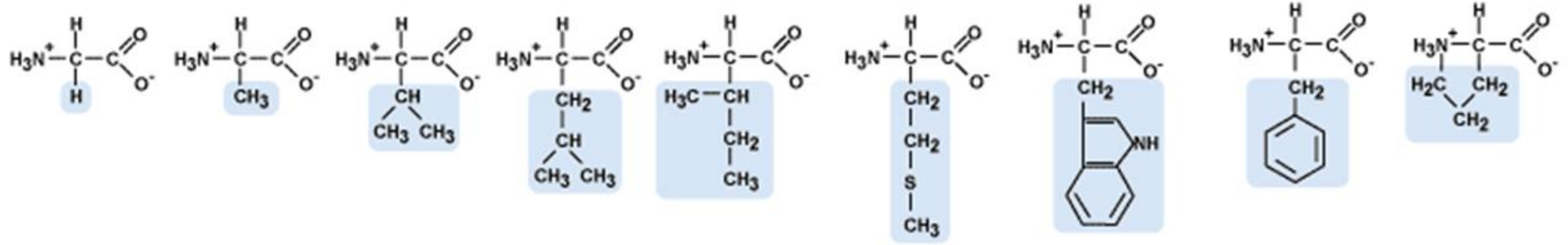
There are 20 different amino acids because there are 20 different side chains.



Amino Acid Monomers

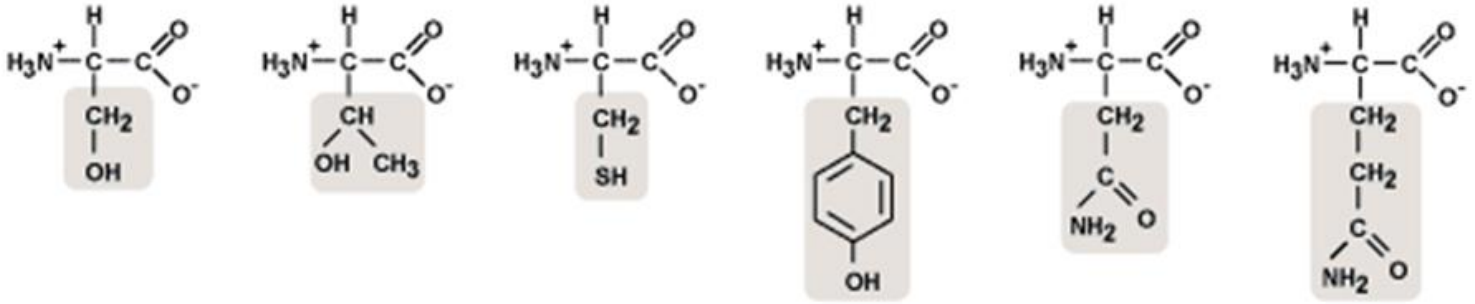


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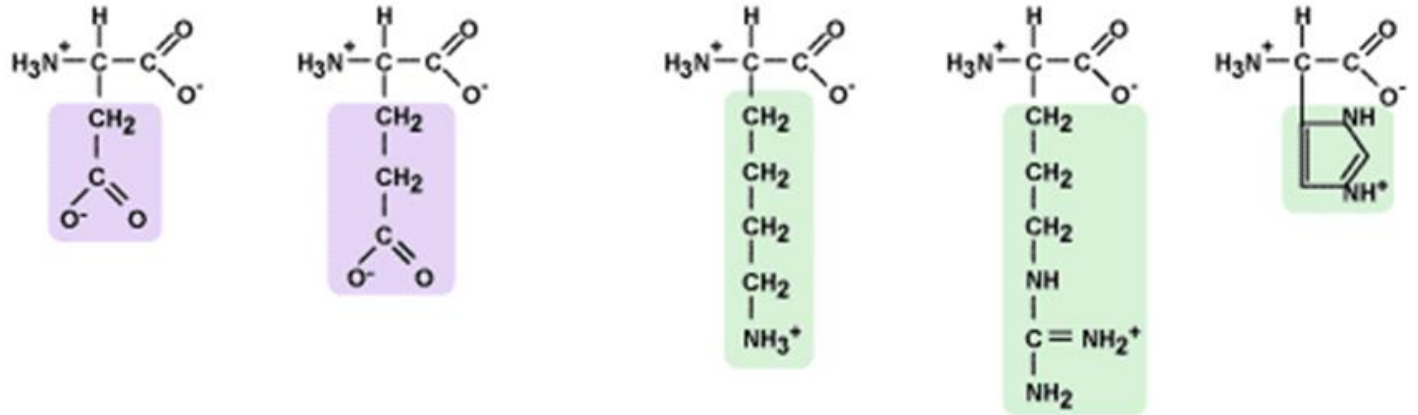
Glycine (Gly) Alanine (Ala) Valine (Val) Leucine (Leu) Isoleucine (Ile) Methionine (Met) Tryptophan (Trp) Phenylalanine (Phe) Proline (Pro)

POLAR



Serine (Ser) Threonine (Thr) Cysteine (Cys) Tyrosine (Tyr) Asparagine (Asn) Glutamine (Gln)

Electrically Charged



Acidic

Aspartic Acid (Asp) Glutamic Acid (Glu)

Lysine (Lys)

Basic

Arginine (Arg) Histidine (His)

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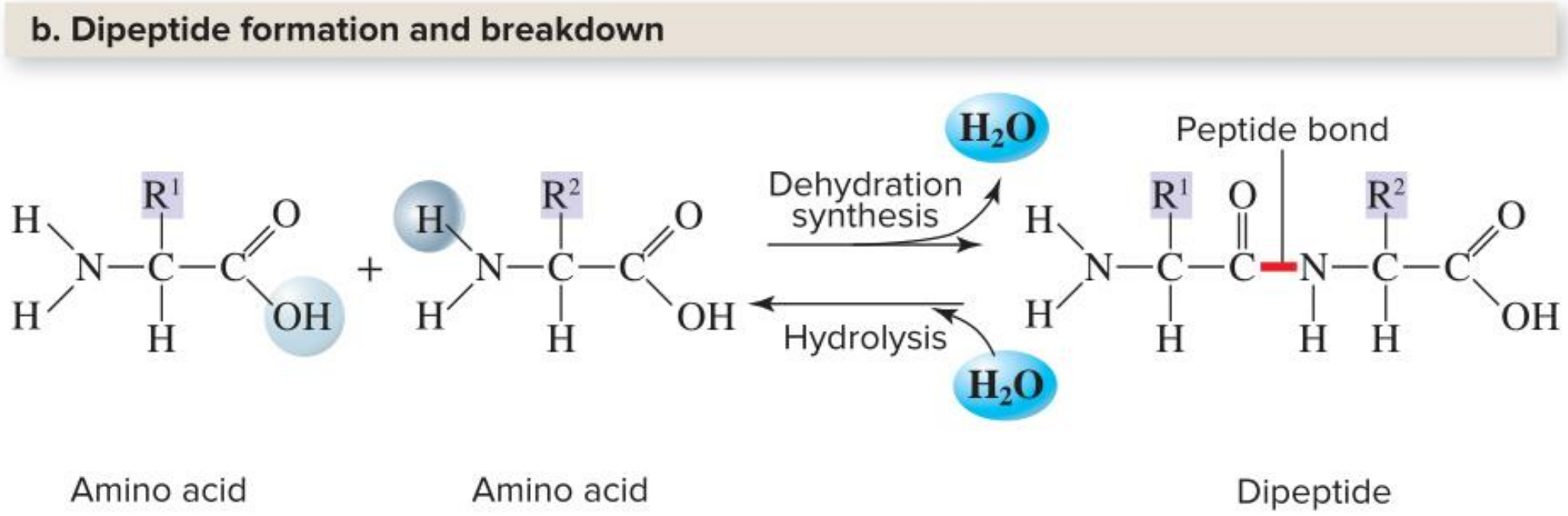
Amino Acid Polymers

Amino acids are linked by covalent bonds called peptide bonds (formed between C and N) (forming directional chains)

A polypeptide is a polymer of amino acids.

Polypeptides range in length from a few to more than a thousand monomers.

Each polypeptide has a **unique** linear sequence of amino acids.



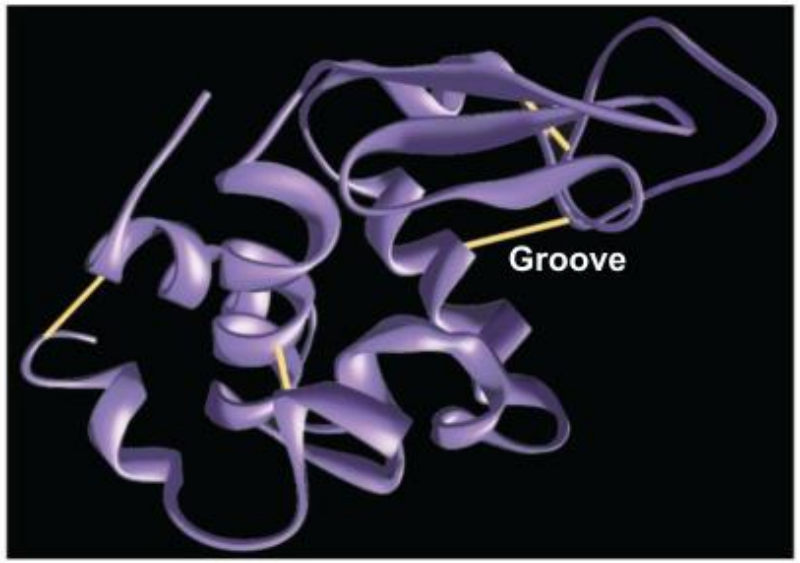
polypeptide backbone does not involve side chains.

Protein Structure and Function

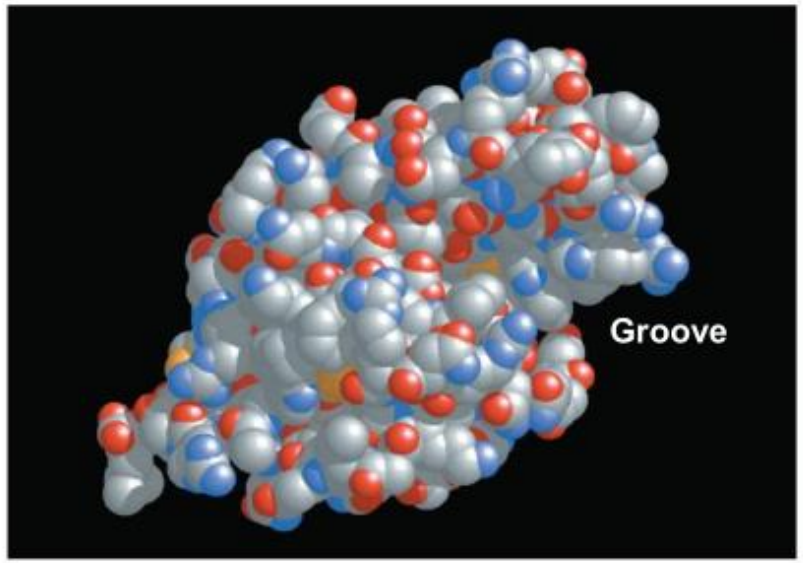
A functional protein is not just a polypeptide chain, but one or more polypeptides precisely twisted, folded, and coiled into a molecule of unique shape.

the amino acid sequence of each polypeptide determines what 3-dimensional structure the protein will have under normal cellular conditions.

A protein's structure determines its function.



(a) A ribbon model

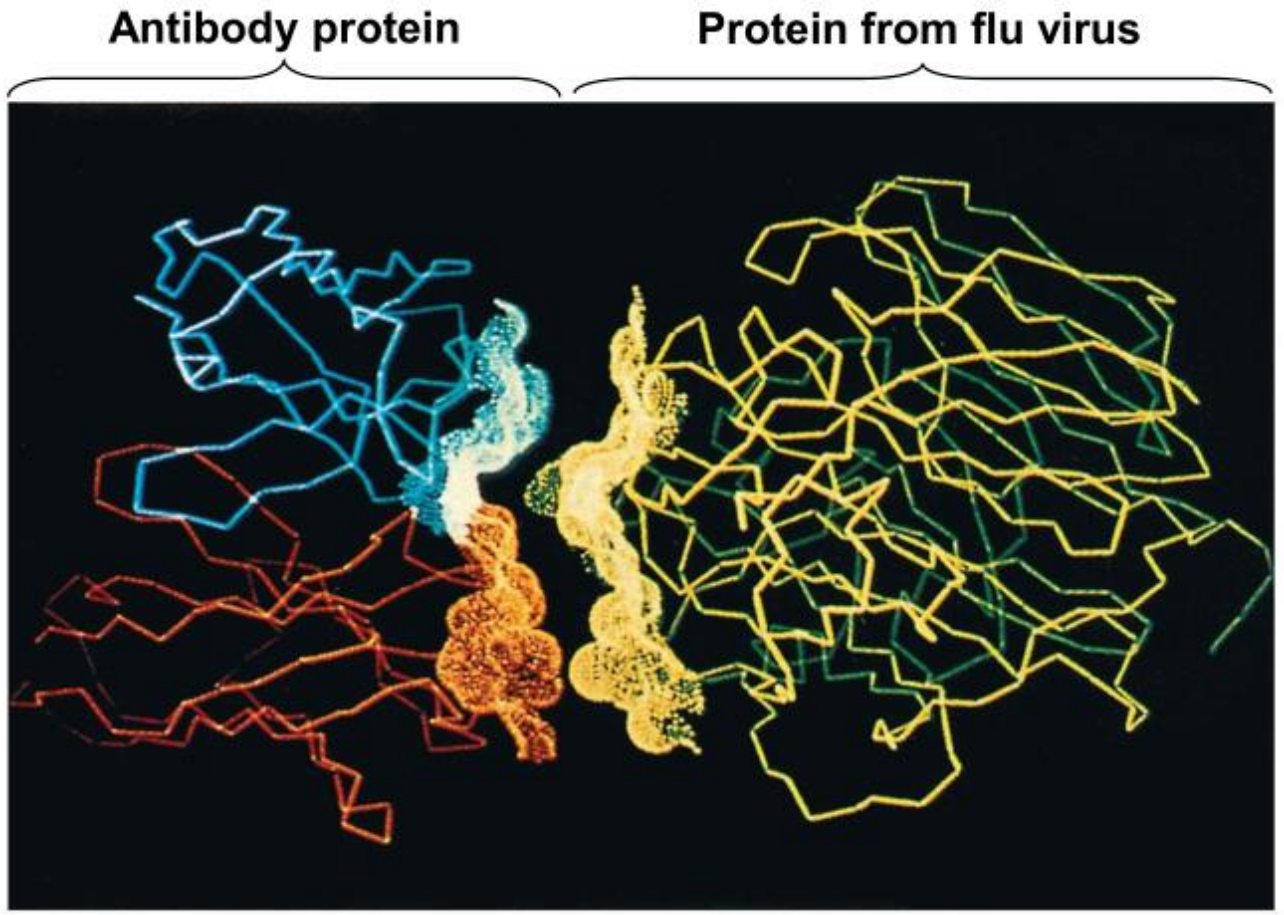


(b) A space-filling model

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Protein Structure and Function

In almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule.



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Four Levels of Protein Structure

The primary structure of a protein is its **unique sequence of amino acids** in the polypeptide chain.

Secondary structure consists of regular coils and folds in the **polypeptide backbone** made by hydrogen bonds.

Tertiary structure is determined by **interactions among various side chains** (R groups)

Quaternary structure results when a protein consists of **multiple polypeptide chains.**

Four Levels of Protein Structure

The primary structure of a protein is its **unique sequence of amino acids** in the polypeptide chain.

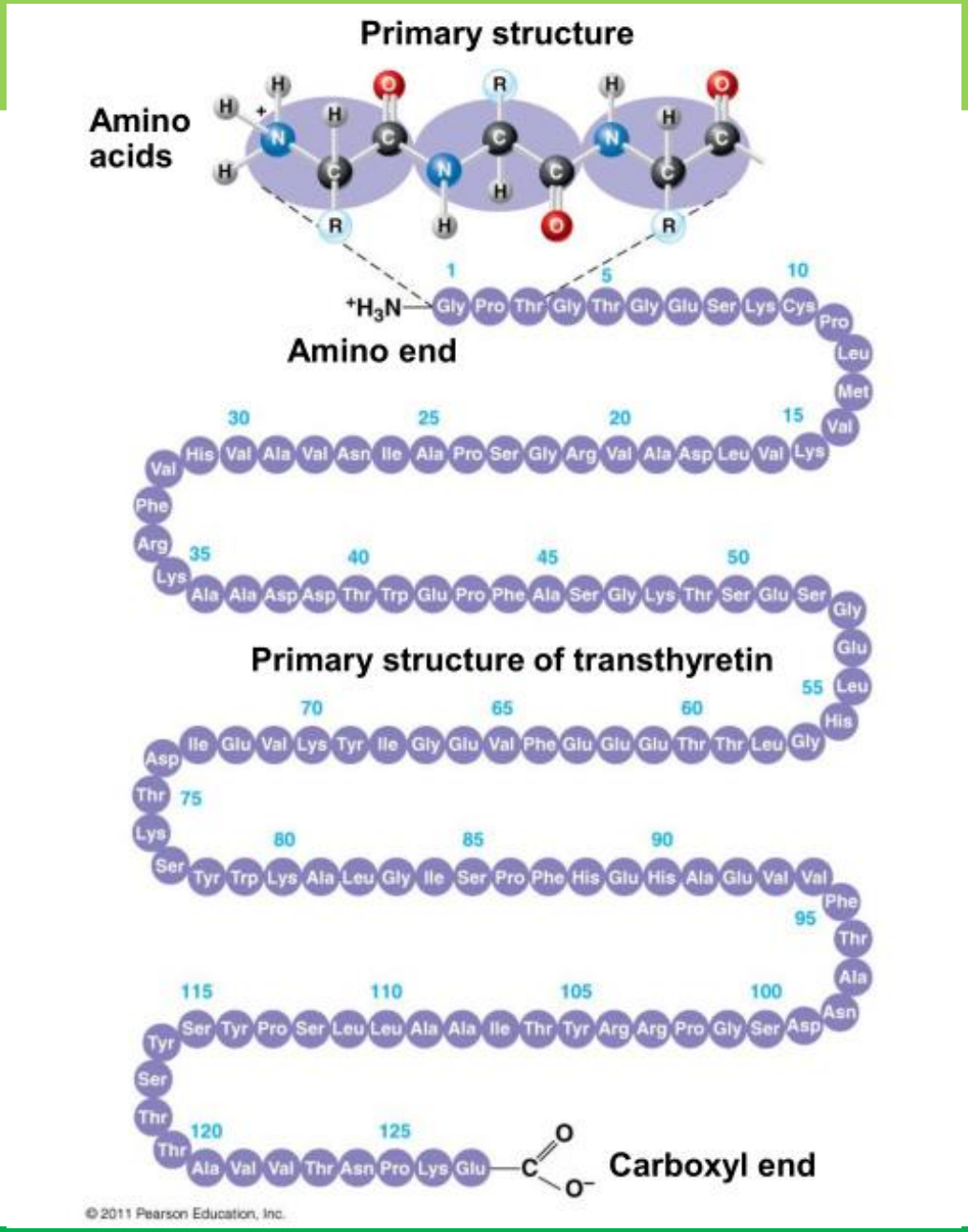
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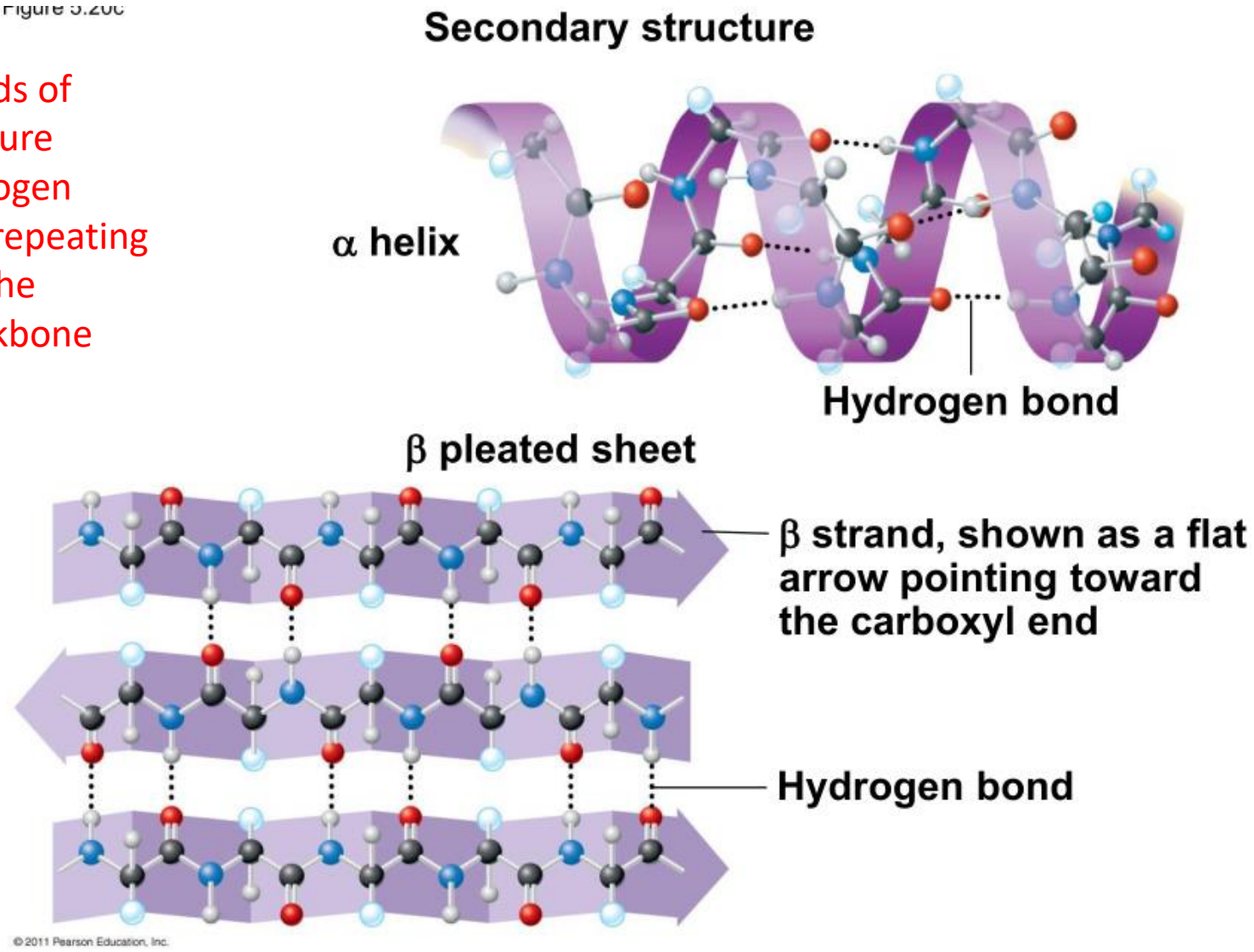
Four Levels of Protein Structure

Primary structure, the sequence of amino acids in a protein, is like the order of letters in a long Word.
Primary structure is determined by inherited genetic information (DNA).

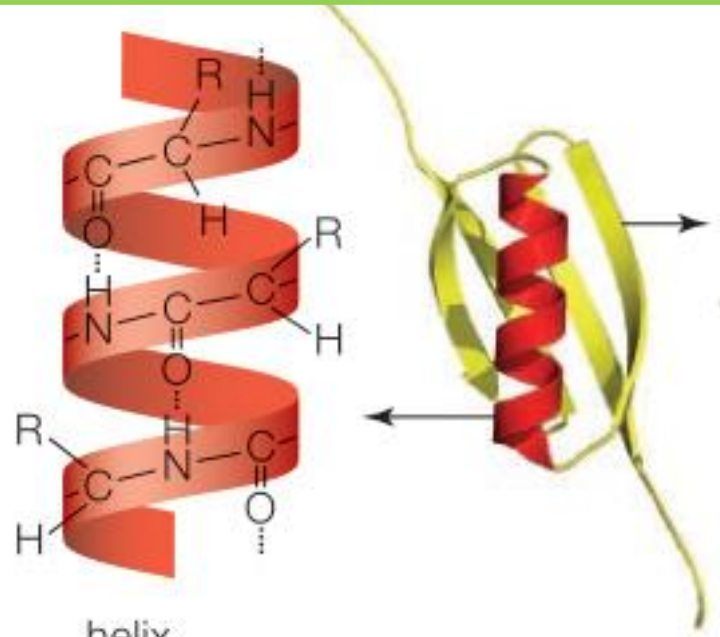


Four Levels of Protein Structure

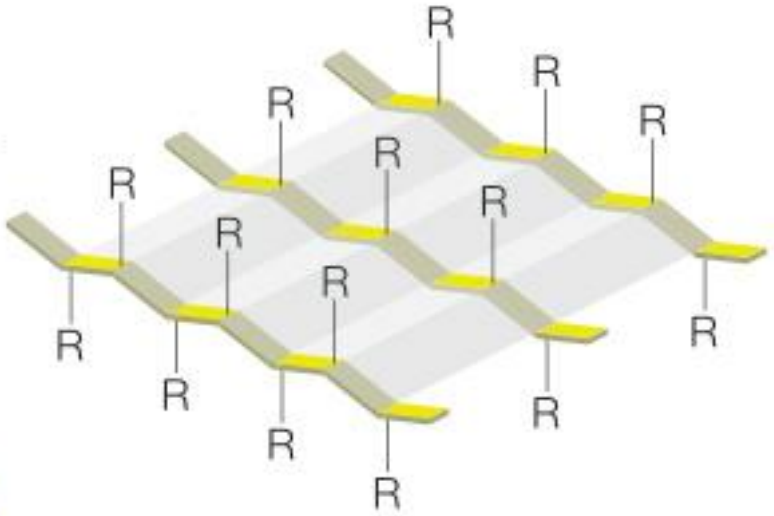
The coils and folds of secondary structure result from hydrogen bonds between repeating constituents of the polypeptide backbone



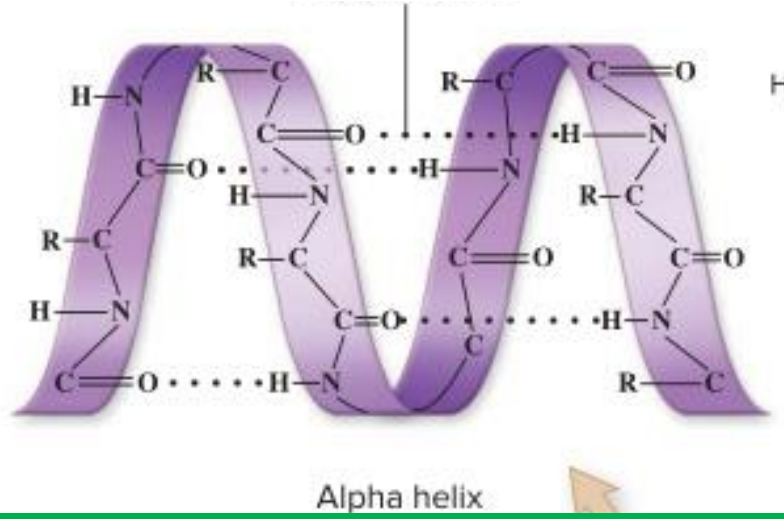
Secondary Structure



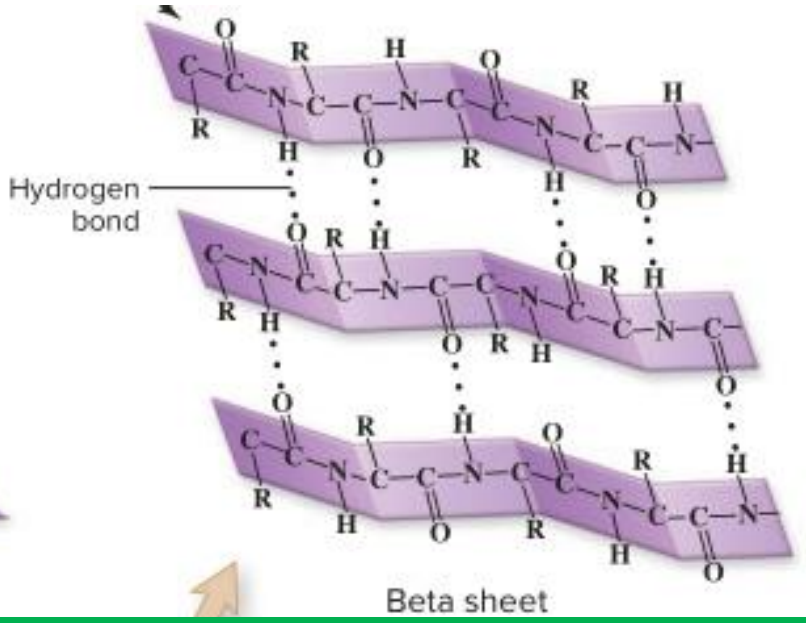
helix



sheet



Alpha helix



Beta sheet

Tertiary structure

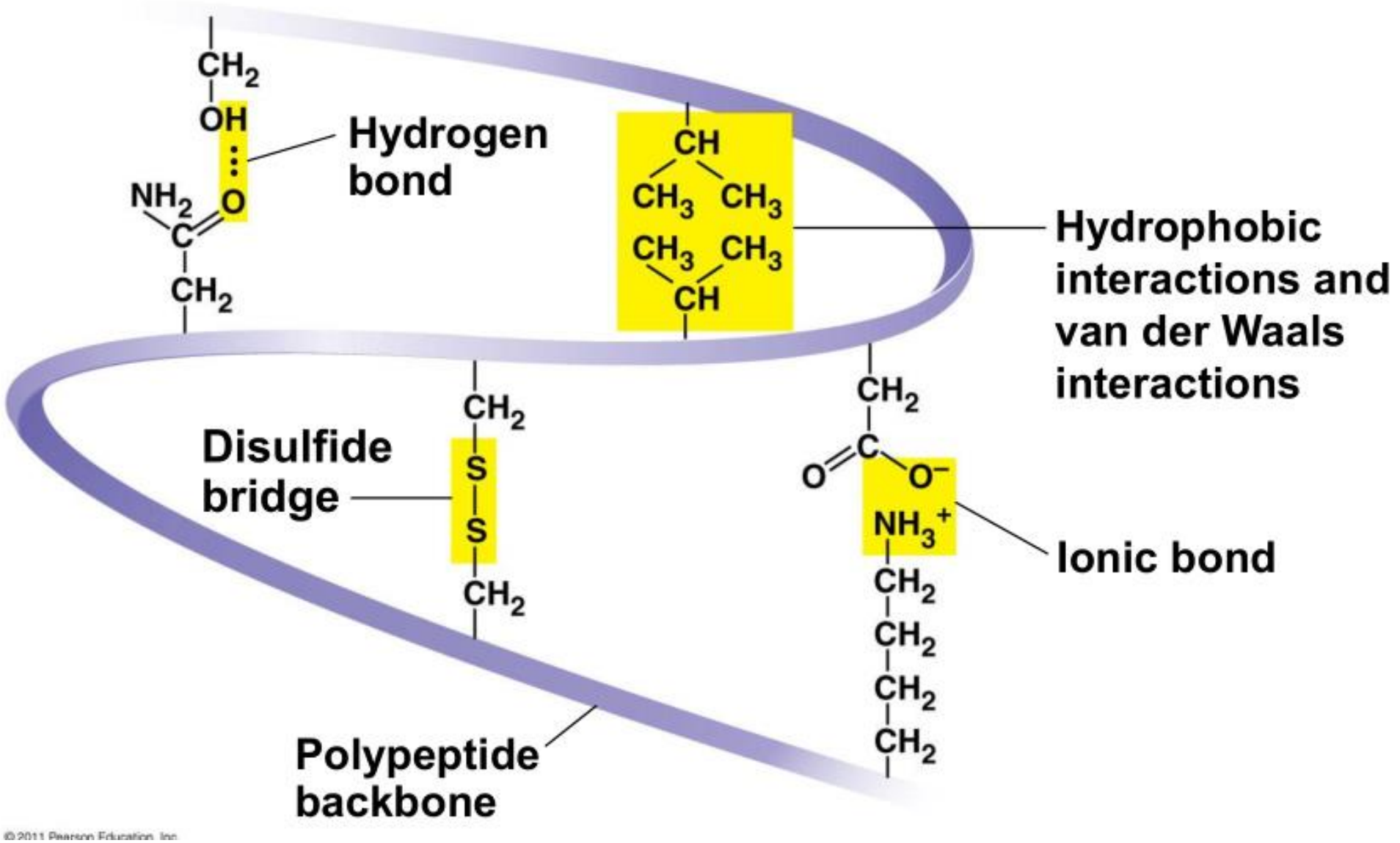
Tertiary structure is determined by **interactions between R groups**, rather than interactions between backbone constituents.

These R group interactions fold the polypeptide into a globular shape.

These interactions between R groups include hydrogen bonds, ionic bonds, hydrophobic interactions, and van der Waals interactions.

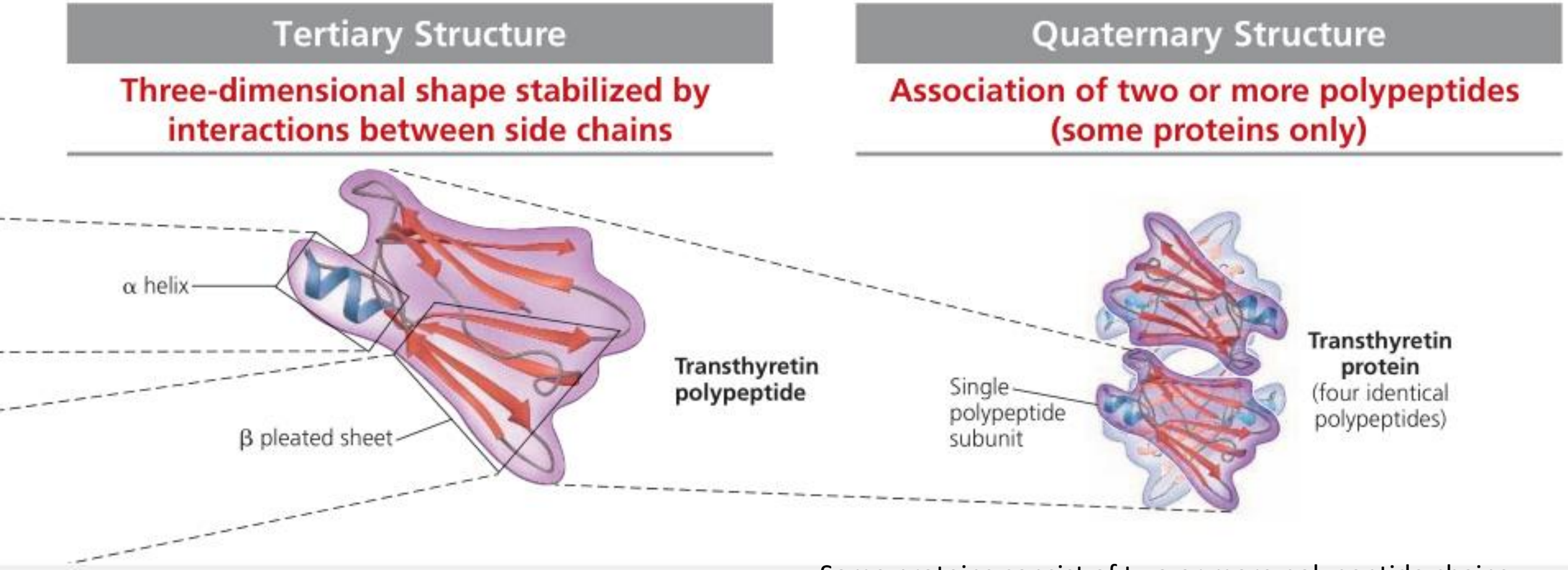
Strong covalent bonds called disulfide bridges may reinforce the protein's structure.

Tertiary structure



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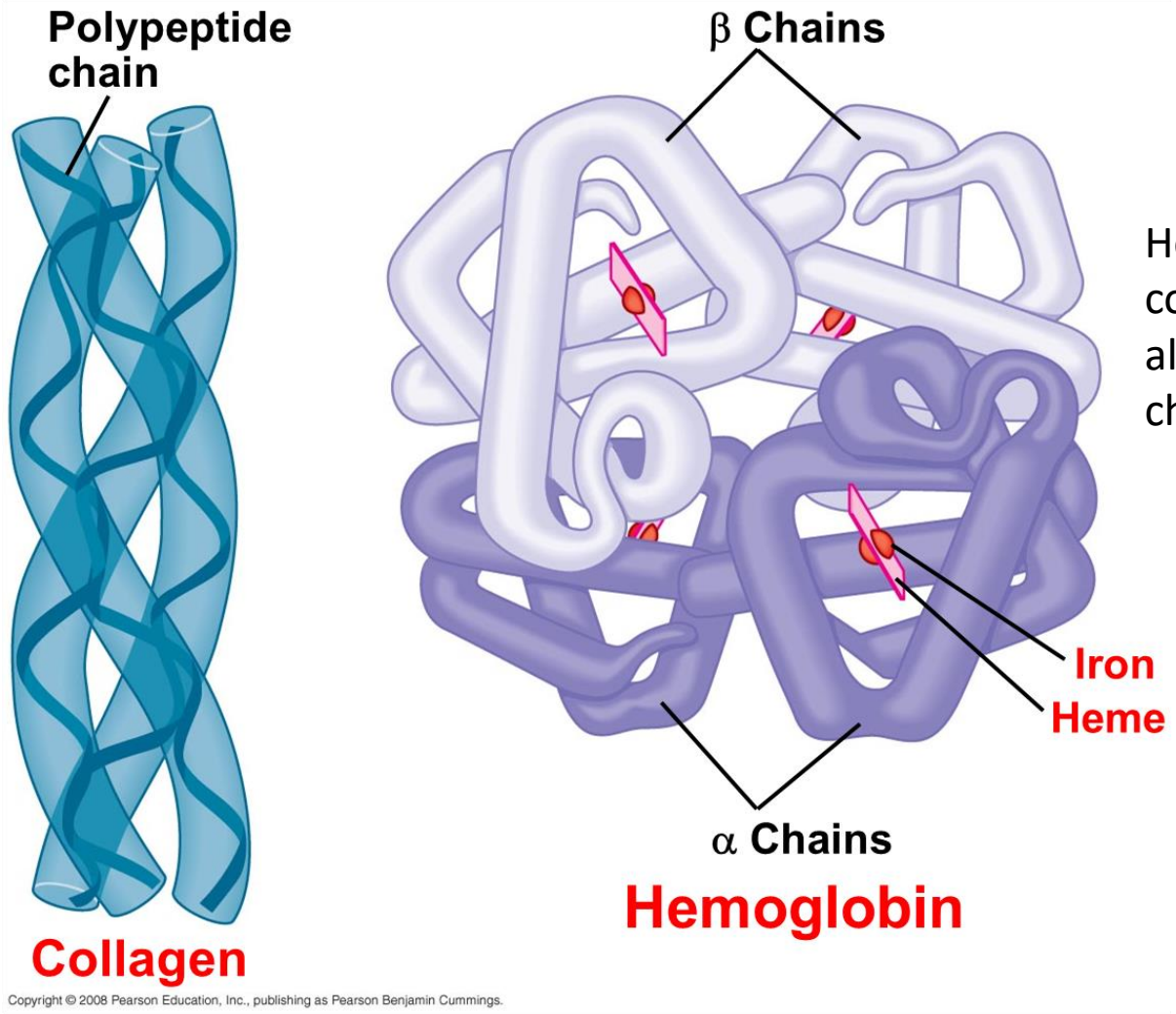
Tertiary structure



Some proteins consist of two or more polypeptide chains aggregated into one functional macromolecule. Quaternary structure is the overall protein structure that results from the aggregation of these polypeptide subunits.

Quaternary structure

Collagen is a fibrous protein consisting of three polypeptides coiled like a rope.



Hemoglobin is a globular protein consisting of four polypeptides: two alpha and two beta chains

Sickle-Cell Disease: A Change in DNA and Primary Structure

A slight change in DNA can change the primary structure (amino acid sequence) of the protein it encodes. This can affect a protein's structure and ability to function.

Sickle-cell disease, an inherited blood disorder, results from a single amino acid substitution in the protein hemoglobin.

In sickle-cell disease, the abnormal hemoglobin molecules tend to aggregate into chains, deforming some of the red blood cells into a sickle shape.

Sickle-Cell

	Primary Structure	Secondary and Tertiary Structures	Quaternary Structure	Function	Red Blood Cell Shape
Normal hemoglobin	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Glu 7 Glu	β subunit	Normal hemoglobin α β α β	Molecules do not associate with one another; each carries oxygen.	10 μm
Sickle-cell hemoglobin	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Val 7 Glu	Exposed hydrophobic region β subunit	Sickle-cell hemoglobin α β α β	Molecules crystallize into a fiber; capacity to carry oxygen is reduced.	10 μm

What Determines Protein Structure?

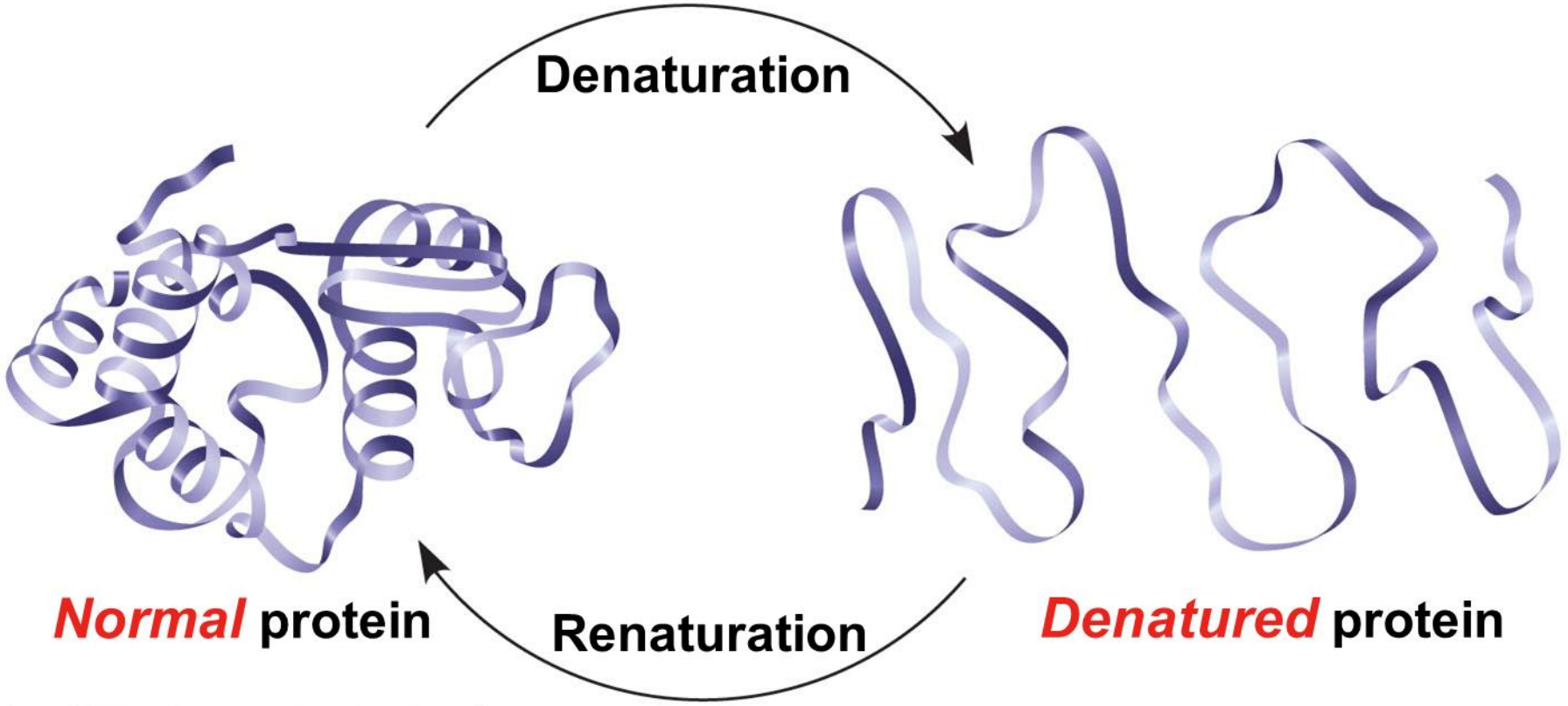
In addition to primary structure, physical and chemical conditions can affect protein structure.

Alterations in pH, salt concentration, temperature, or other environmental factors can cause a protein to loose its native shape.

This loss of a protein's native structure is called denaturation.

A denatured protein is biologically inactive.

What Determines Protein Structure?



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If the denatured protein remains dissolved, it may renature when the chemical and physical aspects of its environment are restored to normal.

The white of an egg becomes opaque during cooking because the denatured proteins are insoluble and solidify. This also explains why excessively high fevers can be fatal: Proteins in the blood tend to denature at very high body temperatures.

Protein Folding in the Cell

It is hard to predict a protein's structure from its primary structure.

Most proteins probably go through several states on their way to a stable structure.

Chaperonins are protein molecules that assist the proper folding of other proteins.

Diseases such as Alzheimer's, Parkinson's, and mad cow disease are associated with misfolded proteins.

Protein Folding in the Cell

Scientists use X-ray crystallography to determine a protein's structure.

Other methods are nuclear magnetic resonance (NMR) spectroscopy and cryo-electron microscopy (cryo-EM) which do not require protein crystallization.

Bioinformatics uses computer programs to predict protein structure from amino acid sequences.

Nucleic acids store, transmit, and help express hereditary information

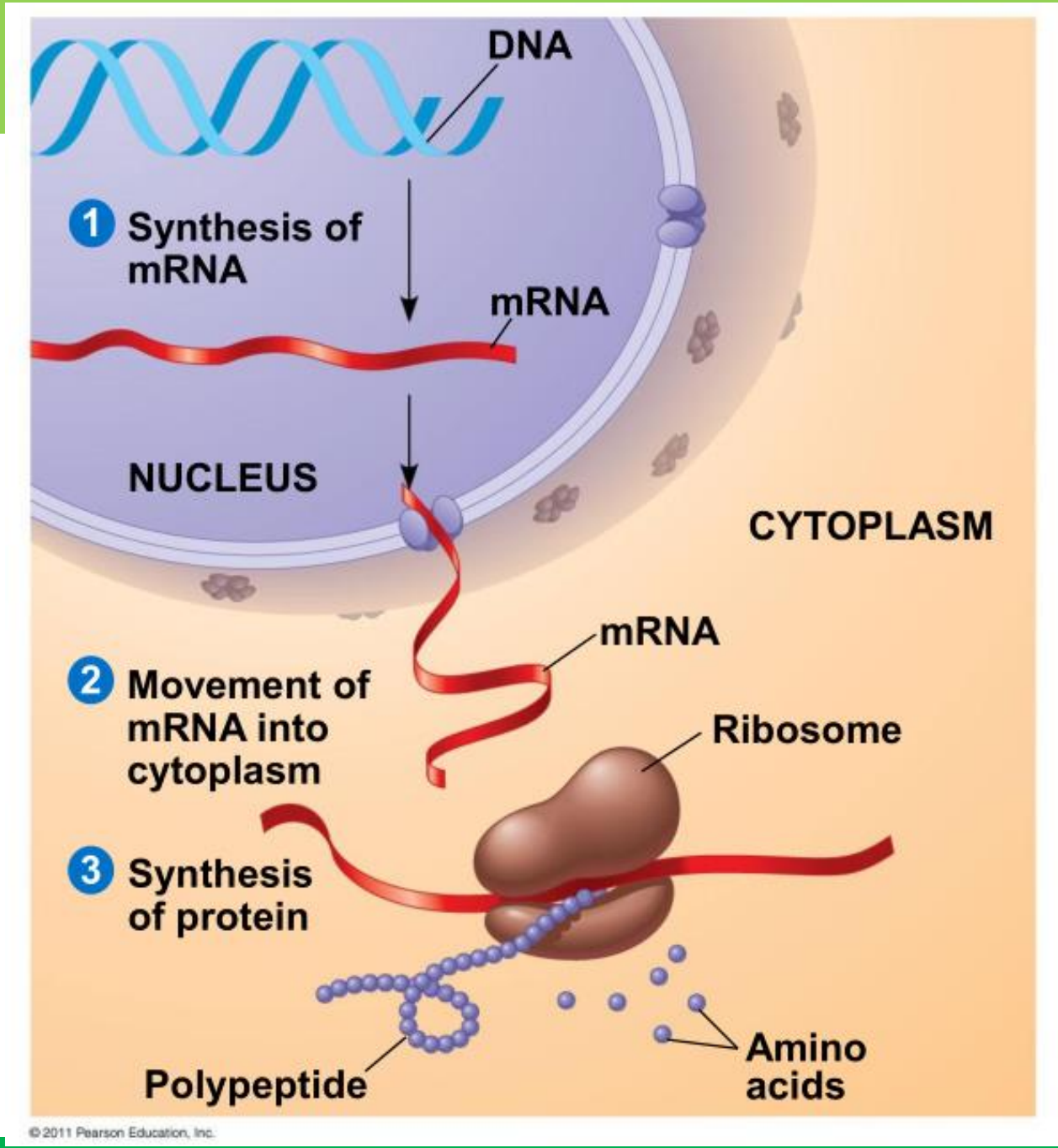
The amino acid sequence of a polypeptide is programmed by a unit of inheritance called a gene
Genes are made of DNA, a **nucleic acid** made of monomers called **nucleotides**

There are two types of nucleic acids:
Deoxyribonucleic acid (DNA)
Ribonucleic acid (RNA)

DNA provides directions for its own replication and the synthesis of messenger RNA (mRNA)

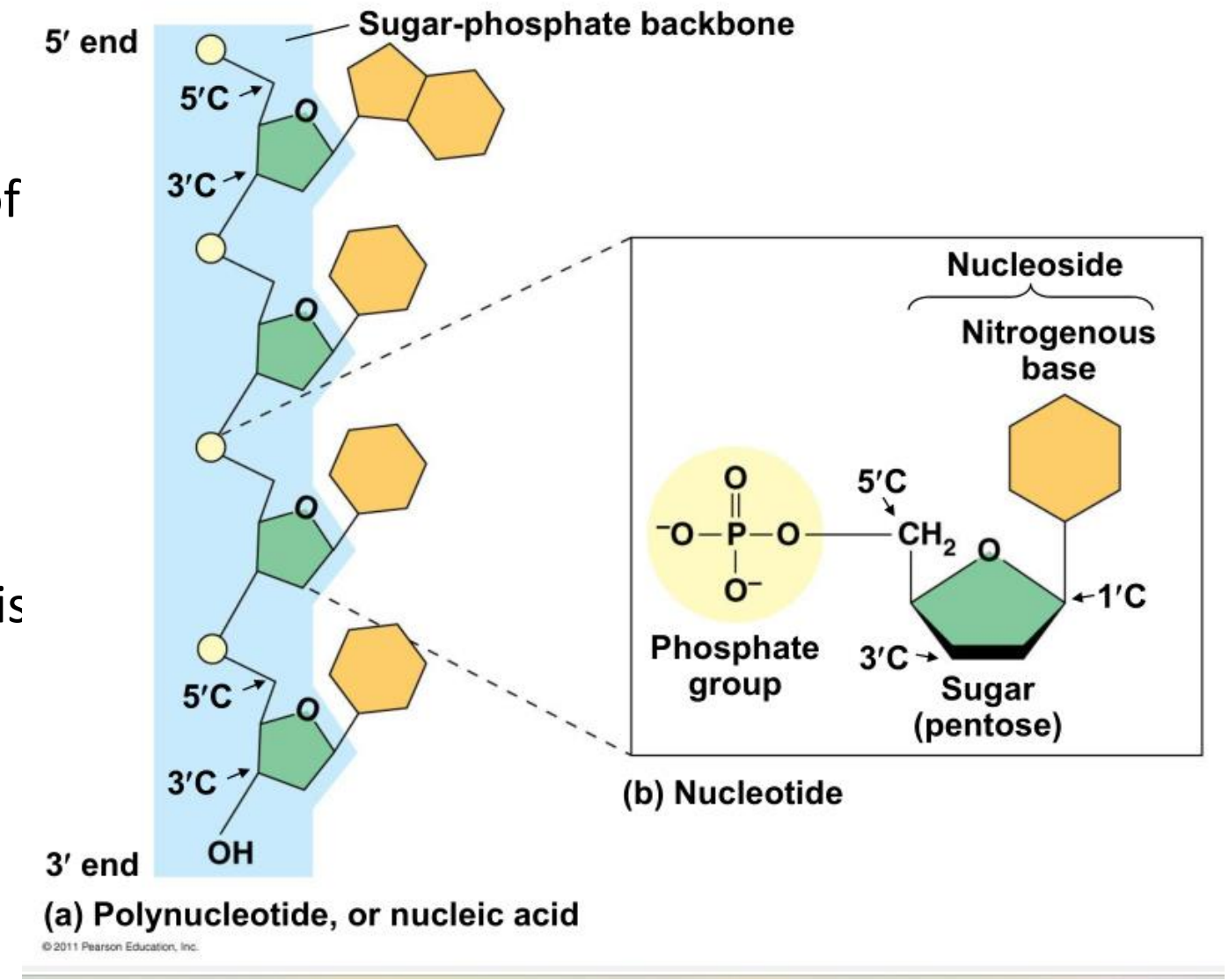
Through mRNA, DNA controls protein synthesis.
This entire process which consists of transcription (synthesis of mRNA in the nucleus), transport of mRNA into the cytoplasm and translation (protein synthesis) is called gene expression.

Gene Expression



The Components of Nucleic Acids

Nucleic acids are polymers called polynucleotides.
Each polynucleotide is made of monomers called nucleotides.
Each nucleotide consists of a nitrogenous base, a pentose sugar, and a phosphate group.
The portion of a nucleotide without the phosphate group is called a nucleoside.

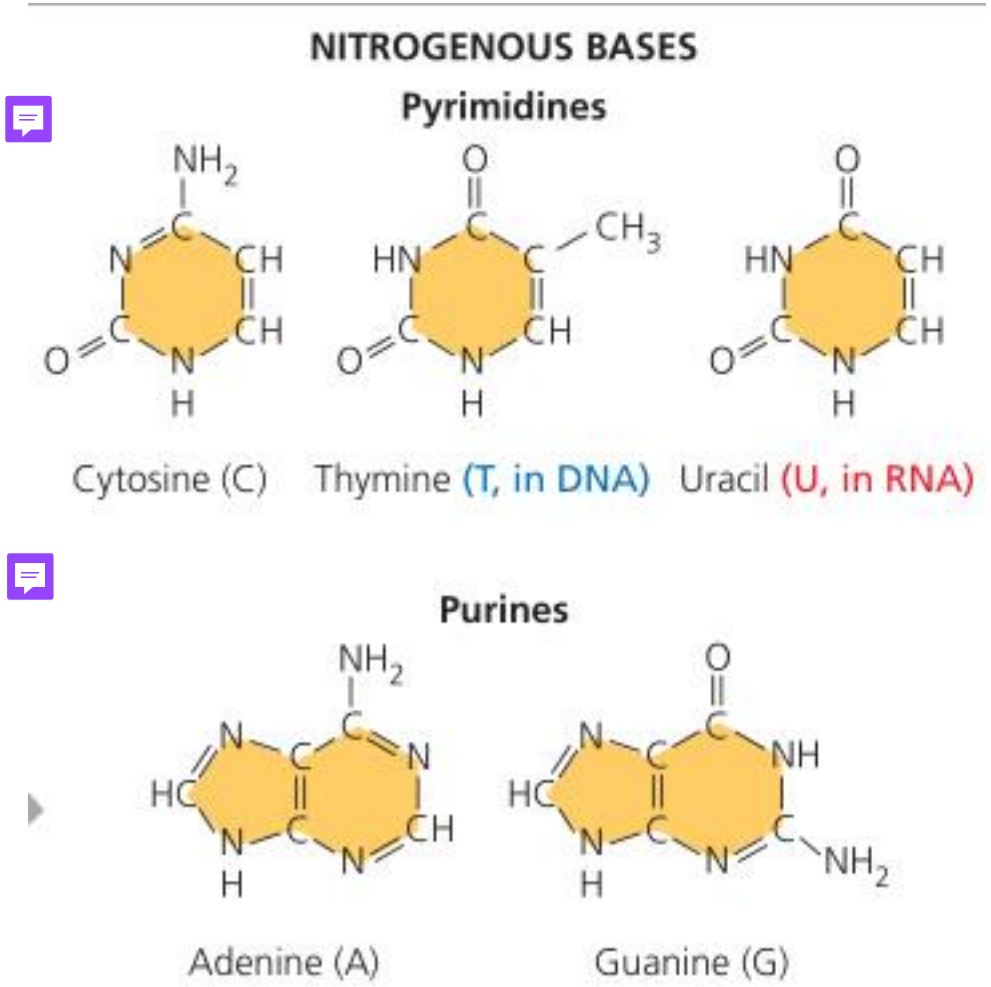


The Components of Nucleic Acids

Nucleoside = nitrogenous base + sugar

There are two families of nitrogenous bases

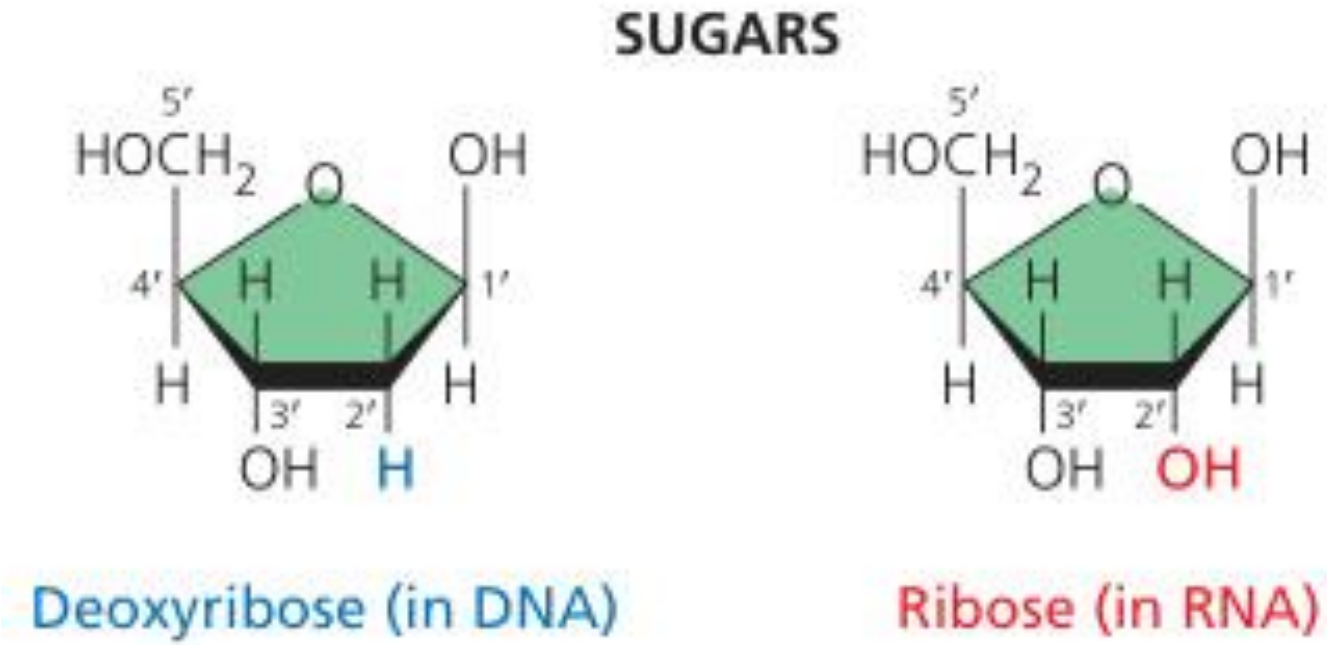
- Pyrimidines (cytosine, thymine, and uracil) have a single six-membered ring
- Purines (adenine and guanine) have a six-membered ring fused to a five-membered ring



The Components of Nucleic Acids

In DNA, the sugar is deoxyribose

In RNA, the sugar is ribose



Nucleotide Polymers

Nucleotide polymers are linked together by dehydration synthesis to build a polynucleotide.

Adjacent nucleotides are joined by covalent bonds called phosphodiester linkages. They form between the –OH group on the 3’ carbon of one nucleotide and the phosphate on the 5’ carbon on the next.

(Directional:One end has a phosphate attached to a 5’ carbon, and the other end has a hydroxyl group on a 3’ carbon)

These links create a backbone of sugar-phosphate units with nitrogenous bases as appendages. (Nitrogenous bases do not involve in covalent bonds)

The sequence of bases along a DNA or mRNA polymer is unique for each gene.

Ex: 5’-AGGTAACTT-3’ , 5’-CGCTTTAAC-3’

The Structures of DNA and RNA Molecules

RNA molecules usually exist as single polypeptide chains

- DNA molecules have two polynucleotides spiraling around an imaginary axis, forming a double helix

- In the DNA double helix, the two backbones run in opposite 5’→ 3’ directions from each other, an arrangement referred to as antiparallel

- One DNA molecule includes many genes

The Structures of DNA and RNA Molecules

The nitrogenous bases in DNA pair up and form hydrogen bonds: adenine (A) always with thymine (T), and guanine (G) always with cytosine (C)

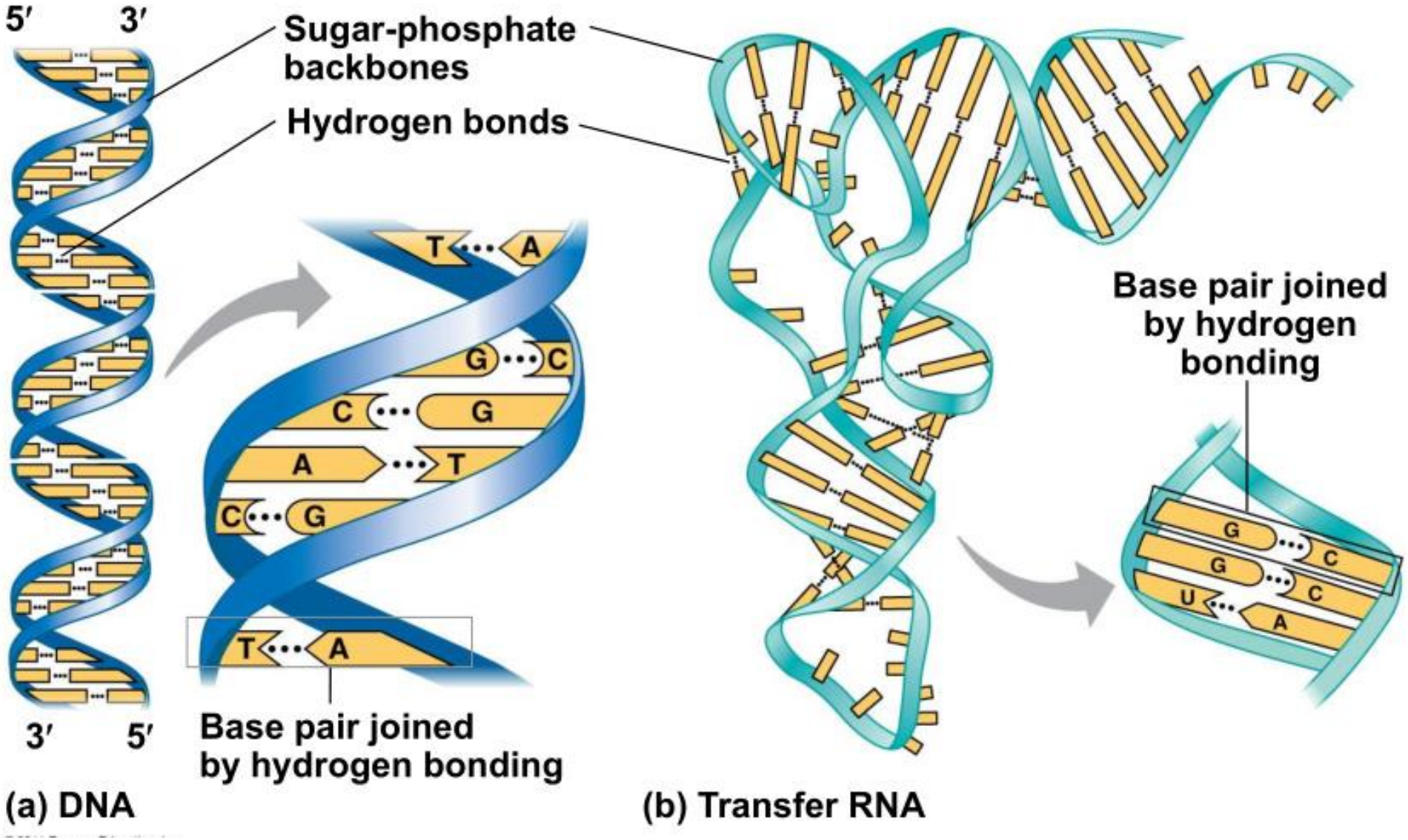
Called complementary base pairing

Hydrogen bonds between pairs of nitrogenous bases hold the two strands together.

Complementary pairing can also occur between two RNA molecules or between parts of the same molecule

In RNA, thymine is replaced by uracil (U) so A and U pair

The Structures of DNA and RNA Molecules



DNA and Proteins as Tape Measures of Evolution

The unique linear sequences of nucleotides in DNA molecules are inherited, passed from parents to offspring.
The linear sequences of nucleotides in DNA
molecules are passed from parents to offspring
Two closely related species are more similar in their DNA and proteins than are more distantly related species
Molecular biology compares DNA sequences and can be used to assess evolutionary kinship.

Genomics and proteomics have transformed biological inquiry and applications

Bioinformatics, the use of computer software and other computational tools that can handle and analyzelarge data sets.

Genomics, the study of an organism's complete set of genes, including their organization, function, and interactions, to understand the genetic information and hereditary traits of that organism.

Proteomics is the study of the complete set of proteins in a biological system, including their structures, functions, interactions, and regulation. It aims to understand the role of proteins in various cellular processes, disease mechanisms, and biological functions.

Using a genomics approach allows us to use gene sequences to identify species and to learn about evolutionary relationships among any two species. This is because all species are related by their evolutionary history, and the evidence is in the DNA sequences. Proteomics—looking at proteins that are expressed—allows us to learn about how organisms or cells are functioning at a given time or in an association with another species.

Review Chart of Organic Molecules

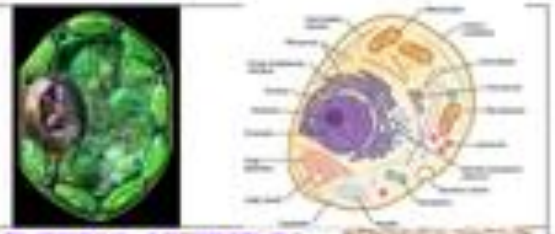
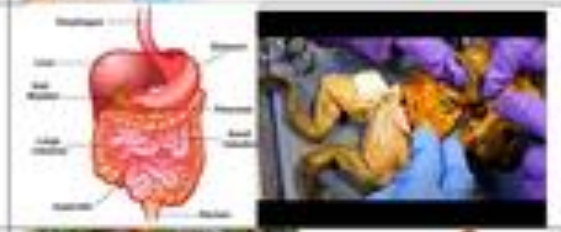
	Monomers or components	Polymer or larger molecule	Type of linkage
Sugars	Monosaccharides	Polysaccharides	Glycosidic linkages
Lipids	Fatty acids	Triacylglycerols	Ester linkages
Proteins	Amino acids	Polypeptides	Peptide bonds
Nucleic acids	Nucleotides	Polynucleotides	Phosphodiester linkages

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Cell: Outline

Cell Components
Nucleus
Ribosomes
Endomembrane system: ER, Golgi, lysosomes
Mitochondria, chloroplasts, peroxisomes
Cytoskeleton

Cells: The Fundamental Units of Life

Cell	Basic structural and functional unit of a living organism	
Tissue	Group of cells with similar structures, working together to perform a shared function	
Organ	Structure made up of a group of tissues, working together to perform specific functions	
Organ System	Group of organs with related functions, working together to perform body functions	
Organism	Living thing performing all seven life processes	

Cells: The Fundamental Units of Life

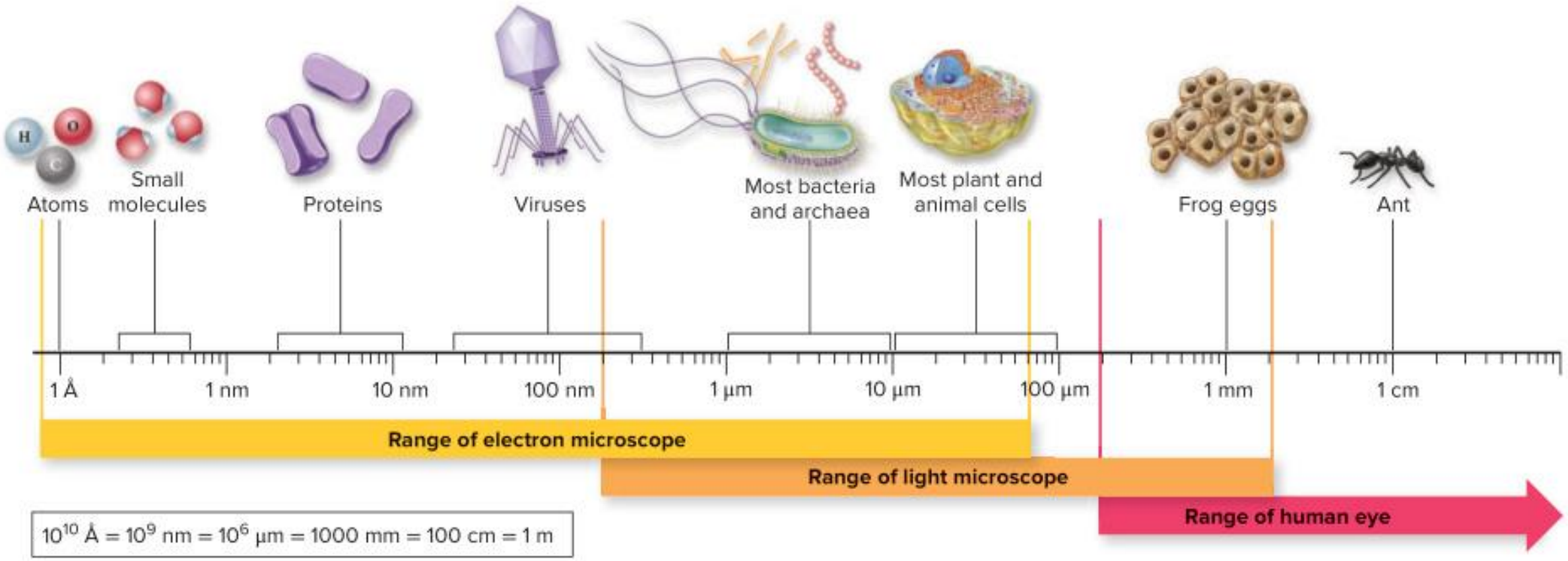
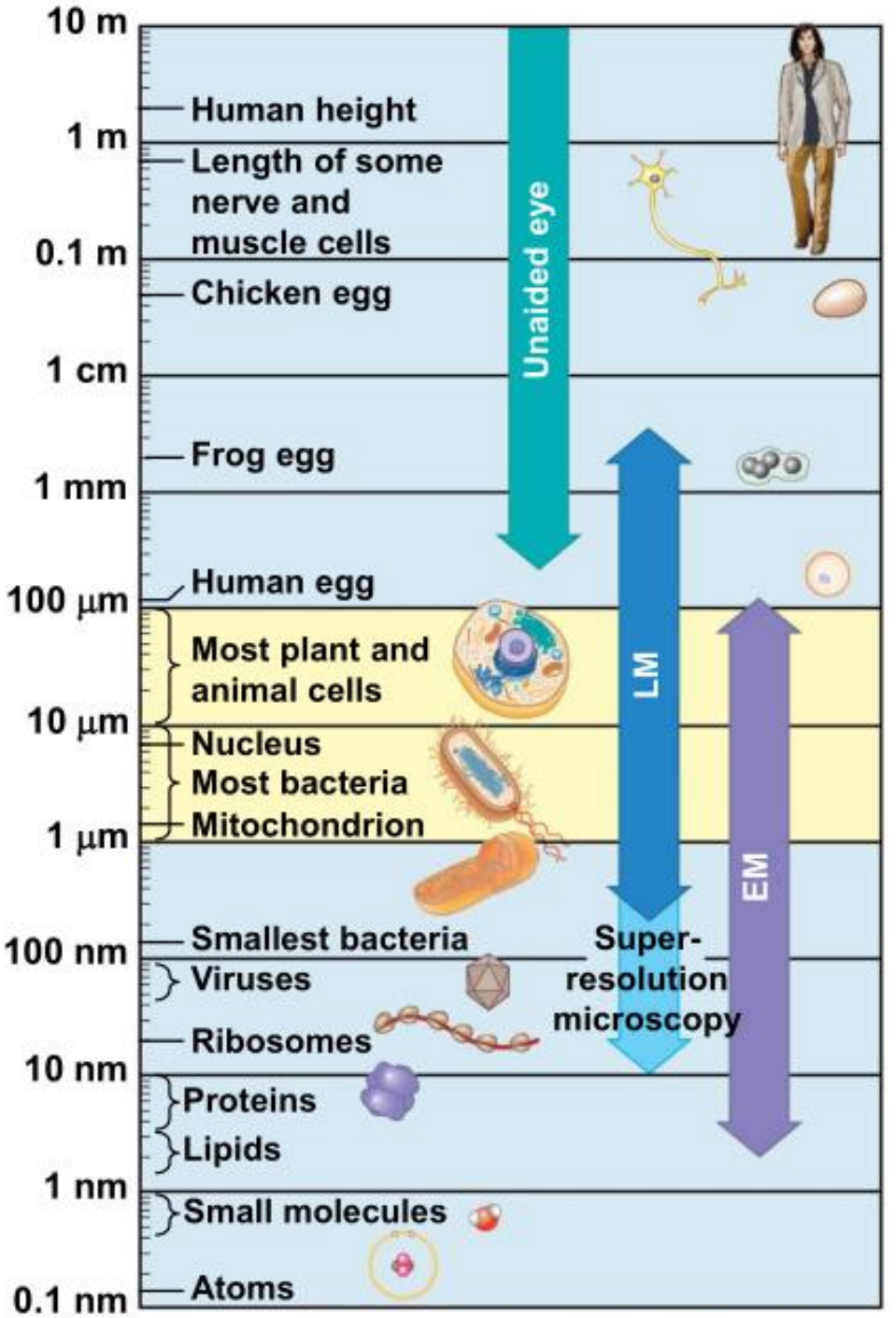


Figure 3.2 **Ranges of Light and Electron Microscopes.** Biologists use light microscopes and electron microscopes to view a world too small to see with the unaided eye. This illustration uses the metric system to measure size (see appendix C). Thanks to the overlapping capabilities of the different microscopes, we can visualize objects ranging in size from large molecules to entire cells.

Hoefnagels Mariëlle. (2020). Biology : concepts and investigations (Fifth). McGraw Hill.

Cells: The Fundamental Units of Life



Urry, L. A., Cain, M. L. 1., Wasserman, S. A., Minorsky, P. V., Reece, J. B., & Campbell, N. A. (2017). Campbell biology. Eleventh edition. New York, NY, Pearson Education, Inc.

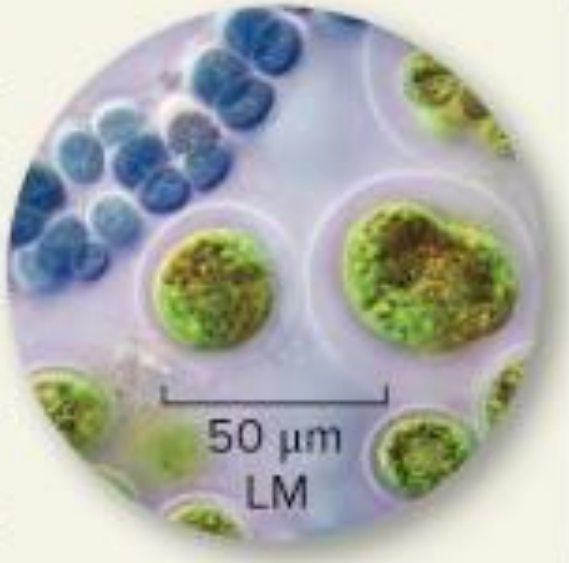
Cells: The Fundamental Units of Life

- The cell is the simplest collection of matter that can be alive.
- Cells are the basic structural and functional units of every organism.
- All organisms are made of cells.
- Cell structure is correlated to cellular function.

Cells: The Fundamental Units of Life

TABLE 3.1 The Cell Theory: A Summary

Components of early cell theory (19th century)
All organisms are made of one or more cells.
The cell is the fundamental unit of life.
All cells come from preexisting cells.
Additional ideas in modern cell theory
All cells have the same basic chemical composition.
All cells use energy.
All cells contain DNA that is duplicated and passed on as each cell divides.



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Comparing Prokaryotic and Eukaryotic Cells

Cells are of two distinct types: prokaryotic and eukaryotic.
Bacteria and Archaea consist of prokaryotic cells.
Eukarya (protists, fungi, animals, and plants) consist of eukaryotic cells.

	Cell Type	Nucleus	Membrane-Bounded Organelles	Membrane Chemistry	Typical Cell Size
Domain Bacteria	Prokaryotic	Absent	Absent	Fatty acids	1-10 µm
Domain Archaea	Prokaryotic	Absent	Absent	Nonfatty acid lipids	1-10 µm
Domain Eukarya	Eukaryotic	Present	Present	Fatty acids	10-100 µm

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Comparing Prokaryotic and Eukaryotic Cells

Basic features of all cells (both prokaryotic and eucaryotic):

Plasma membrane

Semifluid substance called cytosol

Chromosomes (carry genes)

Ribosomes (make proteins)

Prokaryotic Cells

Prokaryotic cells are characterized by having:

No nucleus

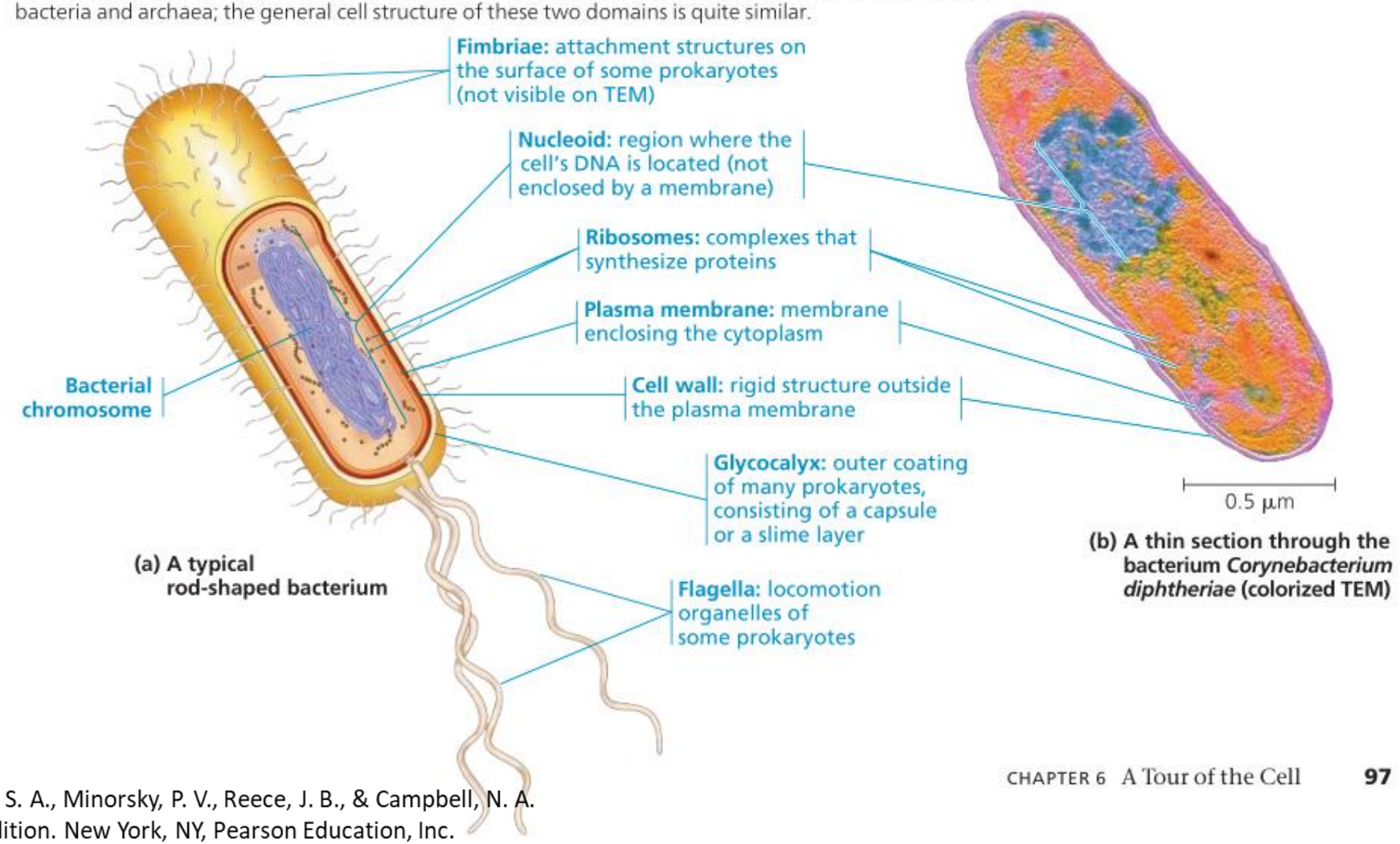
DNA in an unbound region called the nucleoid

No membrane-bound organelles

Cytoplasm bound by the plasma membrane

Prokaryotic Cells

▼ **Figure 6.5 A prokaryotic cell.** Lacking a true nucleus and the other membrane-enclosed organelles of the eukaryotic cell, the prokaryotic cell appears much simpler in internal structure. Prokaryotes include bacteria and archaea; the general cell structure of these two domains is quite similar.



Urry, L. A., Cain, M. L. 1., Wasserman, S. A., Minorsky, P. V., Reece, J. B., & Campbell, N. A. (2017). Campbell biology. Eleventh edition. New York, NY, Pearson Education, Inc.

Eukaryotic Cells

Eukaryotic cells are characterized by having:

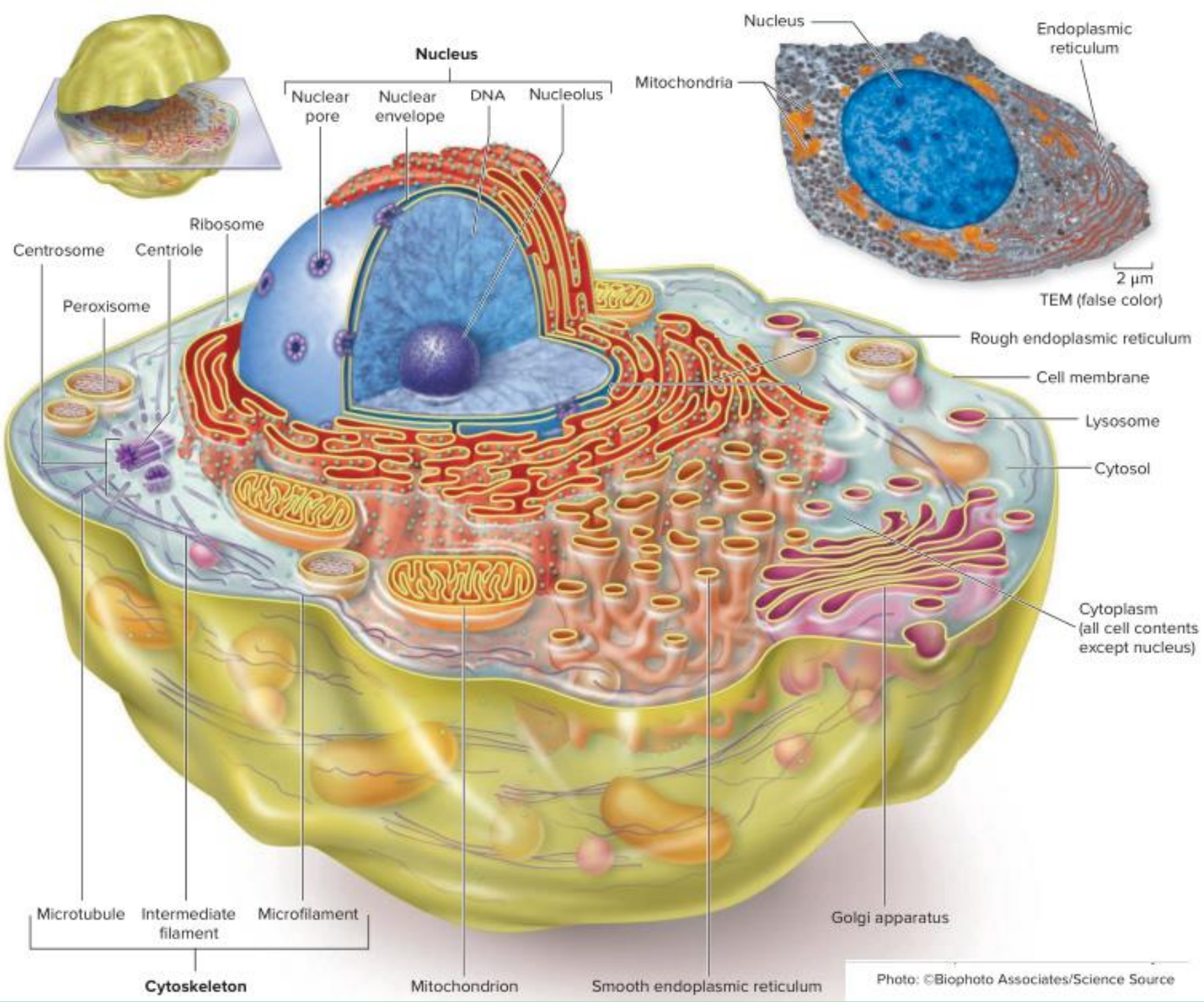
DNA in a nucleus that is bounded by a membranous nuclear envelope.

Membrane-bound organelles.

Cytoplasm in the region between the plasma membrane and nucleus.

Eukaryotic cells are generally much larger than prokaryotic cells.

Eukaryotic Cells



Hoefnagels Mariëlle. (2020). Biology : concepts and investigations (Fifth). McGraw Hill.

The surface area to volume ratio of a cell is critical

Metabolic requirements set upper limits on the size of cells.

The surface area to volume ratio of a cell is critical.

As the surface area increases by a factor of n^2, the volume increases by a factor of $n^{3.}$

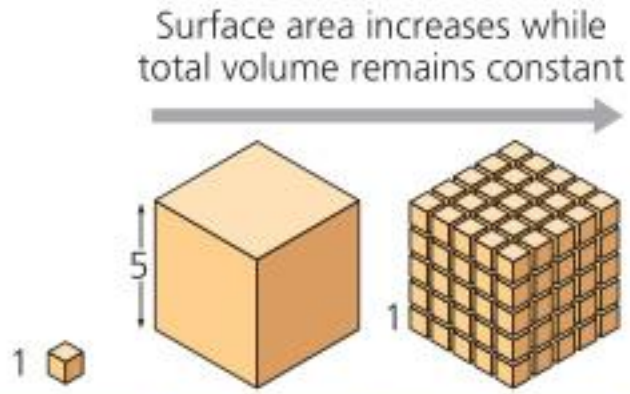
Small cells have a greater surface area relative to volume.

For each square micrometer of membrane, only a limited amount of a particular substance can cross per second, so the ratio of surface area to volume is critical. As a cell increases in size, its surface area grows proportionately less than its volume. Thus, a cell has to be small.

Larger organisms do not generally have larger cells than smaller organisms—they simply have more cells.

The surface area to volume ratio of a cell is critical

▼ Figure 6.7 Geometric relationships between surface area and volume. In this diagram, cells are represented as cubes. Using arbitrary units of length, we can calculate the cell's surface area (in square units, or units^2), volume (in cubic units, or units^3), and ratio of surface area to volume. A high surface area-to-volume ratio facilitates the exchange of materials between a cell and its environment.



Total surface area [(height × width of 1 side) × 6 sides × number of cells]	6 units^2	150 units^2	750 units^2
Total volume [(height × width × length of 1 cell) × number of cells]	1 unit^3	125 units^3	125 units^3
Surface area-to-volume ratio [surface area ÷ volume]	6	1.2	6

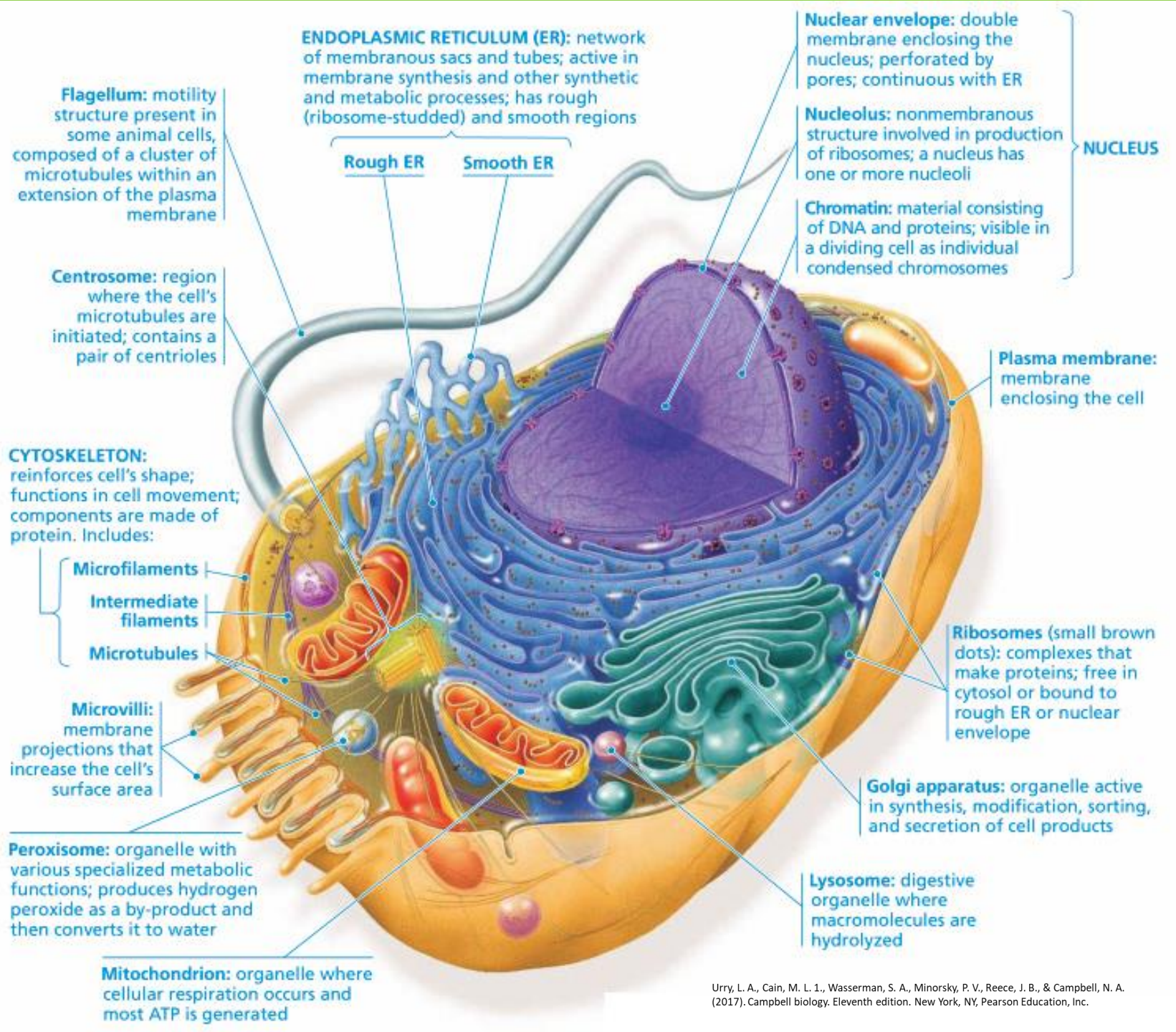
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A sufficiently high ratio of surface area to volume is especially important in cells that exchange a lot of material with their surroundings, such as intestinal cells. Such cells may have many long, thin projections from their surface called microvilli, which increase surface area without an appreciable increase in volume.

Animal cell

A eukaryotic cell has internal membranes that divide the cell into compartments called organelles.

The cell's compartments provide different local environments that support specific metabolic functions, so incompatible processes can occur simultaneously in a single cell. The plasma membrane and organelle membranes also participate directly in the cell's metabolism because many enzymes are built right into the membranes.

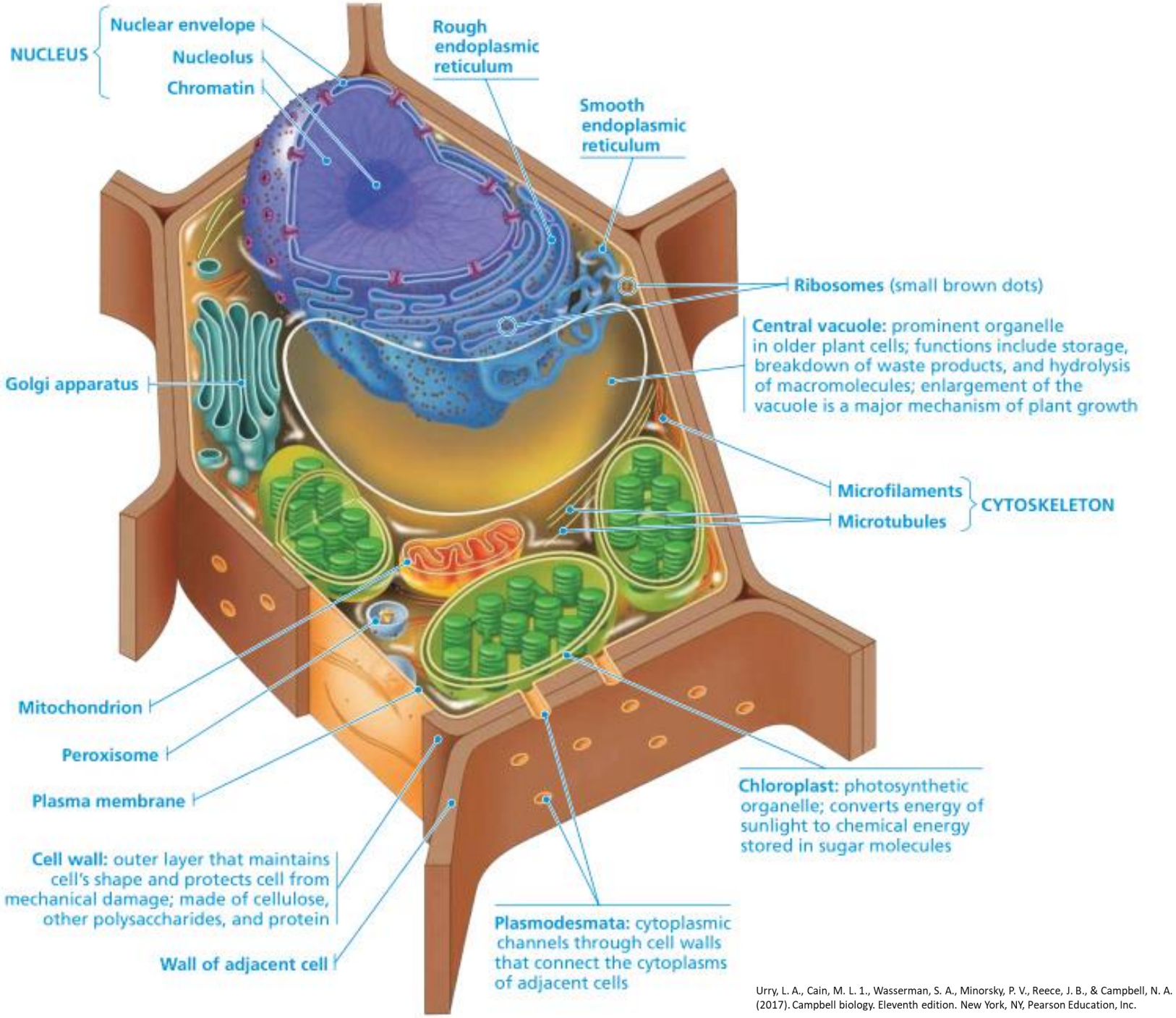


Urry, L. A., Cain, M. L. 1., Wasserman, S. A., Minorsky, P. V., Reece, J. B., & Campbell, N. A. (2017). Campbell biology. Eleventh edition. New York, NY, Pearson Education, Inc.

Plant cell

A eukaryotic cell has internal membranes that divide the cell into compartments called organelles.

The cell's compartments provide different local environments that support specific metabolic functions, so incompatible processes can occur simultaneously in a single cell. The plasma membrane and organelle membranes also participate directly in the cell's metabolism because many enzymes are built right into the membranes.



Urry, L. A., Cain, M. L. 1., Wasserman, S. A., Minorsky, P. V., Reece, J. B., & Campbell, N. A. (2017). Campbell biology. Eleventh edition. New York, NY, Pearson Education, Inc.

Eukaryotic Cells

TABLE 4.3

Some Components of Eukaryotic Cells

Organelles with membranes	
Nucleus	Protects and controls access to DNA
Endoplasmic reticulum (ER)	Makes and modifies new polypeptides and lipids, among other tasks
Golgi body	Modifies and sorts polypeptides and lipids
Vesicle	Transports, stores, or breaks down substances
Mitochondrion	Makes ATP by glucose breakdown
Chloroplast	Makes sugars (in plants and some protists)
Lysosome	Intracellular digestion
Peroxisome	Breaks down fatty acids, amino acids, toxins
Vacuole	Stores, breaks down substances
Organelles without membranes	
Ribosome	Assembles polypeptides
Centriole	Anchors cytoskeleton
Other components	
Cytoskeleton	Contributes to cell shape, internal organization, and movement

Starr C. Taggart R. Evers C. A. & Starr L. (2019).
Biology : the unity and diversity of life (15th ed.). Cengage.

The Nucleus: Information Central

The Nucleus: Information Central

The nucleus contains most of the cell's genes and is usually the most outstanding organelle

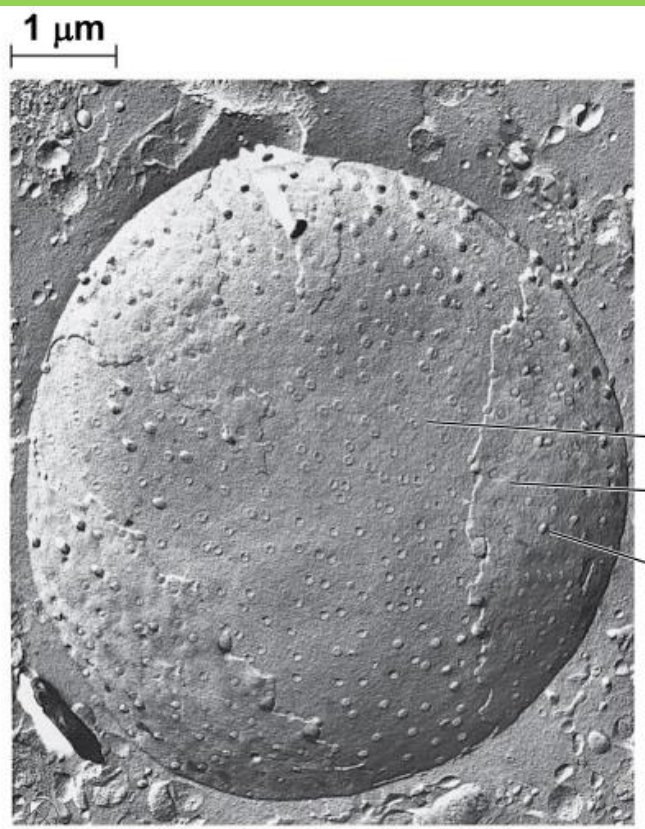
The nuclear envelope encloses the nucleus, separating it from the cytoplasm

The nuclear membrane is a double membrane; each membrane consists of a lipid bilayer

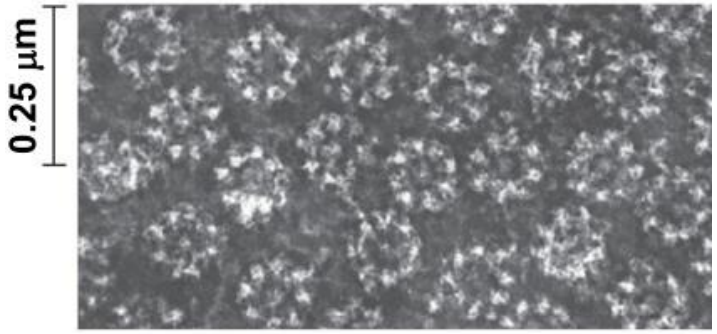
Nuclear pores regulate the entry and exit of molecules from the nucleus

The shape of the nucleus is maintained by the nuclear lamina, which is composed of protein called intermediate filaments.

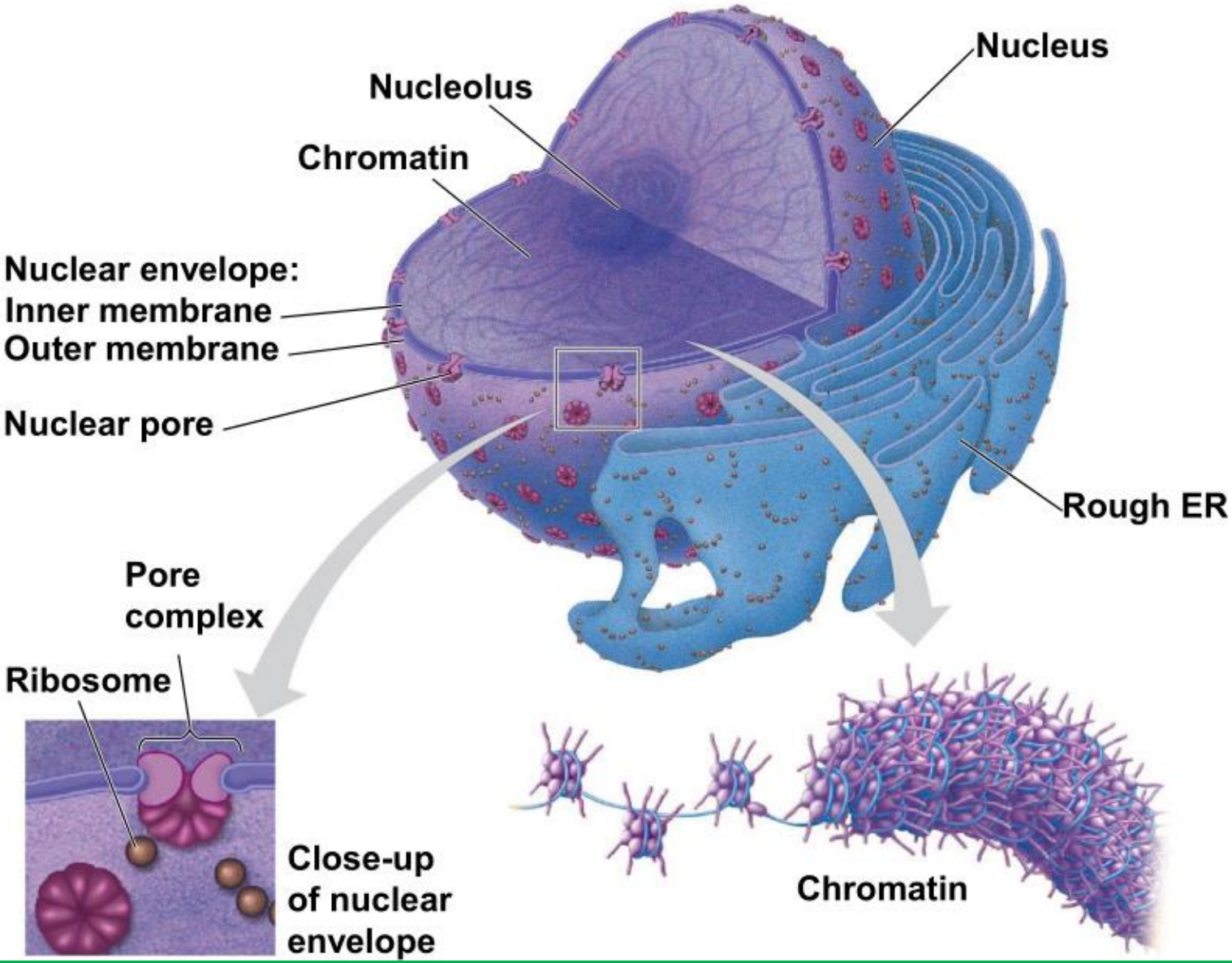
The Nucleus: Information Central



Surface of nuclear envelope (TEM)



Pore complexes (TEM)



The Nucleus: Information Central

- In the nucleus, DNA is organized into discrete units called **chromosomes**
- Each chromosome is one long DNA molecule associated with proteins
- The DNA and proteins of chromosomes together are called **chromatin**
- Chromatin condenses to form discrete chromosomes as a cell prepares to divide
- The **nucleolus** is located within the nucleus and is the site of ribosomal RNA (rRNA) synthesis. Also in the nucleolus, proteins imported from the cytoplasm are assembled with rRNA into large and small subunits of ribosomes.

The Nucleus: Information Central

Components of the Nucleus	
Chromatin	DNA and associated proteins in a cell nucleus
Nucleoplasm	Semifluid interior portion of the nucleus
Nuclear envelope	Double membrane with nuclear pores that control which substances enter and exit the nucleus
Nucleolus	Dense region of proteins and nucleic acid where ribosomal subunits are being produced

Starr C. Taggart R. Evers C. A. & Starr L. (2019).
Biology : the unity and diversity of life (15th ed.). Cengage.

Ribosomes: Protein synthesis machinery

Ribosomes are complexes of ribosomal RNA and protein.
They can be found either freely suspended in the cytosol or attached to the outside of the endoplasmic reticulum or nuclear envelope.

The ribosomes synthesize proteins using the genetic information contained in the DNA.
Once in the cytoplasm, mRNA coming from the nucleus binds to a ribosome, which manufactures proteins.

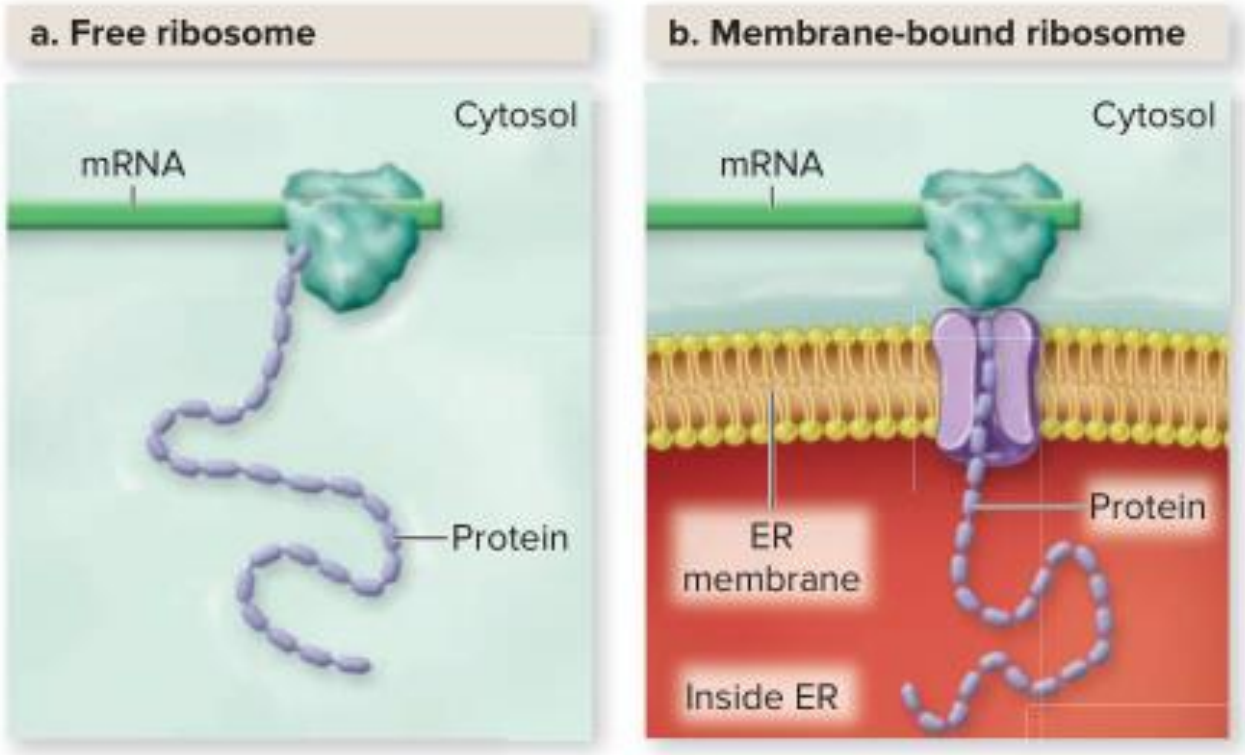
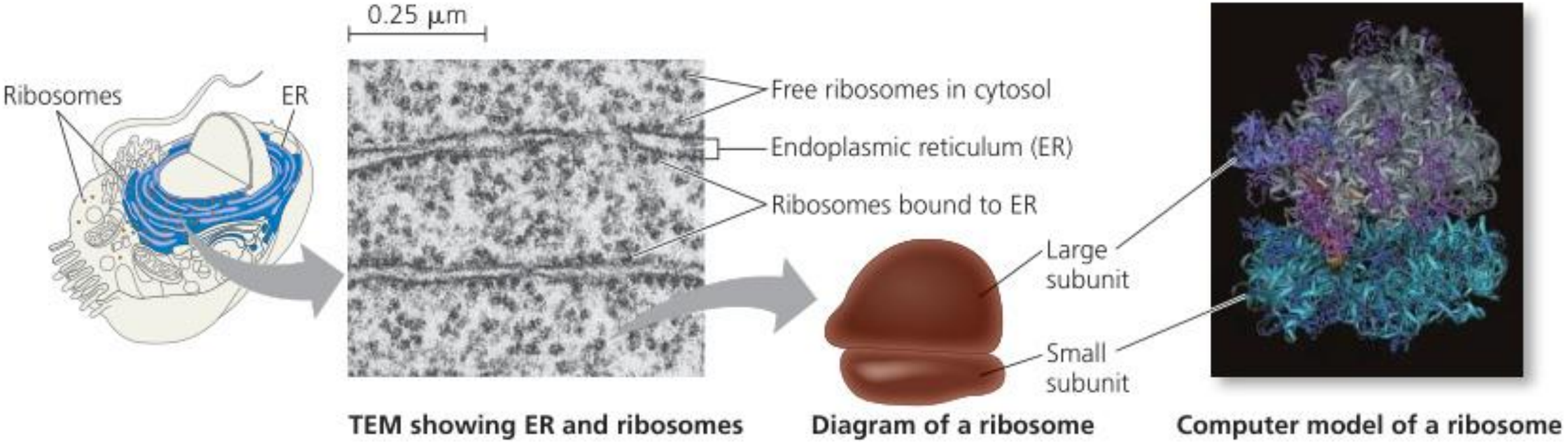


Figure 3.15 Ribosomes. (a) Free ribosomes produce proteins used in the cell's cytosol. (b) Proteins produced by ribosomes attached to the rough ER's membrane are typically used in specialized organelles or in the cell membrane; they may also be secreted outside the cell.

Ribosomes: Protein synthesis machinery

▼ Figure 6.10 Ribosomes. This electron micrograph of a pancreas cell shows both free and bound ribosomes. The simplified diagram and computer model show the two subunits of a ribosome.



The endomembrane system

The endomembrane system regulates protein traffic and performs metabolic functions in the cell

Components of the endomembrane system

1. Nuclear envelope
2. Endoplasmic reticulum
3. Golgi apparatus
4. Lysosomes
5. Vacuoles
6. Plasma membrane

These components are either continuous or connected through transfer by vesicles (sacs made of membrane)

The Endoplasmic Reticulum: Biosynthetic Factory

The endoplasmic reticulum (ER) accounts for more than half of the total membrane in many eukaryotic cells

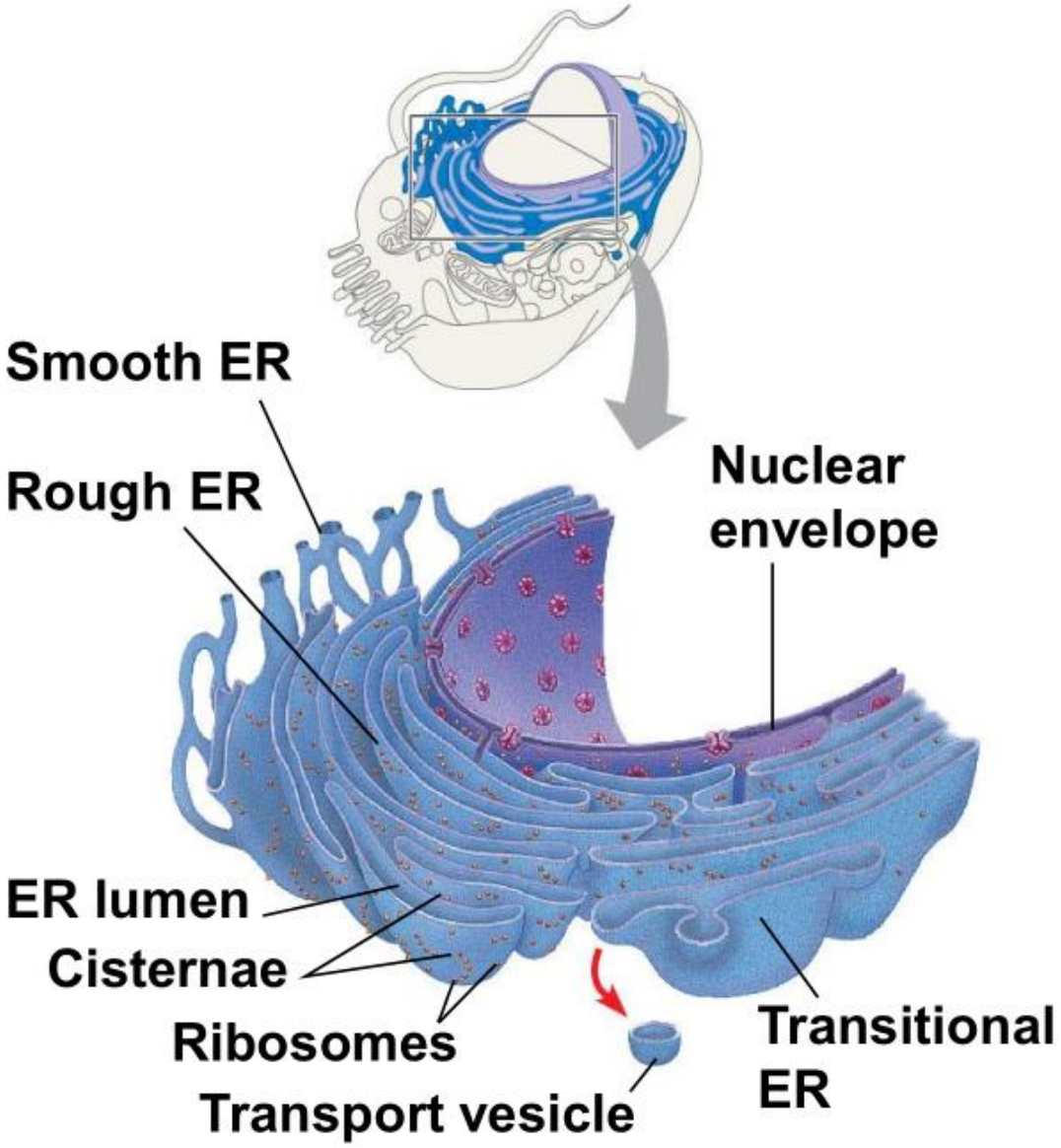
The ER membrane is continuous with the nuclear envelope

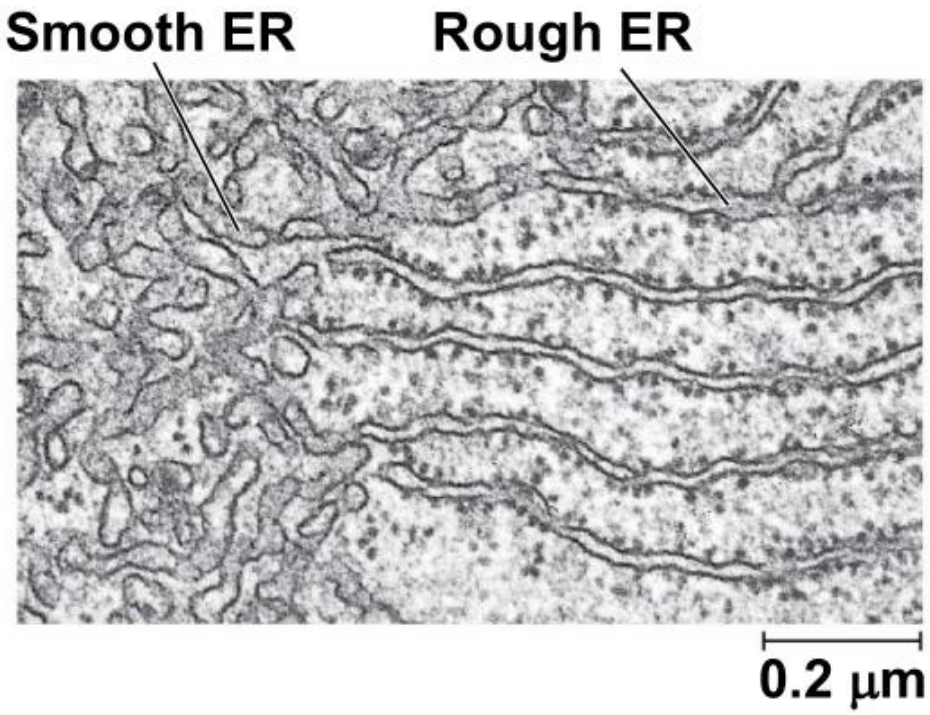
There are two distinct regions of ER:

Smooth ER, which lacks ribosomes

Rough ER, with ribosomes attached to its surface

The Endoplasmic Reticulum: Biosynthetic Factory





Functions of Smooth ER

The smooth ER

- Synthesizes lipids (including oils, steroids, and new membrane phospholipids)
- Releases glucose from glycogen in liver cells.
- Detoxifies drugs and poisons (Detoxification usually involves adding hydroxyl groups to drug molecules, making them more water-soluble and easier to flush from the body.
- Stores calcium ions (In muscle cells, a specialized type of smooth ER stores and delivers the calcium ions required for muscle contraction.)

Functions of Rough ER

The rough ER

- Poly peptides synthesized on the ER bound ribosomes thread into rough ER's interior, where they take on tertiary structure and assemble with other polypeptides and are sometimes glycosylated (addition of carbohydrate groups). The proteins that are incorporated into the membranes and secretory proteins are processed within the rough ER lumen.

- Secretory proteins depart from the ER wrapped in the membranes of vesicles that bud like bubbles from a specialized region called transitional ER. Others become enzymes that stay and carry out various functions of the smooth ER, including synthesis of phospholipids for the cell's membranes.

- Creates and modifies cellular membranes: Rough ER grows in place by adding membrane proteins and phospholipids to its own membrane and portions of it are transferred in the form of transport vesicles to other components of the endomembrane system.

The Golgi Apparatus: receiving, sorting, shipping center

The Golgi apparatus consists of flattened membranous sacs called cisternae

Functions of the Golgi apparatus:

- Modifies products of the ER such as proteins and phospholipids.

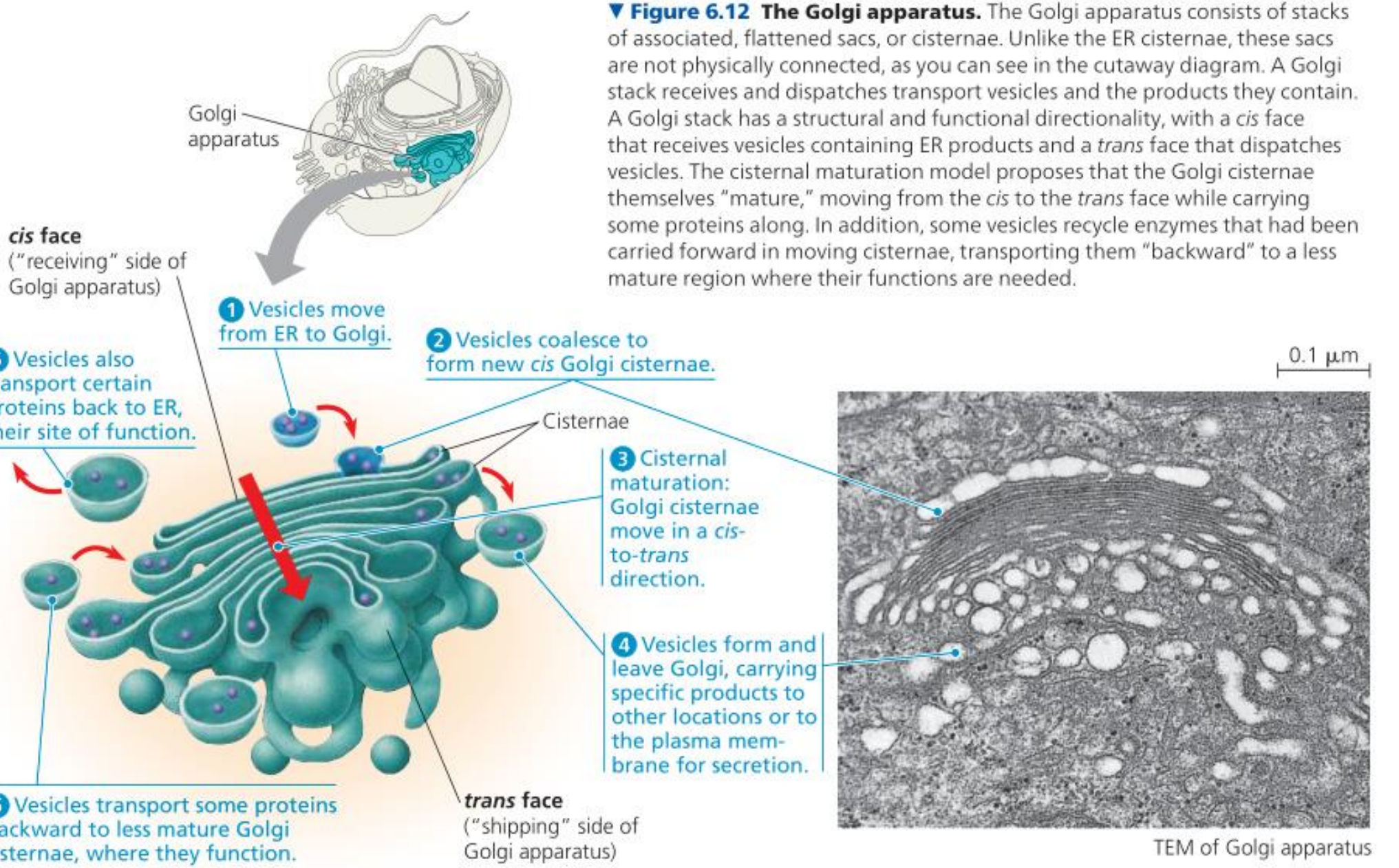
Puts finishing touches on proteins and lipids that have been delivered from ER, for example by attaching sugars, oligosaccharides, or phosphate groups.

The Golgi Apparatus: receiving, sorting, shipping center

- Sorts and packages materials into transport vesicle: Before a Golgi stack dispatches its products by budding vesicles from the trans face, it sorts these products and targets them for various parts of the cell. Molecular identification tags, such as phosphate groups added to the Golgi products, aid in sorting by acting like zip codes on mailing labels.

- Manufactures certain macromolecules in plant cells (polysaccharides such as pectin).

The Golgi



▼ **Figure 6.12 The Golgi apparatus.** The Golgi apparatus consists of stacks of associated, flattened sacs, or cisternae. Unlike the ER cisternae, these sacs are not physically connected, as you can see in the cutaway diagram. A Golgi stack receives and dispatches transport vesicles and the products they contain. A Golgi stack has a structural and functional directionality, with a *cis* face that receives vesicles containing ER products and a *trans* face that dispatches vesicles. The cisternal maturation model proposes that the Golgi cisternae themselves "mature," moving from the *cis* to the *trans* face while carrying some proteins along. In addition, some vesicles recycle enzymes that had been carried forward in moving cisternae, transporting them "backward" to a less mature region where their functions are needed.

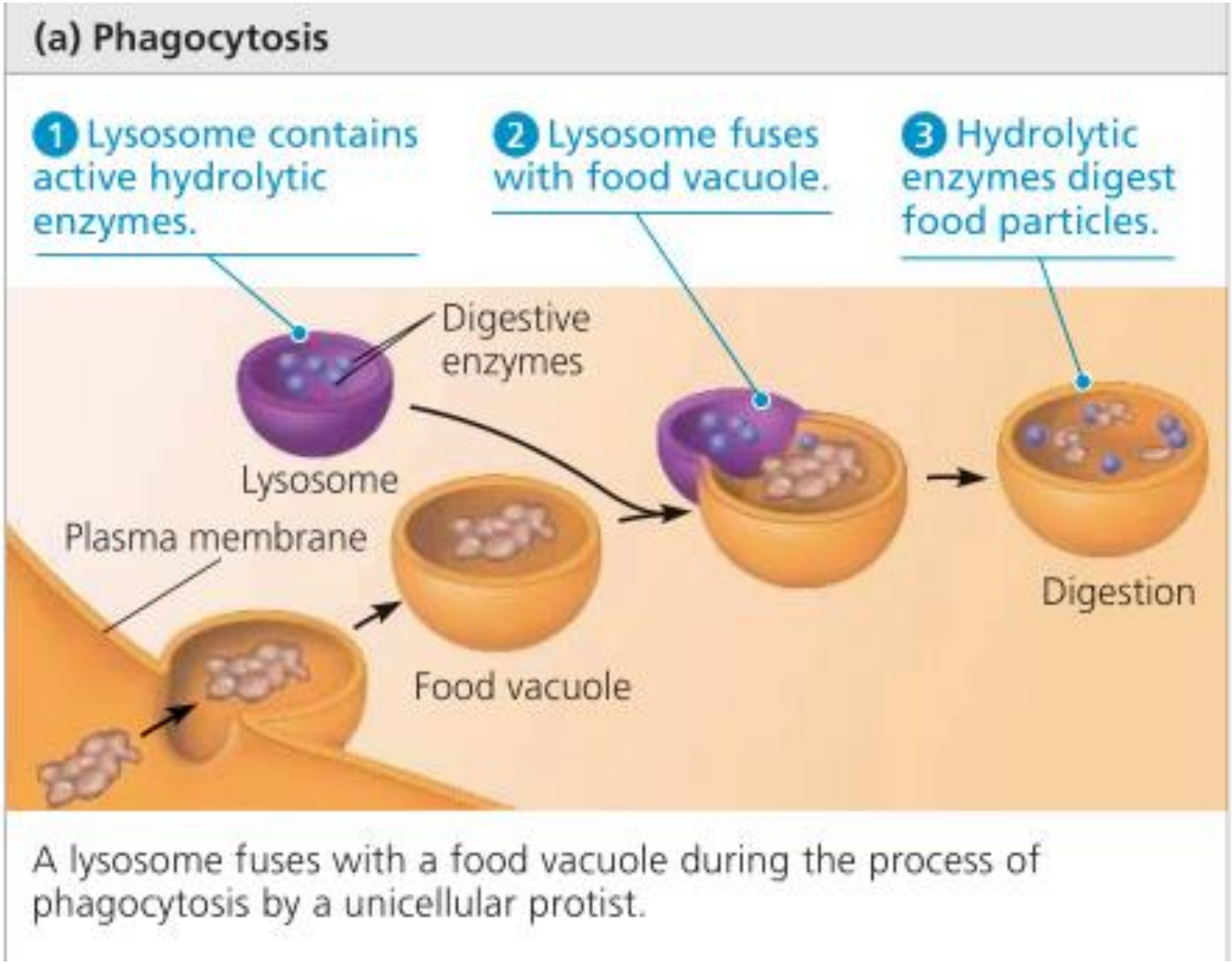
Lysosomes: Digestive Compartments

A lysosome is a membrane bound sac which contains hydrolytic enzymes that can digest proteins, fats, polysaccharides, nucleic acids, captured bacteria, organelles and debris.

Lysosomal enzymes break down the large organic molecules into smaller subunits by hydrolysis, releasing them into the cytosol for the cell to use. They work best in the acidic environment inside the lysosome. Therefore they can’t function in the neutral pH of cytosol.

The three-dimensional shape of lysosomal proteins protects them from digestion by lysosomal enzymes.

Phagocytosis



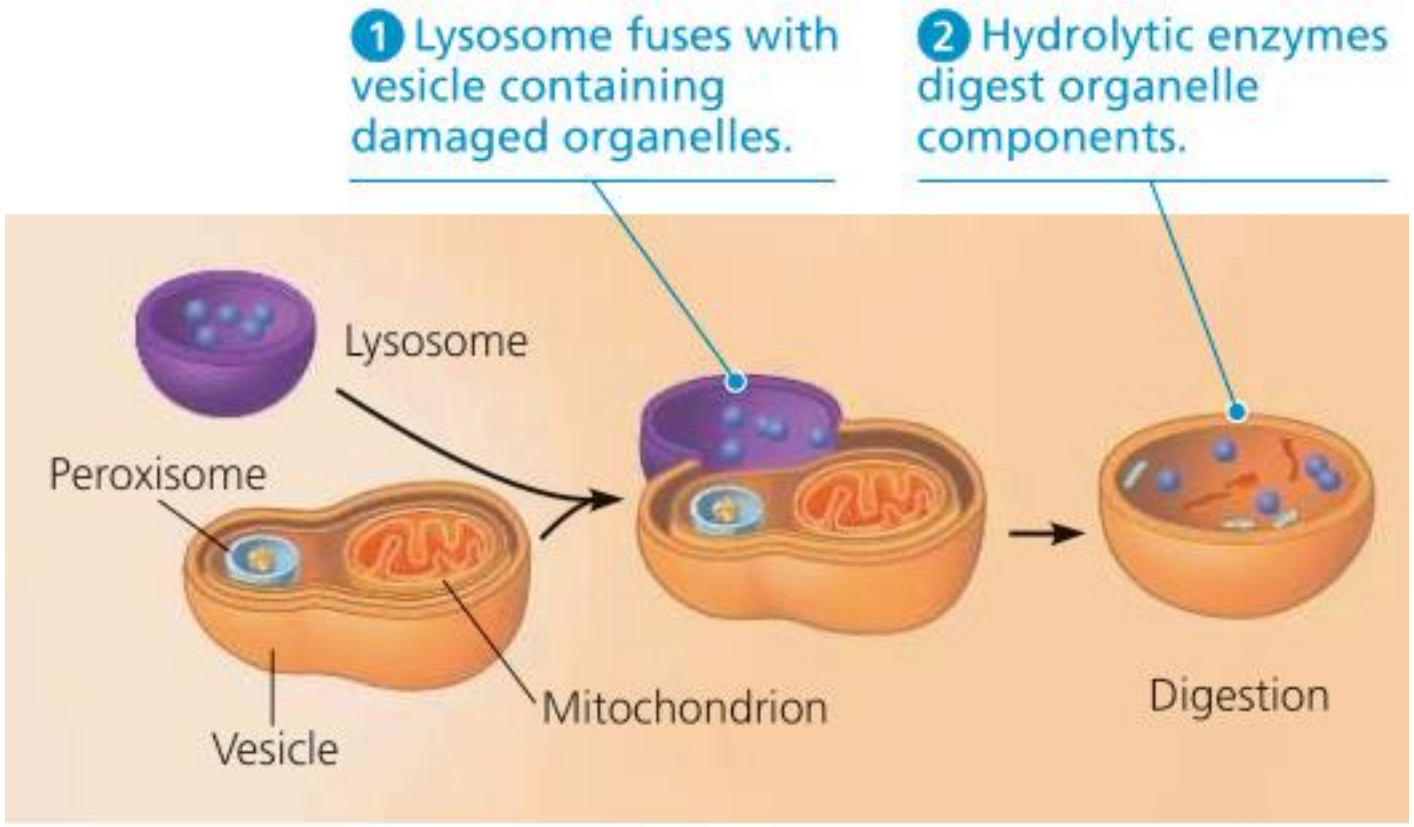
A lysosome fuses with a food vacuole during the process of phagocytosis by a unicellular protist.

Some types of cell (macrophages, amoeba) can engulf another cell by phagocytosis; this forms a food vacuole.

A lysosome fuses with the food vacuole and digests the engulfed molecules.

Digestion products, including simple sugars, amino acids, and other monomers, pass into the cytosol and become nutrients for the cell.

Autophagy



The vesicle with damaged organelles fuses with a lysosome. The organelles are then digested and their components recycled.

Lysosomes also use enzymes to recycle the cell's own damaged organelles and macromolecules such as misfolded proteins, a process called autophagy.

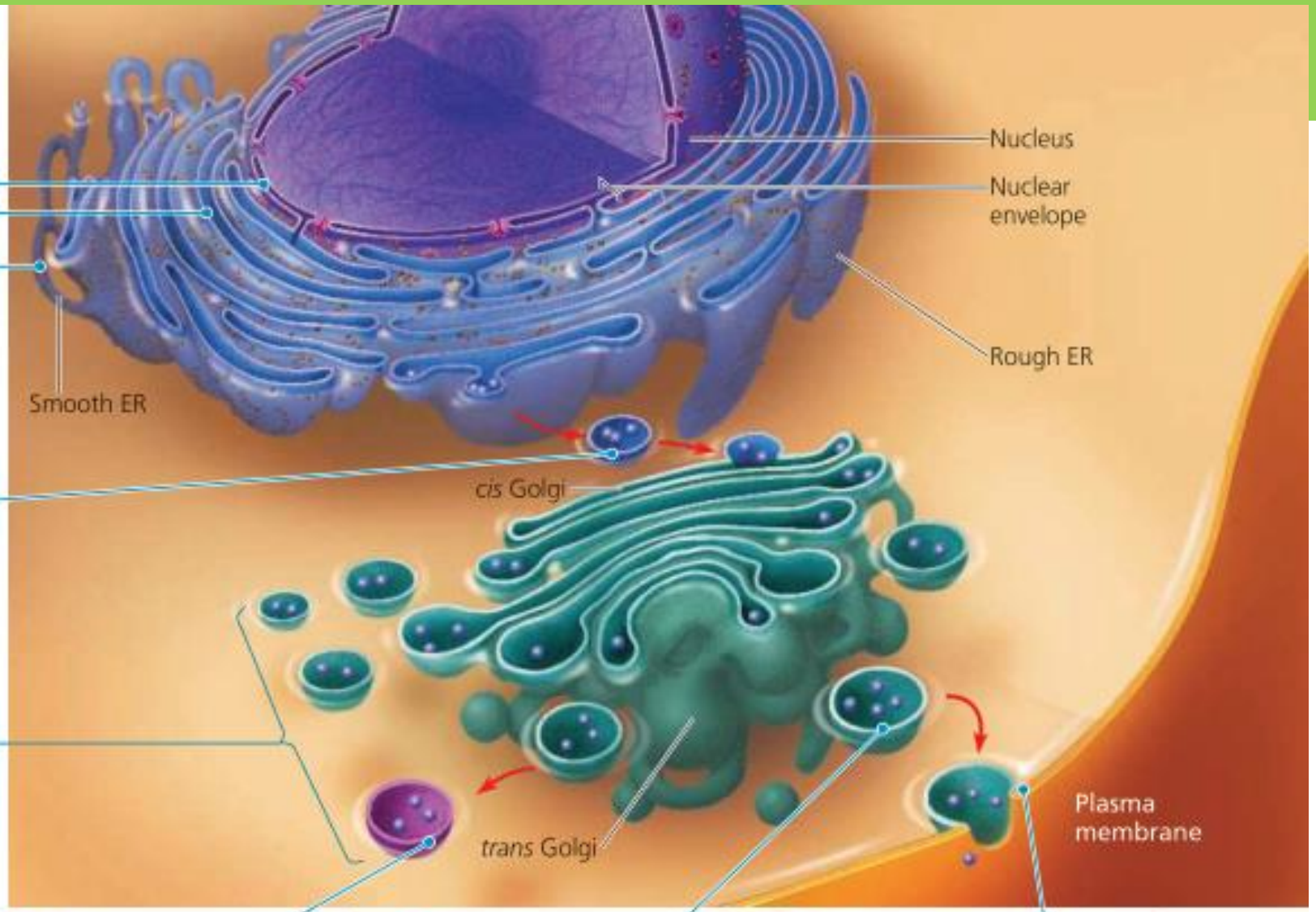
With the help of lysosomes, the cell continually renews itself.
A human liver cell, for example, recycles half of its macromolecules each week.

Vacuoles: Diverse Maintenance Compartments

Vacuoles are large vesicles derived from the endoplasmic reticulum and Golgi apparatus.

The solution inside a vacuole differs in composition from the cytosol because like all cellular membranes, the vacuolar membrane is selective in transporting solutes.

Vacuoles perform a variety of functions in different kinds of cells.
Food vacuoles are formed by phagocytosis.
Contractile vacuoles, found in many freshwater protists, pump excess water out of cells thereby maintaining a suitable concentration of ions and molecules inside the cell.
Central vacuoles, found in many mature plant cells, hold inorganic ions (including potassium and chloride), sugars, pigments and water.
Certain vacuoles in plants and fungi carry out enzymatic hydrolysis like lysosomes.



▲ Figure 6.15 Review: Relationships among organelles of the endomembrane system.
The red arrows show some of the migration pathways for membranes and the materials they enclose.

The Evolutionary Origins of Mitochondria and Chloroplasts

Mitochondria and chloroplasts display similarities with bacteria.

- Have a double membrane (unlike the organelles of the endomembrane system).
- Have their own ribosomes and circular DNA molecules—like bacterial chromosomes. Their DNA programs the synthesis of their own proteins.
- Grow and divide independently of the cell.

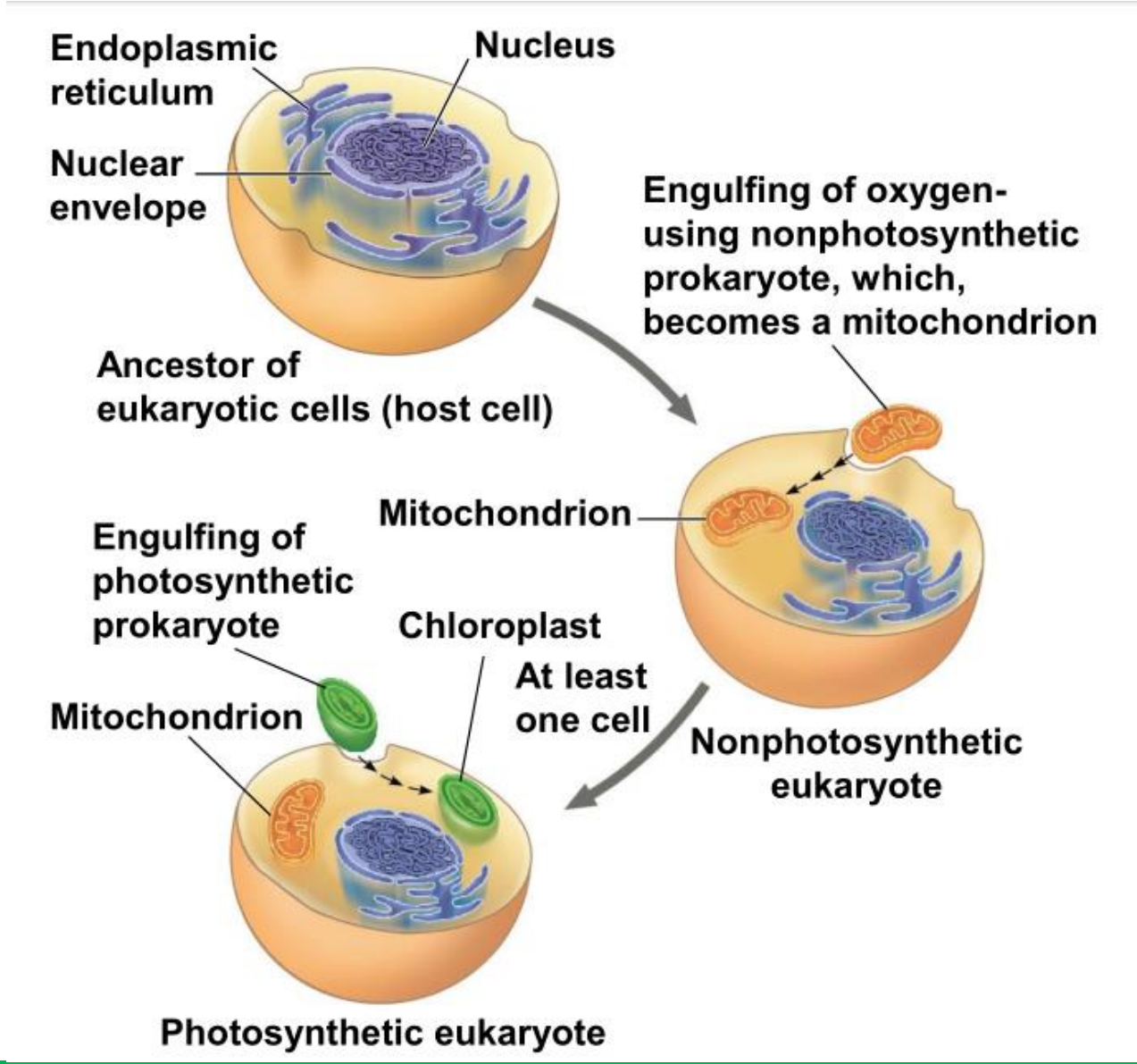
The Evolutionary Origins of Mitochondria and Chloroplasts

The endosymbiont theory of the origins of mitochondria and chloroplasts in eukaryotic cells states that:

- An early ancestor of eukaryotic cells engulfed a nonphotosynthetic prokaryotic cell, which formed an endosymbiont relationship with its host
- The host cell and endosymbiont (a cell living within another cell) merged into a single organism, a eukaryotic cell with a mitochondrion
- At least one of these cells may have taken up a photosynthetic prokaryote, becoming the ancestor of cells that contain chloroplasts

The endosymbiont theory

The endosymbiont theory of the origins of mitochondria and chloroplasts in eukaryotic cells. According to this theory, the proposed ancestors of mitochondria were oxygen-using nonphotosynthetic prokaryotes that were taken into host cells, while the proposed ancestors of chloroplasts were photosynthetic prokaryotes. The large arrows represent change over evolutionary
time; the small arrows inside the cells show the process of the endosymbiont becoming an organelle, also over long periods of time.



Mitochondria and chloroplasts change energy from one form to another

In eukaryotic cells, mitochondria and chloroplasts are the organelles that convert energy to forms that cells can use for work.

Mitochondria are the sites of cellular respiration, the metabolic process that uses oxygen to drive the generation of ATP by extracting energy from sugars, fats, and other fuels.

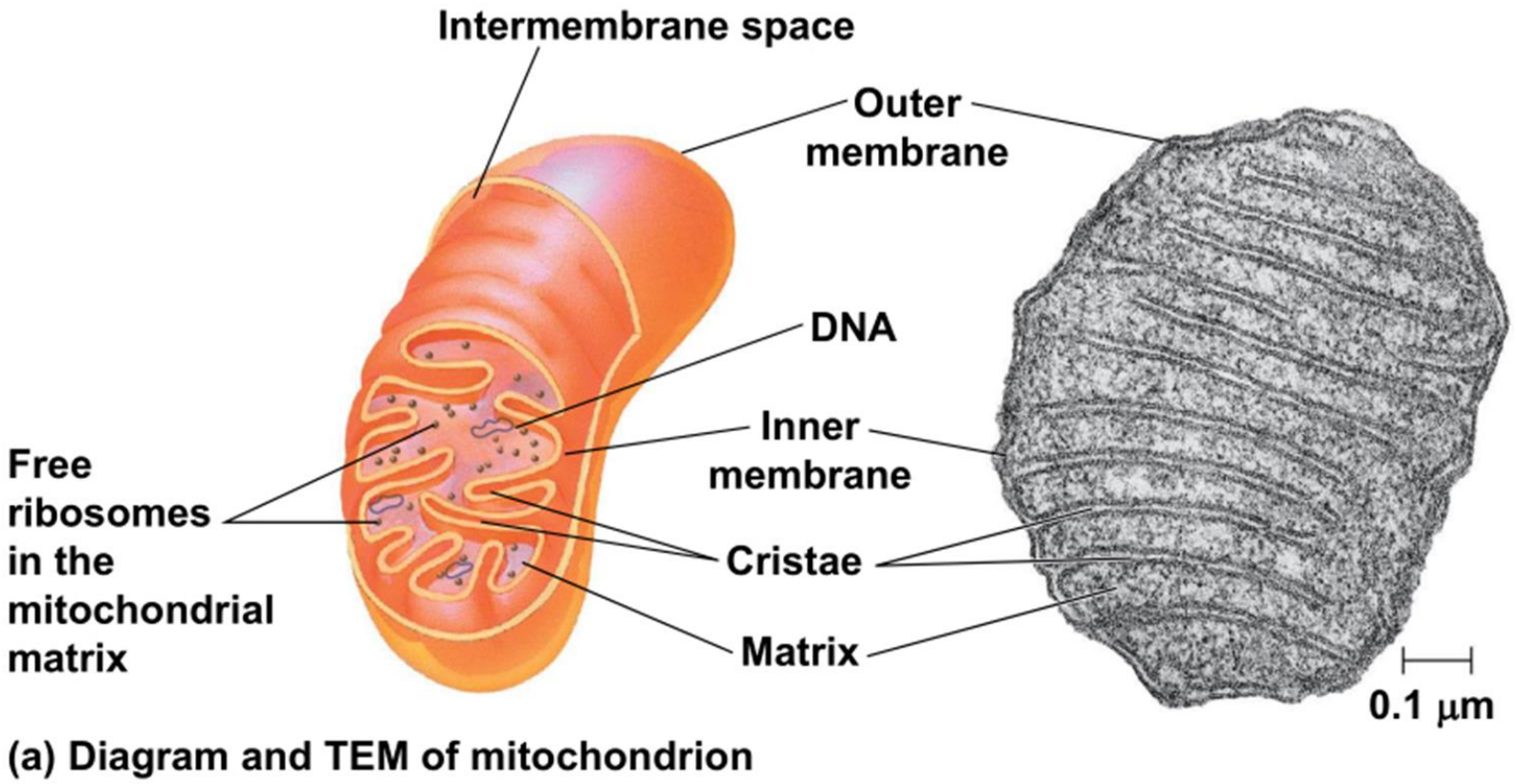
Chloroplasts, found in plants and algae, are the sites of photosynthesis the process that converts solar energy to chemical energy by absorbing sunlight and using it to drive the synthesis of organic compounds such as sugars from carbon dioxide and water.

Peroxisomes are oxidative organelles.

Mitochondria: Chemical Energy Conversion

Almost all eukaryotic cells contain mitochondria.

In general, cells that have a high demand for energy tend to have more mitochondria.



(a) Diagram and TEM of mitochondrion

Mitochondria: Chemical Energy Conversion

Growth, cell division, protein production, secretion, and many chemical reactions in the cytoplasm all require a steady supply of energy.

The nucleotide ATP supply energy for reactions.

Cells require a lot of ATP.

The most efficient way they can produce it is by aerobic respiration, a series of oxygen-requiring reactions that harvests energy from sugars by breaking their bonds.

In eukaryotes, aerobic respiration (cellular respiration) occurs inside mitochondria (singular, mitochondrion).

The structure of a mitochondrion is specialized for carrying out reactions of aerobic respiration.

Mitochondria: Chemical Energy Conversion

Mitochondria is surrounded by **two lipid bilayer** membranes:
The **outer membrane** is smooth.
The **inner membrane** is convoluted, with infoldings called **cristae**.
The outer compartment between these membranes is called **the intermembrane space**.
The inner compartment enclosed by the inner membrane is called **mitochondrial matrix**.
The matrix contains enzymes, mitochondrial DNA and ribosomes.
Enzymes in the matrix catalyze some of the steps of cellular respiration.
Other proteins that function in respiration are built into the inner membrane.
The cristae give the inner membrane a large surface area, thus enhancing the productivity of cellular respiration.
Hydrogen ions accumulate in the outer compartment. The buildup pushes the ions across the inner membrane, into the inner compartment, and this flow drives ATP formation.

Chloroplasts: Capture of Light Energy

The chloroplast is a member of a family of organelles called plastids

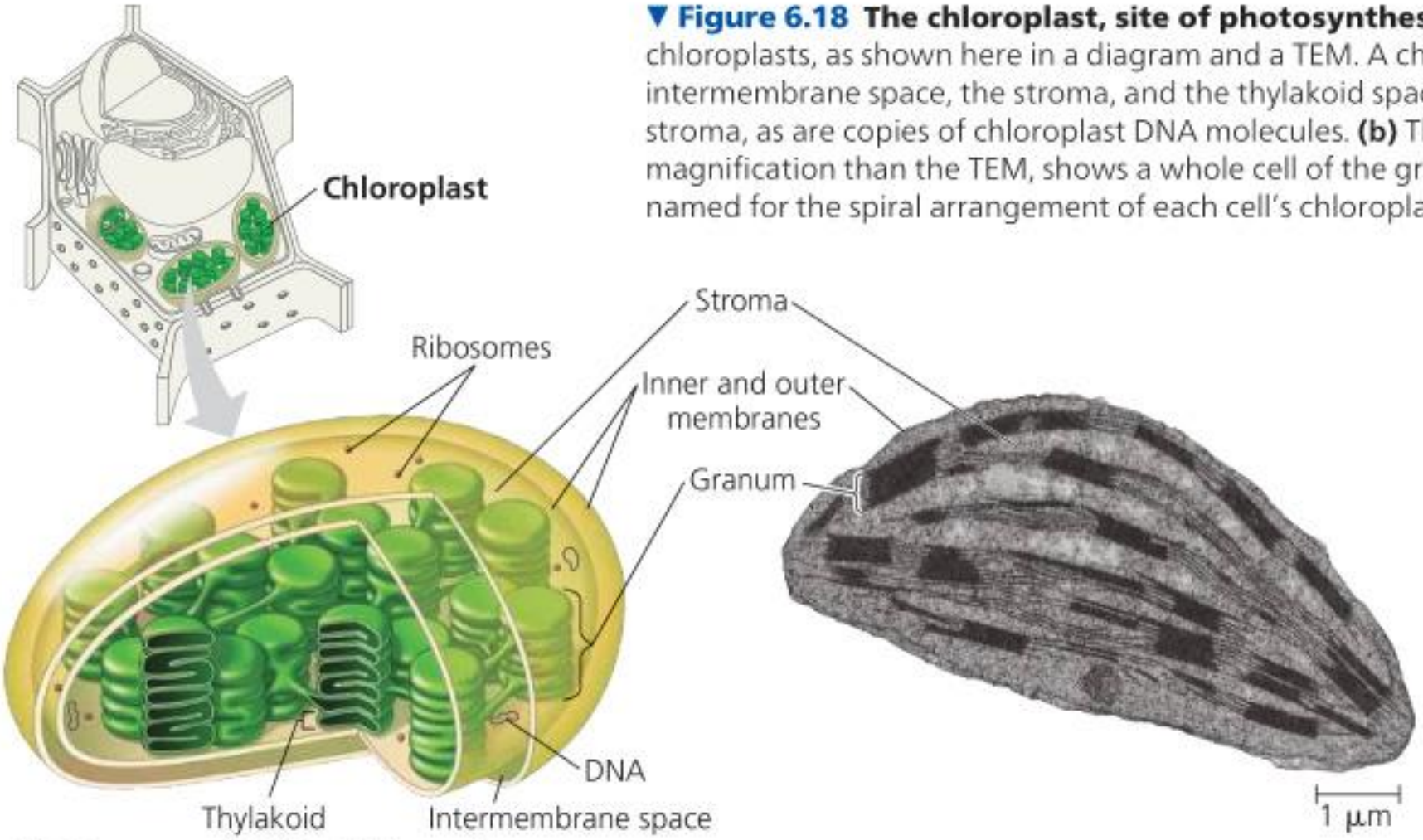
Chloroplasts contain the green pigment chlorophyll, as well as enzymes and other molecules that function in the photosynthetic production of sugar.

Chloroplasts are found in leaves and other green organs of plants and in algae

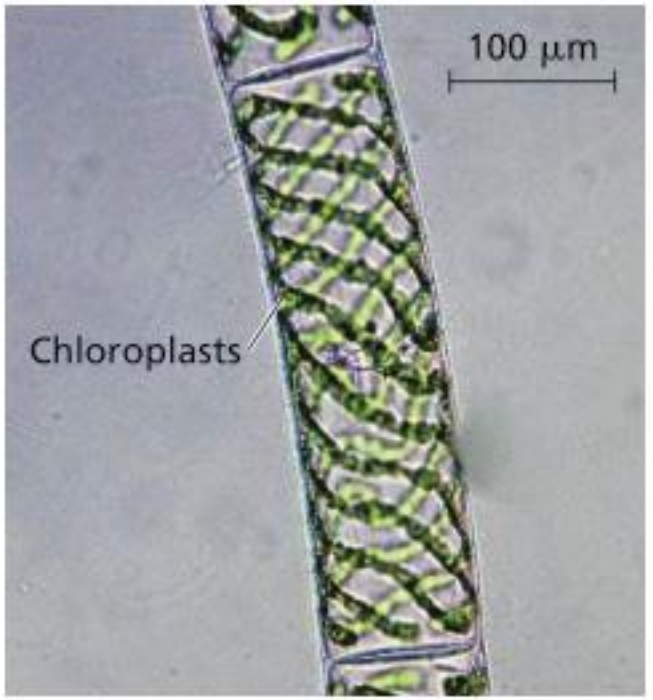
Chloroplast structure includes:
Thylakoids, membranous sacs, stacked to form a **granum**
Stroma, the -fluid outside the thylakoids. Stroma contains the chloroplast DNA, ribosomes and many enzymes.

The membranes of the chloroplast divide the chloroplast space into three compartments: the intermembrane space, the stroma, and the thylakoid space. This compartmental organization enables the chloroplast to convert light energy to chemical energy during photosynthesis.

Chloroplasts: Capture of Light Energy



(a) Diagram and TEM of chloroplast



(b) Chloroplasts in an algal cell

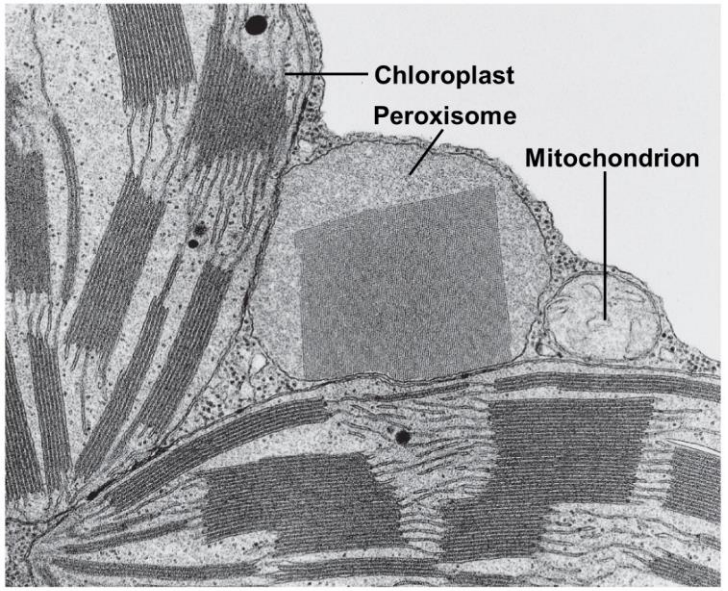
▼ **Figure 6.18 The chloroplast, site of photosynthesis. (a)** Many plants have lens-shaped chloroplasts, as shown here in a diagram and a TEM. A chloroplast has three compartments: the intermembrane space, the stroma, and the thylakoid space. Free ribosomes are present in the stroma, as are copies of chloroplast DNA molecules. **(b)** This light micrograph, at a much lower magnification than the TEM, shows a whole cell of the green alga *Spirogyra crassa*, which is named for the spiral arrangement of each cell's chloroplasts.

Peroxisomes: Oxidation

Peroxisomes are specialized metabolic compartments bounded by a single membrane

Peroxisomes perform reactions with many different functions. Oxygen is used to break down different types of molecules. (fatty acids, alcohol)

Peroxisomes produce hydrogen peroxide as a byproduct which is also toxic and convert it to water.



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The cytoskeleton

- The cytoskeleton (Cell skeleton) is a network of fibers that organizes structures and activities in the cell
- The cytoskeleton extends throughout the cytoplasm
- It organizes the cell's structures and activities, anchoring many organelles

The cytoskeleton

It is composed of three types of constantly assembling and disassembling molecular structures:

Microtubules

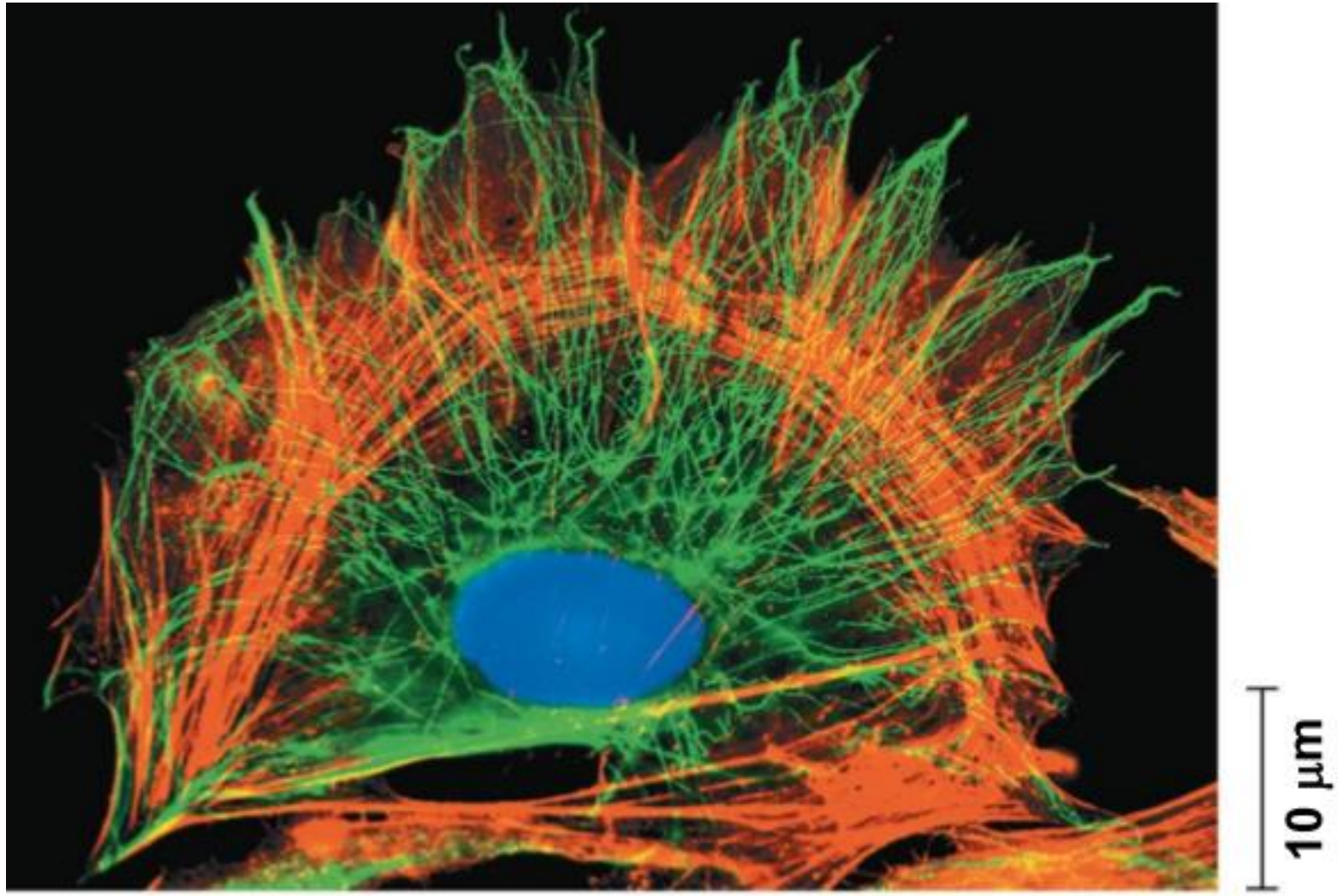
Microfilaments

Intermediate filaments

As shown in this fluorescence micrograph, the cytoskeleton extends throughout the cell.

The cytoskeletal elements have been tagged with different fluorescent molecules: green for microtubules and orange for microfilaments.

Intermediate filaments, is not evident here. (The blue color tags the DNA in the nucleus.)



Functions of the Cytoskeleton: Support, Motility, and Regulation

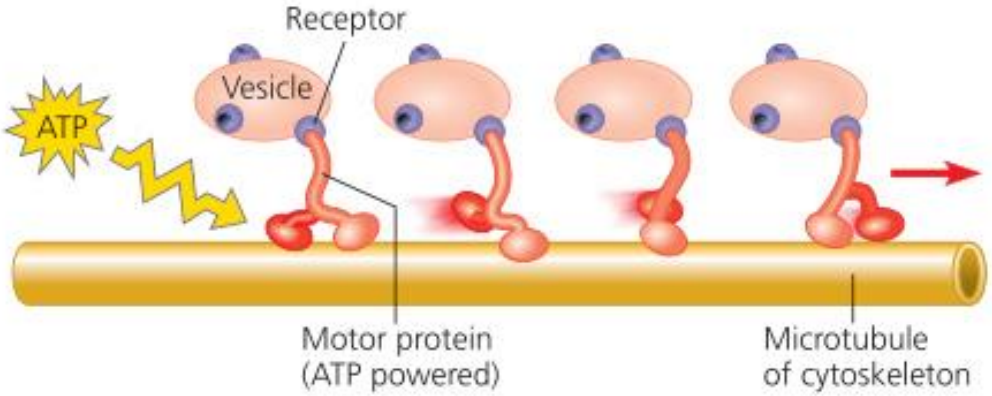
The cytoskeleton helps to support the cell and maintain its shape (Gives the cell structure)

It provides anchorage for many organelles and molecules (It holds them in place)

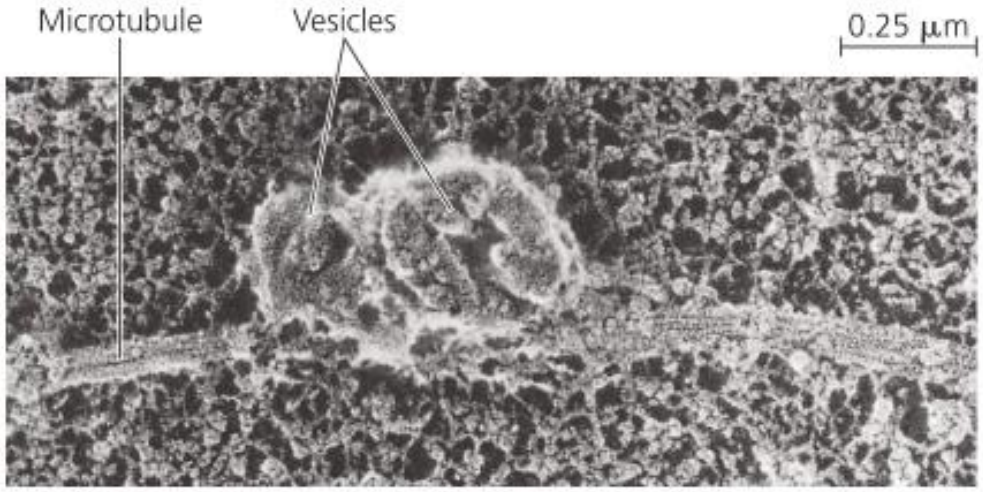
It interacts with motor proteins to produce motility. (motility includes both changes in cell location and movements of cell parts.)

Inside the cell, vesicles and other organelles can “walk” along the tracks provided by the cytoskeleton.

▼ **Figure 6.21 Motor proteins and the cytoskeleton.**

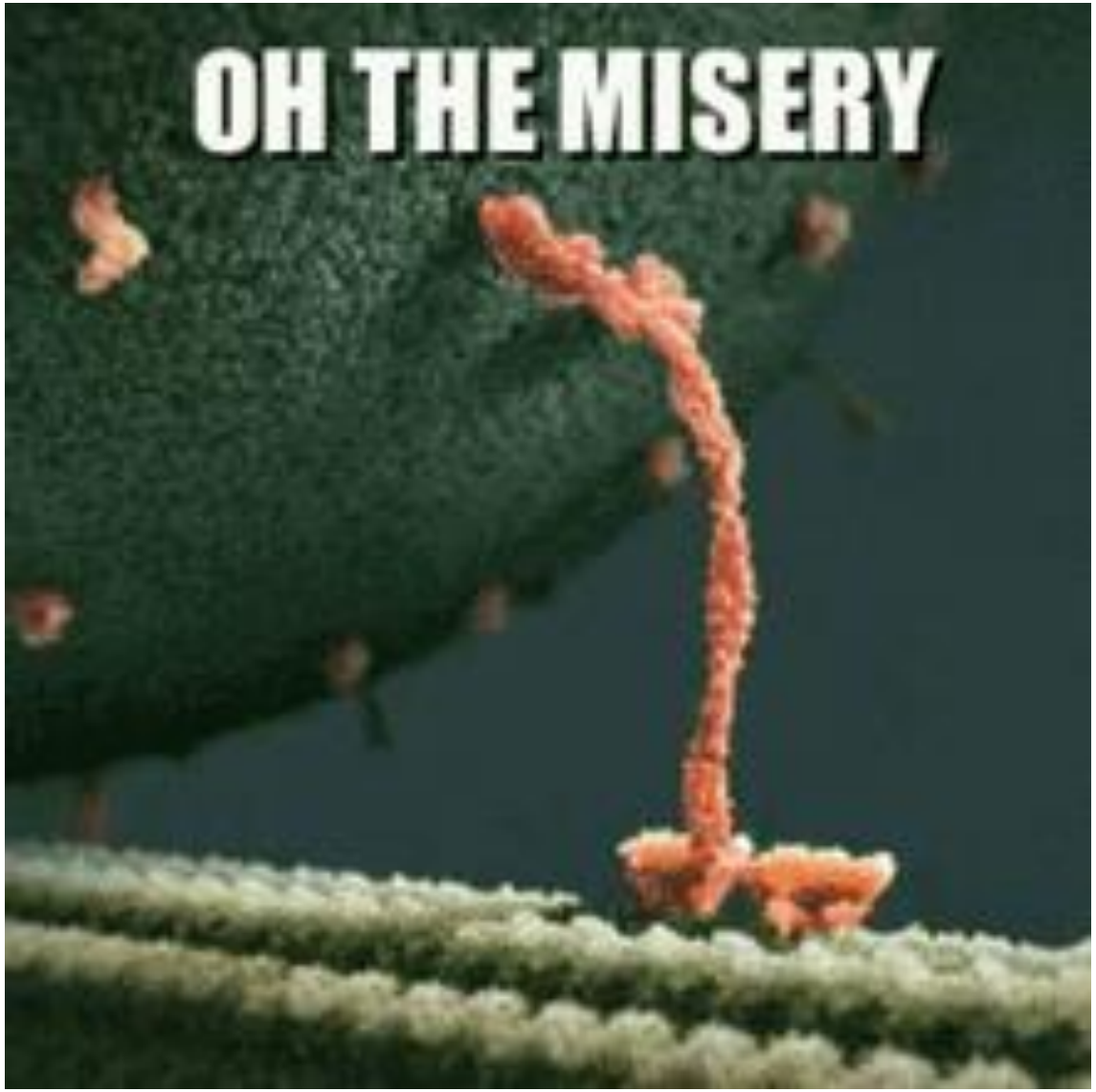


(a) A motor protein that attaches to a receptor on a vesicle can "walk" the vesicle along a microtubule or microfilament.



(b) Two vesicles containing neurotransmitters move along a microtubule toward the tip of a nerve cell extension called an axon (SEM).

The cytoskeleton



Kinesin walking on a microtubule carrying a vesicle.

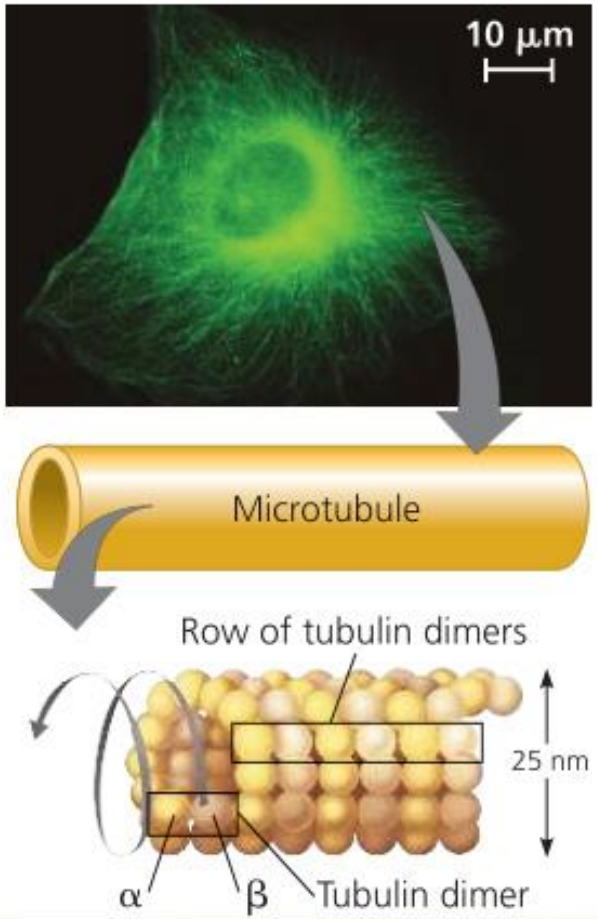
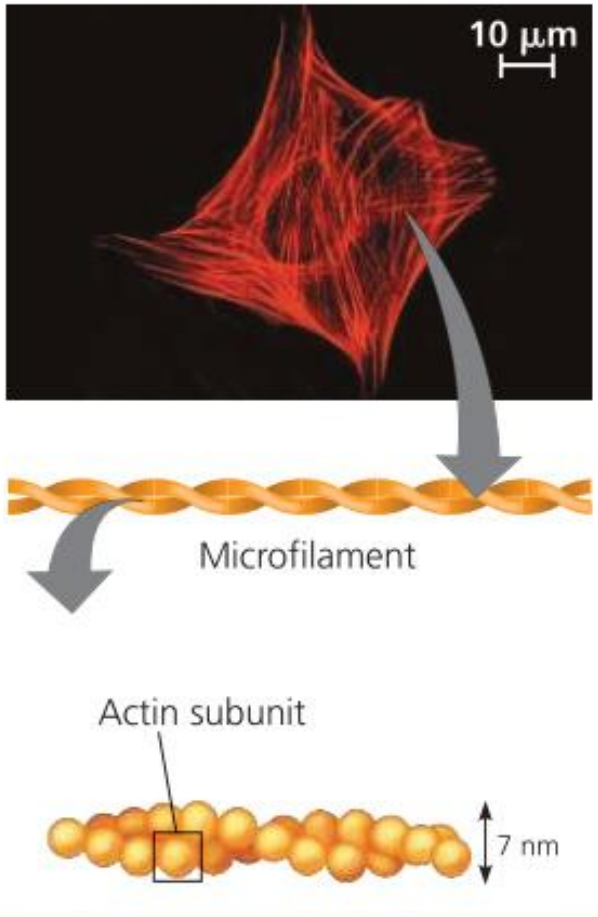
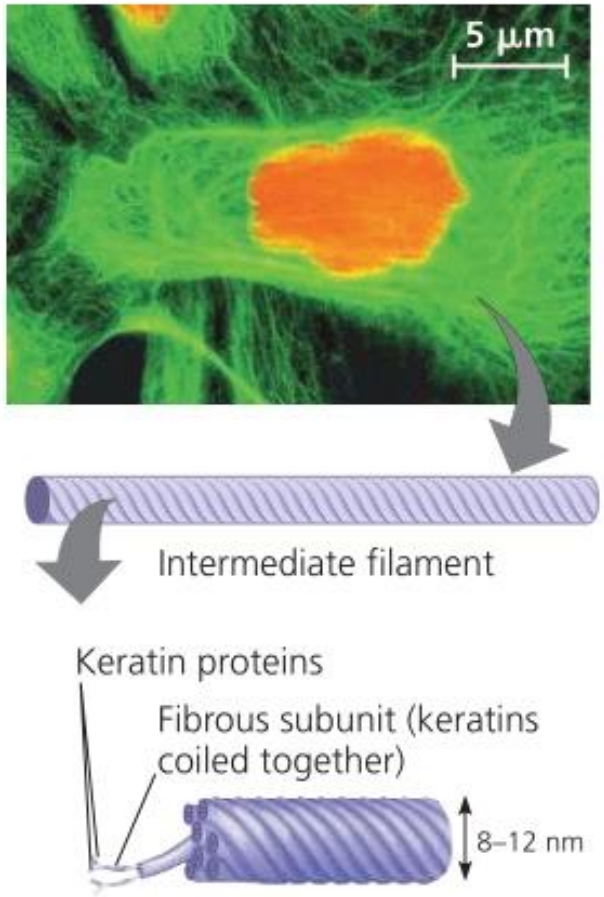
Components of the Cytoskeleton

Three main types of fibers make up the cytoskeleton:

Microtubules are the thickest

Microfilaments (actin filaments) are the thinnest.

Intermediate filaments are fibers with diameters in a middle range.

Property	Microtubules (Tubulin Polymers)	Microfilaments (Actin Filaments)	Intermediate Filaments
Structure	Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables
Diameter	25 nm with 15-nm lumen	7 nm	8–12 nm
Protein subunits	Tubulin, a dimer consisting of an α-tubulin and a β-tubulin	Actin	One of several different proteins (including keratins)
Main functions	Maintenance of cell shape; cell motility; chromosome movements in cell division; organelle movements	Maintenance of cell shape; changes in cell shape; muscle contraction; cytoplasmic streaming (plant cells); cell motility; cell division (animal cells)	Maintenance of cell shape; anchorage of nucleus and certain other organelles; formation of nuclear lamina
Fluorescence micrographs of fibroblasts. Connective tissue cells called fibroblasts are a favorite cell type for cell biology studies because they spread out flat and their internal structures are easy to see. In each fibroblast shown here, the structure of interest has been tagged with fluorescent molecules. In the third micrograph, the DNA in the nucleus has also been tagged (orange).			

Microtubules

Microtubules are hollow rods constructed from globular protein dimers called tubulin

Functions of microtubules

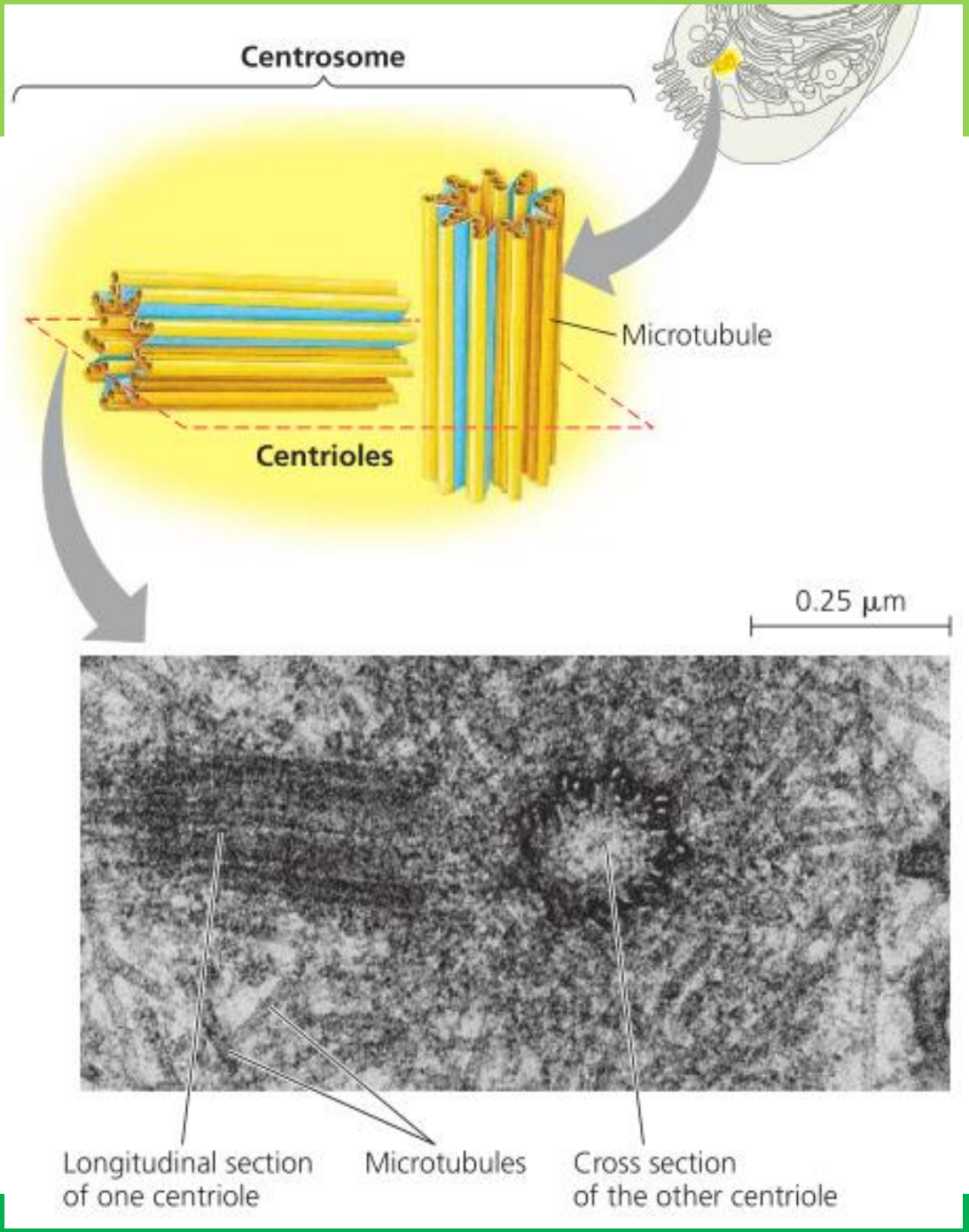
- Shape and support the cell (Resist compression)
- Guide movement of organelles (microtubules guide vesicles from the ER to the Golgi apparatus and from the Golgi to the plasma membrane)
- Separate chromosomes during cell division

Centrosomes and Centrioles

In many cells, microtubules grow out from a centrosome a region that is often located near the nucleus.

The centrosome is a "microtubule-organizing center"

In animal cells, the centrosome has a pair of centrioles, each with nine triplets of microtubules arranged in a ring



Cilia and Flagella

Microtubules control the beating of cilia and flagella, locomotor appendages of some cells.

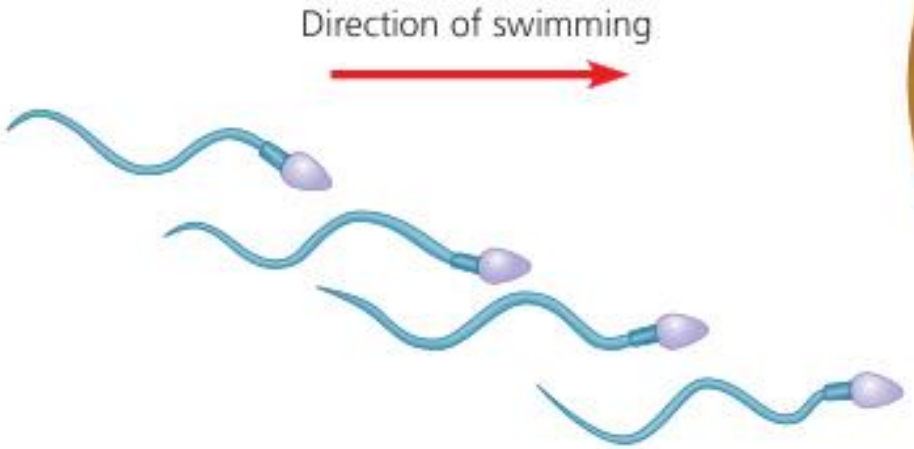
Many unicellular protists are propelled through water by cilia or flagella that act as locomotor appendages, and the sperm of animals, algae, and some plants have flagella.

Cilia and flagella differ in their beating patterns.
A flagellum has a snakelike motion driving a cell in the same direction as the axis of the flagellum.

Cilia have a back-and-forth motion. The rapid power stroke moves the cell in a direction perpendicular to the axis of the cilium. Then, during the slower recovery stroke,the cilium bends and sweeps sideways, closer to the cell surface.

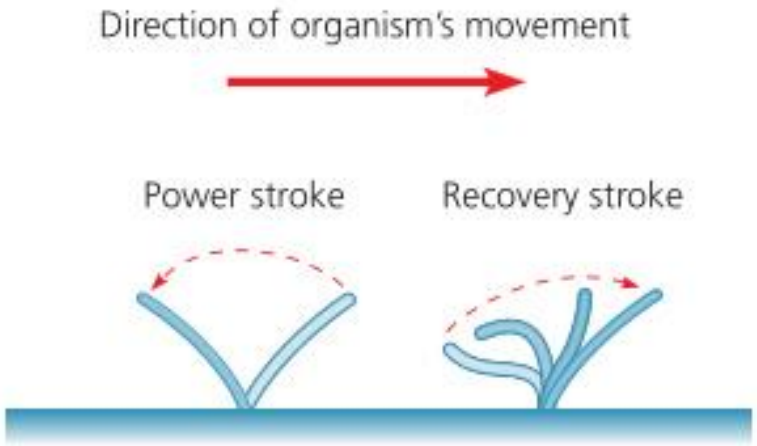
▼ Figure 6.23 A comparison of the beating of flagella and motile cilia.

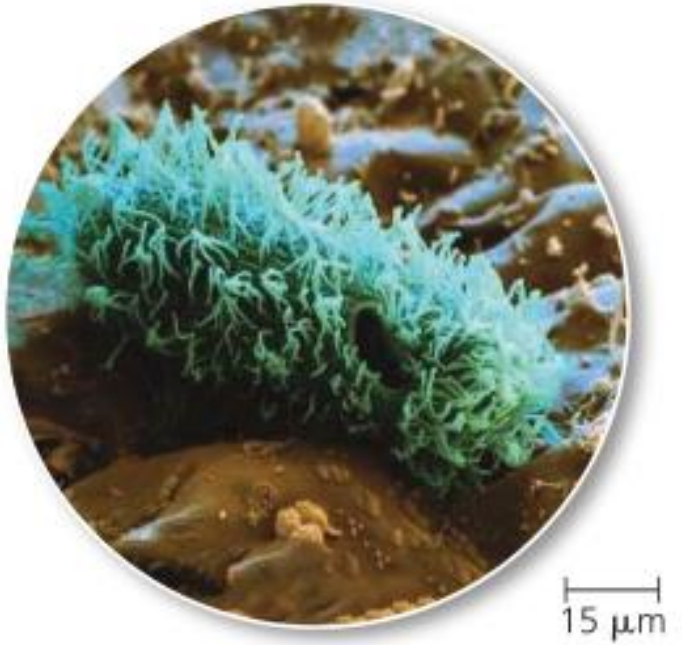
(a) Motion of flagella. A flagellum usually undulates, its snakelike motion driving a cell in the same direction as the axis of the flagellum. Propulsion of a human sperm cell is an example of flagellate locomotion (LM).





(b) Motion of cilia. Cilia have a back-and-forth motion. The rapid power stroke moves the cell in a direction perpendicular to the axis of the cilium. Then, during the slower recovery stroke, the cilium bends and sweeps sideways, closer to the cell surface. A dense nap of cilia, beating at a rate of about 40 to 60 strokes a second, covers this *Colpidium*, a freshwater protist (colorized SEM).





Microfilaments (Actin Filaments)

Microfilaments are thin solid rods, built from actin subunits which are globular proteins.

The structural role of microfilaments is to bear tension, resisting pulling forces within the cell.

In some kinds of animal cells, such as nutrient-absorbing intestinal cells, bundles of microfilaments make up the core of **microvilli**, delicate projections that increase the cell's surface area

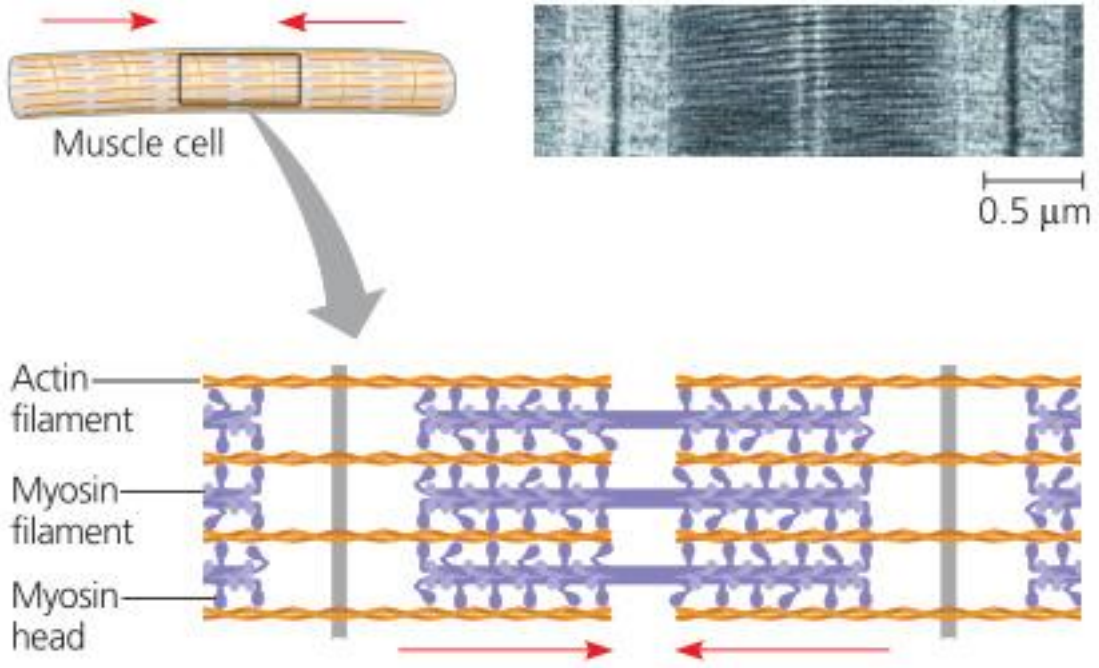
Microfilaments (Actin Filaments)

Microfilaments that function in cellular motility interact with the motor protein myosin

For example, actin and myosin interact to cause muscle contraction, amoeboid movement of white blood cells, and cytoplasmic streaming in plant cells.

The cell crawls along a surface by extending cellular extensions called pseudopodia (from the Greek pseudes, false, and pod, foot) and moving toward them.

▼ **Figure 6.26 Microfilaments and motility.** In these three examples, interactions between actin filaments and motor proteins bring about cell movement.



(a) Myosin motors in muscle cell contraction. The "walking" of myosin projections (the so-called heads) drives the parallel myosin and actin filaments past each other so that the actin filaments approach each other in the middle (red arrows). This shortens the muscle cell. Muscle contraction involves the shortening of many muscle cells at the same time (TEM).

Intermediate Filaments

Intermediate filaments are larger than microfilaments but smaller than microtubules

Intermediate filaments are only found in the cells of some animals, including vertebrates

They support cell shape and fix organelles in place

Intermediate filaments are more permanent cytoskeleton elements than the other two classes

The nucleus typically sits within a cage made of intermediate filaments.
Other intermediate filaments make up the nuclear lamina, which lines the interior of the nuclear membrane.

Extracellular components and connections between cells

Most cells synthesize and secrete materials that are external to the plasma membrane

These extracellular structures include:
Cell walls of plants

The extracellular matrix (ECM) of animal cells

Intercellular junctions

Cell Walls of Plants

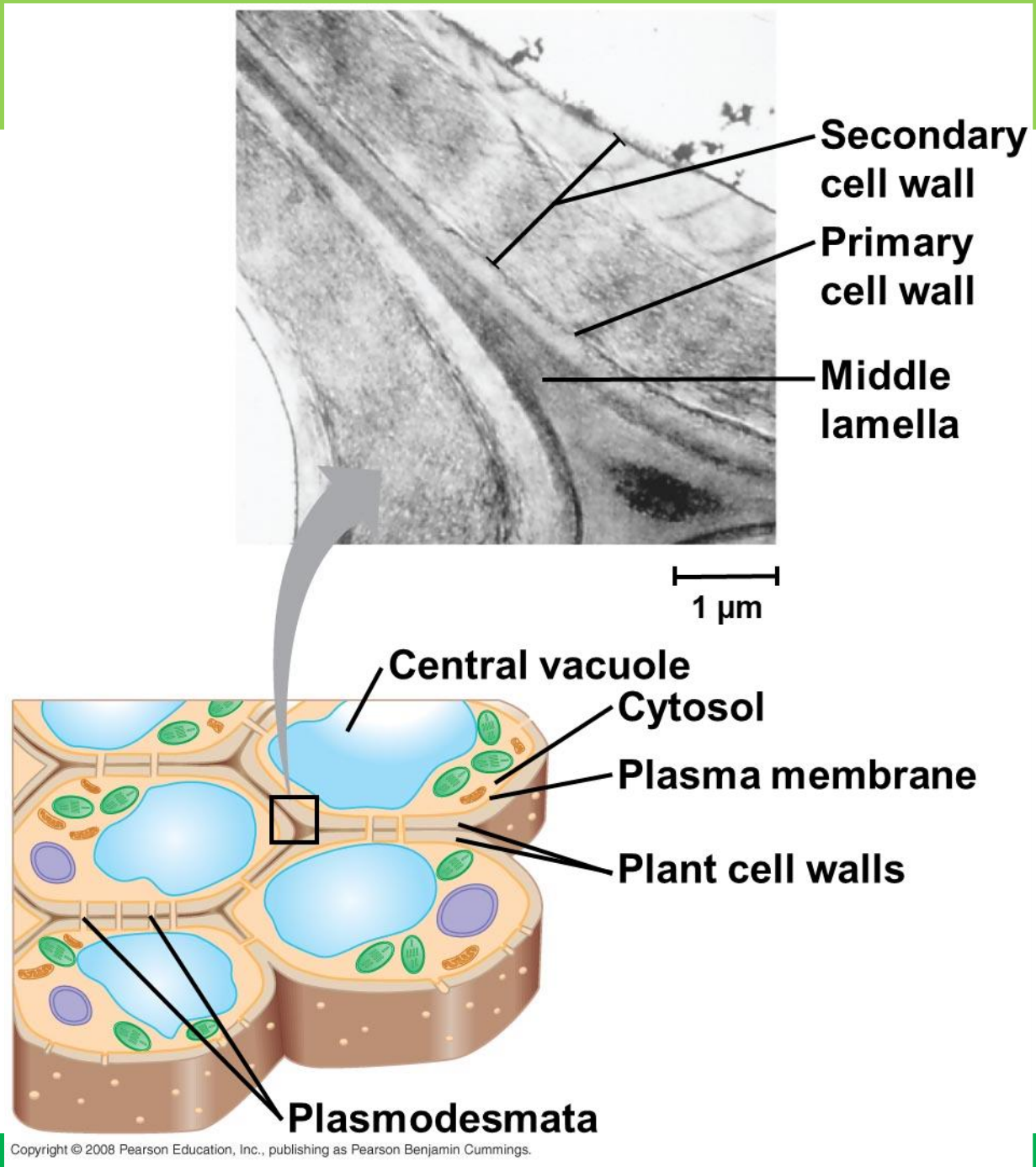
The cell wall is an extracellular structure that distinguishes plant cells from animal cells Prokaryotes, fungi, and some protists also have cell walls.

The cell wall protects the plant cell, maintains its shape, and prevents excessive uptake of water.

Plant cell walls are made of cellulose fibers embedded in other polysaccharides and protein.

Cell Walls of Plants

Plant cell walls may have multiple layers:
Primary cell wall: relatively thin and flexible
Middle lamella: thin layer between primary walls of adjacent cells
Secondary cell wall (in some cells): added between the plasma membrane and the primary cell wall. The secondary wall, often deposited in several laminated layers, has a strong and durable matrix that affords the cell protection and support. (Wood)



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The Extracellular Matrix (ECM) of Animal Cells

Animal cells lack cell walls but are covered by an elaborate extracellular matrix (ECM).

The ECM is made up of glycoproteins (proteins with covalently bonded carbohydrates) such as collagen, proteoglycans, and fibronectin secreted by the cells.
(collagen accounts for about 40% of the total protein in the human body.)

ECM proteins bind to receptor proteins in the plasma membrane called integrins.
Integrins are in a position to transmit signals between the ECM and the cytoskeleton and thus to integrate changes occurring outside and inside the cell.

By communicating with a cell through integrins, the ECM can regulate a cell's behavior.

The Extracellular Matrix (ECM) of Animal Cells

Functions of the ECM:
structurally supports and organizes tissues

Support: provides mechanical support to tissues and organs. It is especially important in connective tissues, such as bone, cartilage, and tendons, where it gives strength and elasticity.

Adhesion: allows cells to adhere to the surrounding tissue, facilitating cell migration, organization, and maintenance. This is important in processes like wound healing and tissue development.

Regulation: The ECM contains signaling molecules and receptors that can influence cell behavior. These signals can regulate cell growth, differentiation, and migration.

Intercellular Junctions

Neighboring cells in tissues, organs, or organ systems often adhere, interact, and communicate through direct physical contact.

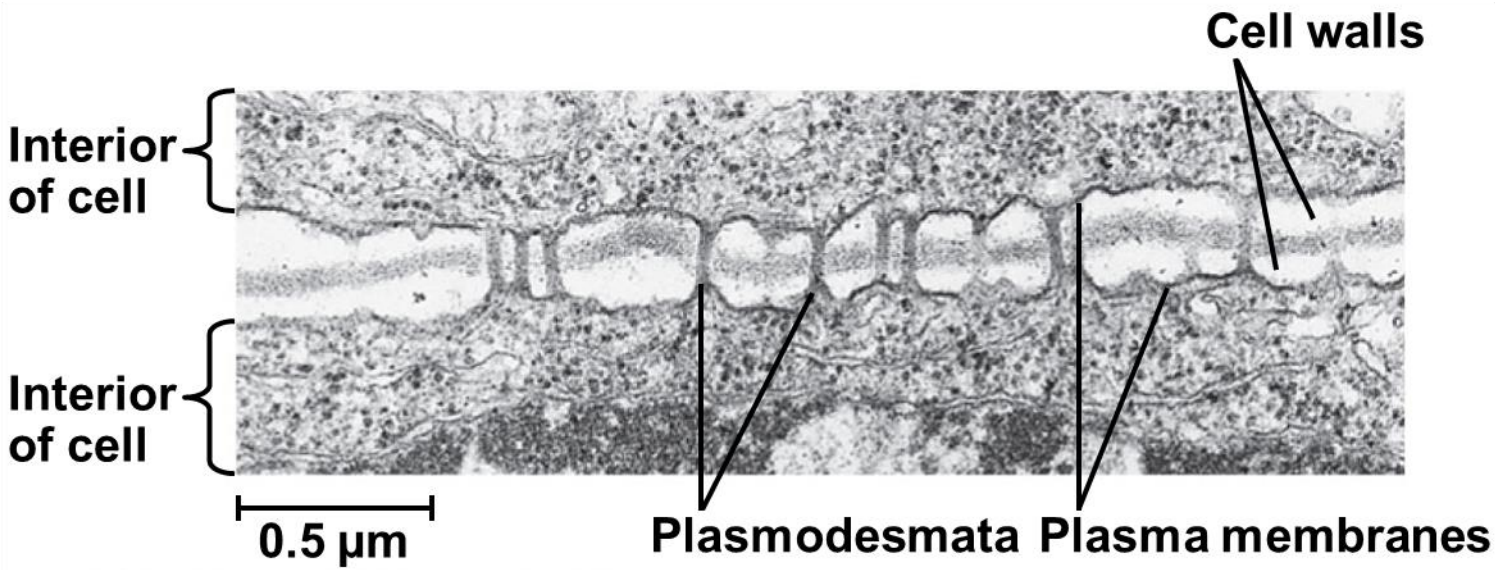
Intercellular junctions facilitate this contact.

There are several types of intercellular junctions
Plasmodesmata
Tight junctions
Desmosomes
Gap junctions

Plasmodesmata in Plant Cells

Plasmodesmata are channels that connect adjacent cells. They are essentially “tunnels” in the cell wall.

Water and small molecules can move from one cell to another via a thin strand of cytoplasm that passes through each channel.



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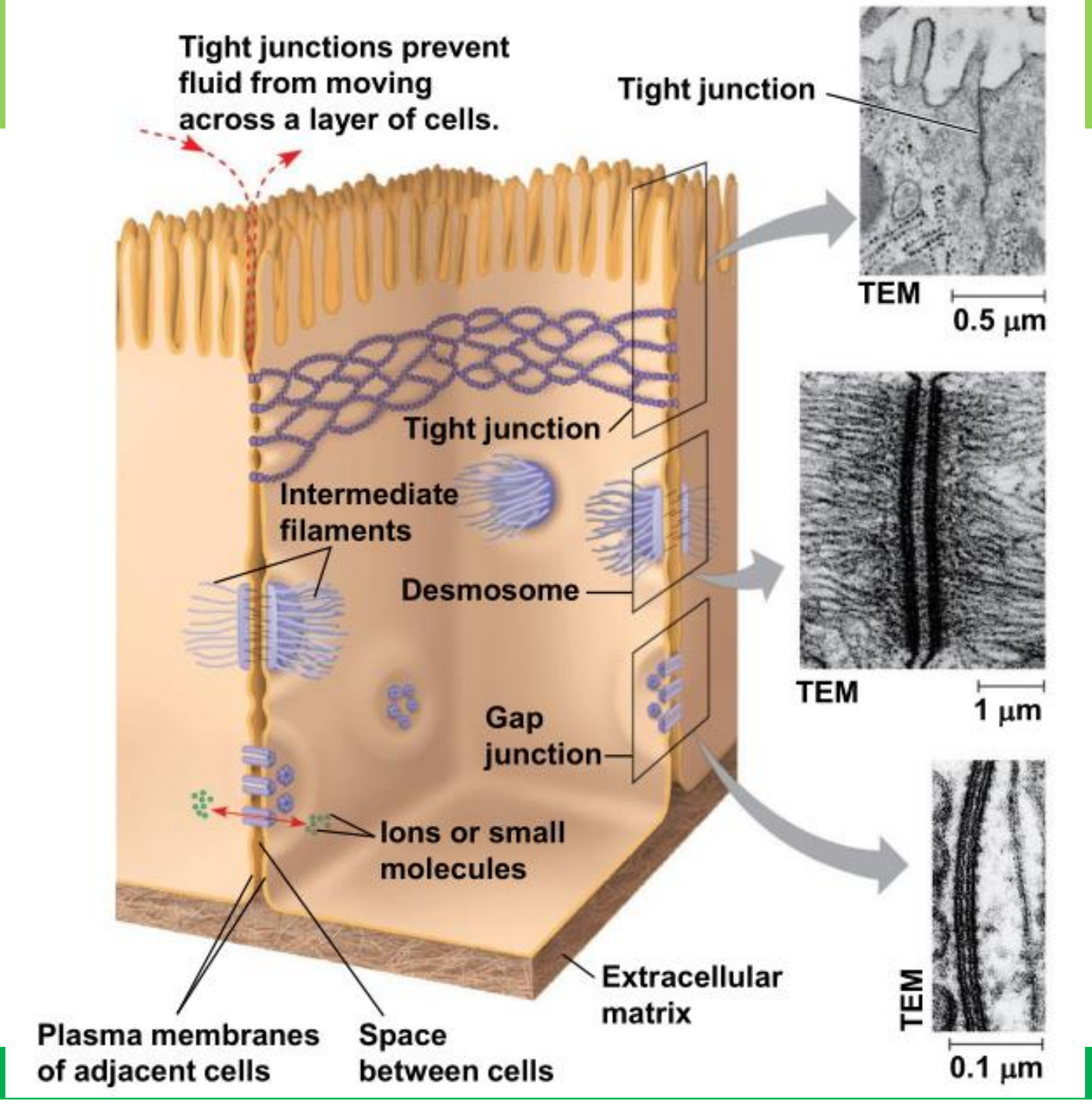
Tight Junctions, Desmosomes, and Gap Junctions in Animal Cells

At **tight junctions**, membranes of neighboring cells are pressed together, preventing leakage of extracellular fluid. For example, tight junctions between skin cells make us watertight.

Desmosomes (anchoring junctions) fasten cells together into strong sheets. Intermediate filaments made of sturdy keratin proteins anchor desmosomes in the cytoplasm. Desmosomes attach muscle cells to each other in a muscle.

Gap junctions (communicating junctions) provide cytoplasmic channels between adjacent cells Gap junctions consist of membrane proteins extending from the membranes of the two cells. These proteins create pores through which ions, sugars, amino acids, and other small molecules may pass.

All are especially common in epithelial tissue which lines the external and internal surfaces of the body.


Tight junctions prevent fluid from moving across a layer of cells.
Tight junction
TEM
0.5 µm
Tight junction
Intermediate filaments
Desmosome
TEM
1 µm
Gap junction
Ions or small molecules
TEM
0.1 µm
Extracellular matrix
Plasma membranes of adjacent cells
Space between cells